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Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV

By
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Abstract

Over ninety AI models have been built to predict sepsis in real time, yet they fail in practice: Wang et al. (2025) report a clinical Utility Score falling from +0.381 internally to -0.164 externally while AUROC barely moves. This thesis asks whether that failure belongs to the model, to the sepsis label, or to the analyst’s discretionary choices.

Using MIMIC-IV v3.1 (65,078 adult ICU stays) with external validation on eICU-CRD v2.0, three pre-registered Sepsis-3 label variants are crossed with five model classes, two split designs and two evaluation metrics, under stay-level cluster-bootstrap intervals with Holm and Benjamini–Hochberg correction. Because all three variants share one culture-plus-antibiotic anchor, the conjunction is additionally split into its limbs (including a deterioration label using no treatment timestamp) and re-timed under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset rule.

The results invert the study’s premise. Model class dominates the decomposition, moving AUROC by 0.1499 against the label’s 0.0123 on an interval excluding zero; none of the four testable hypotheses is confirmed and two fail in the direction opposite to the one predicted. An interpretable hazard model is not outperformed by gradient-boosted trees, leading them by 0.0165 AUROC, and random splits do not inflate AUROC relative to temporal ones. Discrimination degrades on transport under every label, but the Sepsis-3 anchor is recoverable in only 1.55% of eICU-CRD stays, so the degradation is reported as a direction and not as a magnitude.

The label’s consequence lies elsewhere. Under the primary label no model exceeds a Utility Score of 0.031 at any threshold, or 0.111 under any pre-registered variant, on a scale bounded by 1; the two narrowest readings agree at only $\kappa = 0.267$ on which patients are septic, so the label fixes the cohort and the denominator of every clinical quantity computed on it. And the pre-treatment window contains zero Sepsis-3 events, proved here to follow from the conjunction-based onset rule rather than discovered in the data. Re-timing the *same* labelled stays by the standard rule places 71 onsets before treatment begins; the treatment-independent label places 1,362 there.

Better models cannot repair a label whose *membership* is constituted by the actions it is meant to anticipate, and whose onset *timing* decides whether anticipation is measurable at all.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; label definition;

variance decomposition; clinical utility; TRIPOD+AI.

Declaration

I declare that this thesis is my own work, that it has not been submitted for any other academic award, and that all sources consulted are acknowledged in the text and listed in the references. The research question, the pre-registered hypotheses, the study design and the interpretation of the findings are my own, as are the decisions recorded as Protocol Amendments 1–8 and the judgements about what the evidence does and does not support.

Use of Generative AI

During this project I applied the PAIR framework to use generative AI responsibly.

- **Problem:** building a complete pipeline for real-time sepsis onset prediction required cohort extraction and labelling under several Sepsis-3 readings, person-hour panel construction, model implementation across five model classes, an external-validation arm, a clinical-metrics and inference layer, and the documentation of all of it.
- **AI:** I used Claude to discuss implementation choices, to refactor early prototypes into the twelve-stage structure presented here, to audit that code for defects, and to produce documentation drafts.
- **Interaction:** all generated output was critically reviewed, modified and adapted to fit this study’s design. The research question, the pre-registered hypotheses, the label definitions, the model choices, the configuration values, the decisions recorded as Protocol Amendments 1–8 and the reported results are my own.
- **Reflection:** this approach improved development speed and documentation quality while keeping the experimental design, the analysis and the final conclusions my own work.

Generative AI was *not* used to generate data, to produce or alter any result, or to decide what the results mean. Every quantity this study estimates

(every AUROC, interval, p-value, count, rate and calibration statistic) is written into this document mechanically from the pipeline's own output tables by `thesis_constants.R` (Appendix A.3), and a quantity the pipeline did not produce typesets as a visible marker rather than as a plausible number. Three classes of number are typed rather than generated, and none of them is a result of this study: figures quoted from prior work, each carrying its citation; thresholds and design constants fixed by the pre-registration before any modelling, such as the equivalence margin and the correction levels; and, in running prose, values rounded for readability from the generated macro that states them exactly in the adjacent table. The analytical claims, the hypothesis verdicts and the limitations are mine, and I take full responsibility for the content of this thesis.

Signed: _____

Date: _____

Abdul Ahad

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The open-source R and Python communities, in particular the maintainers of `data.table`, `arrow`, `nnet`, `xgboost`, `sandwich`, and `pmsampsize`, are acknowledged for the packages that made this pipeline possible.

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Chapter 1

Introduction

1.1 Context and Motivation

Sepsis is a life-threatening condition caused by the body’s extreme response to infection [1]. It kills about eleven million people every year, nearly one in five deaths worldwide [2]. In the ICU, every hour of delayed treatment raises the risk of death [3]. An early warning system that detects sepsis before doctors recognise it could save lives by starting treatment sooner.

This potential has driven major research investment. Wang et al. [4] reviewed ninety-one published real-time sepsis prediction models and found a median internal AUROC of 0.811, suggesting strong performance. But the same review found that the clinical Utility Score collapsed from +0.381 internally to −0.164 externally. A negative score means the model’s alerts cause more harm than benefit. AUROC did not detect this collapse because it only measures whether a model can rank patients correctly, not whether its alerts are timely or useful. This gap between apparent accuracy and real-world usefulness has also been observed in deployment: Wong et al. [5] validated the Epic Sepsis Model v2 prospectively across 227,091 encounters at four US health systems and found encounter-level AUROC between 0.82 and 0.92 alongside positive predictive value of 0.13 to 0.26: 21 to 35 alerts investigated for every sepsis case identified.

The key question is therefore not “which model is best?” but something more fundamental: *how much of a model’s reported performance comes from the model itself, and how much comes from how the study was designed: how sepsis was defined, how data was split, and which metric was reported?*

Cohen et al. [6] showed on MIMIC-III that varying only the Sepsis-3 definition changed the identified cohort from 867 to 2,178 patients, and that the label choice affected performance as much as the model choice. Kamran et al. [7] showed that published models partly learn to predict when clinicians start treatment rather

than when patients deteriorate. Together, these findings suggest that much of the reported performance may come from the researcher’s design choices, not from the AI model.

This thesis tests that hypothesis directly. Instead of fixing the label, the split, and the metric and asking which model wins, it treats all three as **sources of variance** and measures how much each one contributes to the spread in reported performance.

1.2 Limitations of the Conventional Approach

The standard approach to sepsis prediction follows a simple formula: choose one Sepsis-3 definition, use one train/test split, report AUROC, and compare models. Most of the ninety-one models reviewed by Wang et al. [4] follow this pattern.

This approach has four key problems:

1. **The sepsis label is not fixed.** Cohen et al. [6] showed that different ways of applying Sepsis-3 to the same database can create very different patient groups (867–2,178 cases). The label choice can change results as much as the model choice.
2. **The data split can inflate results.** Guo et al. [8] found that sepsis prediction suffers the largest performance drop of any clinical task when tested on data from a different time period. Random splits let future patterns leak into training, making models look better than they are.
3. **AUROC hides the real failure.** Wang et al. [4] showed that a model can keep a high AUROC while its clinical usefulness drops to harmful levels. The failure is invisible under AUROC alone.
4. **The label captures doctor behaviour.** Kamran et al. [7] showed that models may learn when doctors suspect sepsis and begin treatment, not when the patient’s condition actually worsens.

A study that fixes all of these choices and compares only models measures one source of variation and leaves three others unmeasured. Whether those three are of comparable size to the one that is reported is an open question rather than a settled one, and it cannot be answered by a literature that never varies them. This thesis fills in the missing cells of the comparison and estimates each contribution with an interval attached, so that the ordering between them is established rather than assumed.

1.3 Research Question and Pre-Registered Hypotheses

Research question. *In real-time sepsis prediction on MIMIC-IV, how much of the reported performance variation comes from (a) the choice of model, and (b) the researcher’s decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?*

Six hypotheses were pre-registered before any of the modelling reported here:

- H1 (Label dominance)** Changing the sepsis definition affects AUROC at least as much as changing the model.
- H2 (Coefficient instability)** At least one clinical feature changes its effect direction, or changes in magnitude by a factor of two or more, across label variants.
- H3 (Treatment leakage)** Restricting evaluation to before treatment begins reduces AUROC by at least 0.03.
- H4 (Split optimism)** Time-based testing gives at least 0.02 lower AUROC than random testing.
- H5 (Metric divergence)** The Utility Score declines more than AUROC when moving from internal to external validation. This is tested, for the reasons given in Section 5.10.7, on the AUROC limb alone.
- H6 (Model-class null)** The primary statistical model is not outperformed by gradient-boosted trees by more than 0.02 AUROC. The pre-registered statement is two-sided; only the upper side is testable against these data, as Section 5.10.8 sets out.

H1 and H6 are in deliberate tension: if H1 is confirmed, the label is the main source of variation; if H6 is rejected, the model also matters. Both outcomes are informative.

These six form a single confirmatory family and are tested with Holm–Bonferroni control of the family-wise error rate. Every analysis reported in this thesis carries an explicit status (confirmatory, exploratory or descriptive), and the status is not assigned in the prose: it is held in a machine-readable analysis register that the inference stage writes from the same objects it corrects, and reproduced in Table 5.32. Alongside H1–H6 the register marks a further nine analyses confirmatory because their estimand was fixed before any modelling, among them the internal performance estimates, calibration, decision-curve

net benefit, per-patient detection and the external validation; most of these are estimation rather than testing and so take no multiplicity correction. The equity analyses are exploratory: §9 of the protocol fixed their scope but not the levels, the reference or the direction of any contrast, so they are corrected by Benjamini–Hochberg within their own families. Section 3.14 sets out the plan and states plainly that it was written after the analyses had been run.

1.4 Contributions

1. **C1 (Label variance).** Label sensitivity itself is established by Cohen et al. [6] and Dutta et al. [9], and neither the survival setting nor the multi-variant design is new. What is new, to the best of our knowledge, is the combination: three Sepsis-3 readings applied to a discrete-time competing-risks model on MIMIC-IV, with the label spread and the model-class spread measured on the same test rows and each carrying an interval, so that the two can be compared as estimands rather than as ranges of point estimates (Section 2.1 states the search strategy and the bound on this claim).
2. **C2 (Treatment leakage separated: membership from onset timing).** A proof that a conjunction-based Sepsis-3 onset rule, the rule used here and widely in the MIMIC-IV literature, makes pre-treatment prediction structurally impossible, together with the measurement that the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule applied to the identical stays recovers 71 pre-treatment onsets at 0.8081 AUROC. Kamran et al.’s [7] concern is confirmed for label *membership* and narrowed for label *timing*.
3. **C3 (Utility ceiling documented).** Swept across the whole threshold grid rather than scored at a conventional cut-off, no model exceeds 0.031 under the primary label or 0.111 under any pre-registered variant (a few per cent above the strategy of issuing no alerts, on a scale whose upper bound is 1), and reaching even that costs tens of false alerts per true one. This extends Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison and settles it against the operating-point objection.
4. **C4 (Coefficient instability).** 6 of the 22 shared clinical covariates are unstable across label variants, 2 of them by direction, supporting Lauritsen et al.’s [10] framing-instability finding on MIMIC-IV. Referring each flagged covariate to its own cluster-robust standard error narrows this to 3 whose movement exceeds its estimation noise, all of them indicators of whether a test was ordered (Section 5.10.4).

5. **C5 (Equity-by-label analysis).** Subgroup performance disparity and subgroup variation in who gets labelled are both reported by Wang et al. [11], though for mortality prediction among patients already identified as septic rather than for onset prediction. What is new, to the best of our knowledge, is measuring the two on the *same* strata in the same onset-prediction cohort under common false-discovery control, which is what shows that only one of them is precisely enough estimated to support a claim (Section 2.1).
6. **C6 (Reproducible pipeline).** A pre-registered, seed-fixed pipeline following TRIPOD+AI [12] and PROBAST+AI [13] reporting standards.
7. **C7 (Definitional vs. empirical label variance separated).** The Sepsis-3 conjunction is split into its two limbs and each is labelled separately, so that the part of the label effect guaranteed by the definition can be told apart from the part discovered in the data. Two results follow. The onset *time* under every pre-registered Sepsis-3 variant is fixed entirely by the treatment anchor, the organ-dysfunction criterion filters which stays are labelled and moves no onset by a single hour, while a treatment-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance. And the treatment anchor alone is about as predictable as the Sepsis-3 conjunction built on top of it, so the difficulty lies not in the conjunction but in when the label is stamped. That second result is model-dependent and is reported as such: the anchor-attributable fraction is $\phi_{\text{anchor}} = 1.041$ [0.990, 1.097] for the primary model, an interval including 1, and 1.099 [1.037, 1.164] for the gradient-boosted comparator, which excludes it. The same comparison shows the empty pre-treatment window of C2 to be a property of the onset rule, not of the patients: a treatment-independent label records 1,362 onsets there at 0.8498 AUROC, and Sepsis-3 itself records 71 once its onset is timed as the standard specifies (Section 5.6).

1.5 Thesis Structure

Chapter 2 reviews the evidence that motivated this study’s design. Chapter 3 describes the data, labels, models, and evaluation plan. Chapter 4 explains the pipeline architecture and its twelve scripts. Chapter 5 presents all results, organised around the six hypotheses. Chapter 6 interprets them, bounds what each estimate can support, and states the limitations. Chapter 7 draws conclusions and outlines future work. Appendix A gives the source-code repository and the procedure for reproducing every reported number.

Chapter 2

Literature Review

This chapter establishes the evidence base for the study and evaluates it critically. It is organised around the questions the field has not settled rather than around a catalogue of studies, because the design decisions taken in Chapter 3 are responses to specific unresolved disputes, not to a consensus.

Four such disputes structure what follows. The first is taxonomic: the field routinely partitions itself into “statistical” and “machine-learning” traditions, and Section 2.2 argues that this partition does not carve the problem at its joints. The second concerns the outcome itself: what counts as sepsis onset, and whether it is a stable enough object to support the interpretable effect estimates that motivate statistical modelling (Sections 2.3 and 2.4). The third concerns validation: what the published external-validation record actually shows, and why discrimination and clinical utility tell opposite stories about it (Sections 2.5 and 2.6). The fourth is methodological, and is genuinely open in the statistical literature rather than merely under-reported: no survival formulation for ICU event data satisfies all of its own assumptions, and Section 2.7 takes a position on which set of violations is acceptable.

Sections 2.8 and 2.9 then treat two narrower questions (which rule-based comparator is worth beating, and whether subgroup validity has been assessed at all), before Section 2.10 draws the threads together into the argument that motivates this study’s research question, and Section 2.11 maps each design decision onto the evidence that warrants it.

A fuller treatment of the same corpus, covering fifty sources including several whose relevance is contextual rather than design-bearing, is available in the companion critical review *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV*. The sources that bear on the decisions taken here are engaged with in this chapter.

2.1 Search Strategy

The literature search was conducted to identify published work on real-time sepsis prediction, with particular attention to label sensitivity, validation methodology, evaluation metrics, and subgroup validity. Table 2.1 documents the protocol.

What kind of review this is. This is a **structured narrative critical review, not a PRISMA-conformant systematic review**. Searches were executed to a documented protocol and every included source was verified against a primary record, but title-and-abstract screening was not conducted against a full de-duplicated yield, and no second reviewer independently screened. The distinction matters for the novelty claims in Section 1.4: those claims are stated as “to the best of our knowledge” and are bounded by the coverage this method achieves, which is deliberate depth on the methodological literature that constrains this study rather than exhaustive coverage of the MIMIC-IV sepsis corpus. An exhaustive systematic review could surface a prior study combining the two analyses in contribution C5; this search did not.

Table 2.1: Literature search strategy. This search supports the novelty claims for contributions C1 and C5 stated in Section 1.4. Because the review is narrative rather than systematic, those claims are stated as “to the best of our knowledge” and are bounded by the coverage above.

Element	Detail
Databases searched	PubMed/MEDLINE, Embase, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, Google Scholar, Cochrane Library, ScienceDirect, SpringerLink, Wiley Online Library, Nature Portfolio, JAMA Network, BMJ Journals, arXiv/medRxiv
Searches executed	35 structured searches across the domains listed under “Search terms”
Date range	Primary sepsis-prediction literature: January 2015 – July 2026. Methodological and definitional foundations (e.g. pooled logistic regression, decision-curve analysis) were searched without a date limit, and the earliest source retained dates from 1990

Element	Detail
Search terms	(sepsis OR “sepsis prediction” OR “sepsis detection”) AND (MIMIC OR “critical care” OR “intensive care”) combined with each of: “label definition”, “label sensitivity”, “onset definition”, “utility score”, “clinical utility”, “fairness” OR “equity” OR “disparity” OR “bias”, “subgroup analysis”, “competing risks” OR “discrete-time survival”
Inclusion criteria	(1) Peer-reviewed journal article, conference paper, or preprint on a recognised server (arXiv, medRxiv); (2) Develops or evaluates a real-time (hourly or sub-hourly) sepsis prediction model; (3) Uses EHR data from an ICU population; (4) Reports at least one quantitative performance metric
Exclusion criteria	(1) Non-English language; (2) Editorials, commentaries, or letters without original results; (3) Models for sepsis severity staging or mortality only (not onset prediction); (4) Studies on paediatric or neonatal populations exclusively
Verification standard	Every included source was checked against a publisher record, PubMed or PMC record, or institutional repository entry, with journal, volume, year and DOI confirmed. Every numerical claim attributed to a source was located in that source’s own abstract, results text or publisher-hosted record, never inferred from a secondary citation
Total included	50 sources in the companion critical review (46 newly verified in this pass, 4 carried forward from the base review of 19 sources); 45 are cited in this thesis
Excluded at verification	4 items encountered in secondary citation lists could not be verified to the standard above and were excluded; 1 further item is retained in the companion review with an explicit note of incomplete metadata and is not cited in this thesis

Figure 2.1 summarises the identification, verification and inclusion process. It follows the reporting logic of PRISMA 2020 [14], an auditable count at every stage, but is presented as a search-and-verification flow rather than as a PRISMA flow diagram, because the screening stage a PRISMA diagram reports was not

performed in this form.

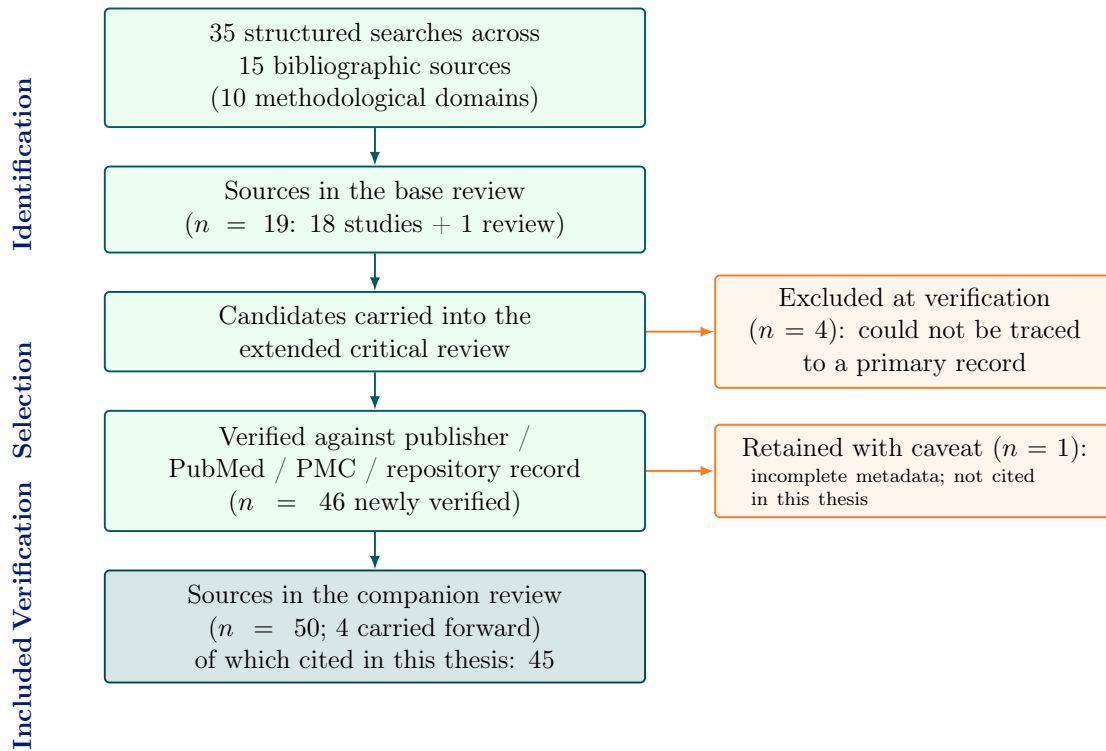


Figure 2.1: Search, verification and inclusion flow. Counts are auditable at every stage, following the reporting logic of PRISMA 2020 [14], but this is not a PRISMA flow diagram: no title-and-abstract screening pass over a full de-duplicated yield was performed, and screening was not duplicated by a second reviewer. The review is narrative and critical, and the novelty claims it supports are bounded accordingly (Section 1.4).

What the search establishes about novelty, and what it does not. Two contributions rest on search-based evidence, and both are claims about a *combination* of analyses rather than about either analysis on its own. Each component has clear precedent; what the search did not identify is any study performing them together.

1. **C1 (Label variance under a dynamic survival model).** Label sensitivity itself is well established: Johnson et al. [15], Cohen et al. [6] and Dutta et al. [9] all demonstrate it, and the last two are the direct precedents for this study. Cohen et al. include a survival model among the families they compare, so the survival setting is not itself new either. What the search did

not identify is a study applying three or more Sepsis-3 readings to a *discrete-time competing-risks* model on MIMIC-IV with the label spread reported as an estimand carrying an interval, alongside the model-class spread measured on the same test rows. The claim is that combination, not label sensitivity as such.

2. **C5 (Performance disparity and label sensitivity across the same subgroups).** Subgroup performance disparity in sepsis cohorts is reported by Wang, Li, Naidech and Luo [11], and subgroup variation in *who gets labelled* is reported in the same paper. It is important to be exact about what that study is: it predicts **mortality among patients already identified as septic**, not sepsis onset. It therefore cannot serve as the prior art against which an onset model's subgroup performance is judged, and the search did not identify any onset-prediction study reporting subgroup discrimination at all. Studies examining label sensitivity for onset prediction (Cohen et al., Dutta et al.) do not stratify by demographic subgroup. The novel element is the conjunction (subgroup discrimination *and* subgroup label sensitivity, measured on the same subgroups, in the same cohort, under common false-discovery control), and not either half.

A bounded claim about clinical utility. Wang et al. [4] reviewed ninety-one real-time sepsis prediction models. Of these, only a minority reported the PhysioNet Utility Score or an equivalent net-benefit metric; the majority reported only AUROC and/or AUPRC. The claim that published models do not demonstrate good clinical utility is therefore based on the subset for which a utility or net-benefit metric was available: among those, the median external Utility Score was -0.164 , indicating net harm. For models reporting only AUROC, clinical utility could not be assessed, and no claim about their utility is made here.

Key Concepts

Several domain-specific terms recur throughout this chapter:

- **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV): a freely available, de-identified database of approximately 74,000 ICU admissions at Beth Israel Deaconess Medical Center, Boston, covering 2008–2019 [16].
- **Sepsis-3**: the current international consensus definition of sepsis as life-threatening organ dysfunction caused by a dysregulated host response to

infection, operationalised through a combination of suspected infection (antibiotic order plus culture draw) and a rise in the Sequential Organ Failure Assessment (SOFA) score of ≥ 2 points [1].

- **AUROC** (Area Under the Receiver Operating Characteristic curve): a discrimination metric that measures a model's ability to rank positive cases above negative cases, ranging from 0.5 (random) to 1.0 (perfect separation). It is invariant to any monotone transformation of the score, and therefore cannot see miscalibration; it does not account for alert timing or false-alarm cost.
- **Utility Score**: a net-benefit metric proposed by the PhysioNet/Computing in Cardiology Challenge [17] that rewards early, true alerts and penalises late alerts and false alarms, producing values that can be negative when a model causes more harm than benefit.
- **NEWS2** (National Early Warning Score 2): a standardised clinical scoring system aggregating six physiological parameters (respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, level of consciousness, and temperature) into a single risk score used for bedside deterioration screening.

Research Questions for the Review

The review is structured around four questions that directly inform the study's design:

- RQ1.** How much does the choice of sepsis label definition affect model performance, and is this effect comparable in magnitude to the choice of model architecture?
- RQ2.** What methodological pitfalls (validation strategy, metric selection, treatment-time leakage) most distort reported performance in the sepsis prediction literature?
- RQ3.** What statistical framework is appropriate for modelling hourly ICU event data when the standard proportional-hazards assumption is violated?
- RQ4.** Does the existing equity literature on sepsis prediction account for label-level variation across demographic subgroups, or only model-level variation?

2.2 The Field’s Usual Partition Is Not the Load-Bearing One

Reviews of this literature conventionally divide it into a classical statistical tradition and a machine-learning tradition, and the division usually carries an implicit evaluative charge: interpretable models on one side, accurate models on the other. The evidence does not support the partition, and treating it as real leads to a specific analytical error that this study is designed to avoid.

The boundary cases dissolve it. The Artificial Intelligence Sepsis Expert [18], routinely classified as a machine-learning system, is a modified Weibull–Cox proportional-hazards formulation: the leading “machine-learning” sepsis model of its generation is built on a survival-analytic backbone. TREWScore [19], a Cox model and the canonical *statistical* benchmark for this task, is described throughout the machine-learning literature as an early machine-learning early-warning system, and its commercial successor is marketed as one [20]. Cohen et al. [6] place tree-based, deep-learning and survival models in a single comparison frame because the distinction is not load-bearing for the question they ask. And TRIPOD+AI [12], the governing reporting standard, covers “regression or machine learning” under one checklist, having concluded that the reporting failures are common to both.

The partition that *is* load-bearing separates models by what they assume about time.

- **Static models** take a snapshot (baseline, or day one, or the worst value in the first 24 hours), and issue one prediction. The great majority of the MIMIC-IV statistical corpus sits here: a candidate biomarker or composite index is regressed on mortality through an almost invariant pipeline of variable selection, Cox and logistic regression, splines, a nomogram, and ROC, calibration and decision curves. These are *prognostic* instruments and cannot in principle perform real-time onset detection.
- **Dynamic models** consume a stream and update a risk estimate as it arrives. TREWScore, the Artificial Intelligence Sepsis Expert, the PhysioNet Challenge entries [17], Moor et al.’s deep temporal model [21], and landmark and joint models all sit here. Whether the update rule is a partial likelihood, a recurrent network or a boosted hazard is a secondary question.

Reframing the taxonomy this way corrects a claim that the statistics-versus-ML framing tends to generate: that statistical methods have neglected onset prediction. They have not. Henry et al. [19] fitted a Cox model for septic-shock onset on MIMIC-II; Fagerström et al. [22] fitted one on MIMIC-III using Henry et al.’s exact inputs and target as the baseline for an LSTM; and Cohen et al.’s survival arm

does so on MIMIC-III again. What is genuinely absent is narrower and harder: a properly specified, assumption-tested, label-sensitivity-analysed, temporally validated dynamic model for *onset* on MIMIC-IV.

The Fagerström result also bounds a common claim about model class. Christodoulou et al. [23], reviewing 71 studies drawn from 927 screened, found no performance advantage for machine learning over logistic regression in clinical prediction, and observed potential bias in the validation procedures of 68% of them. That is the strongest available warrant for treating a classical specification as a serious contender. But its scope is binary outcomes on tabular data, and sepsis onset prediction is not that problem: it is a temporal problem over irregularly sampled multivariate streams in which the quantity of interest is a lead time. On precisely that axis Fagerström et al. report the deep model detecting patients up to twenty hours earlier than the Cox model at comparable sensitivity and specificity. The defensible position is therefore narrower than “statistics matches machine learning”: it is that the equivalence is established for static risk stratification and is contested for the temporal early-warning task. H6 in this study is a direct test of that narrower proposition, on one dataset, under one label, with the split and the evaluation held fixed: the conditions Christodoulou et al. find are usually absent.

Two further consequences follow for design. Moor et al.’s systematic review [24] identifies low comparability and reproducibility as the field’s principal defect, and singles out case-control matching as a concrete trap: when a control patient’s “onset time” is chosen near discharge while a case is aligned near onset, the two are trivially separable and apparent performance is inflated. Fleuren et al. [25] reach a compatible conclusion from a meta-analysis of diagnostic test accuracy, attributing the between-study heterogeneity that prevents pooling to methodological rather than clinical variation: heterogeneity that Cohen et al. later show is partly label variation. This study therefore performs no case-control matching at all: every at-risk person-hour contributes, which removes the alignment decision from the analyst’s hands.

2.3 The Sepsis Label Is Not a Single Object

The common assumption that “the MIMIC-IV sepsis cohort” is one well-defined group is wrong, and the evidence against it is now strong enough to treat as an established property of the problem rather than a curiosity.

Johnson et al. [15] applied five retrospective sepsis identification algorithms, including the Sepsis-3 clinical criteria, to 11,791 qualifying ICU admissions drawn from 23,620, and found agreement across them that the authors themselves characterise as only acceptable (Cronbach’s $\alpha = 0.40\text{--}0.62$). These algorithms are the

very queries that MIMIC-based studies run to construct their cohorts. Different studies claiming to study the same disease in the same database are therefore, measurably, not studying the same patients.

Cohen et al. [6] established what this costs a prediction model, and their design is the direct precedent for this thesis. Working in MIMIC-III, they held the data fixed and varied only the reading of the Sepsis-3 onset rule across three defensible interpretations of the same consensus definition. The number of identified septic admissions ranged from 867 to 2,178, a factor of 2.5, purely as a function of that reading. They then trained representative models spanning tree-based, deep-learning and survival families under each definition. Holding the definition fixed, the best method gained 1–5% AUROC over its rivals; holding the method fixed and varying the onset definition produced swings of 0–6%. The analyst’s discretionary reading of the label moved performance at least as much as the entire choice of modelling paradigm.

Dutta et al. [9] confirm this is not a MIMIC artefact, and the corroboration is unusually strong because almost nothing about the setting is shared. A locally trained gradient-boosted model generating predictions every fifteen minutes across nine acute-care hospitals was evaluated on 198,494 encounters against three electronically computable definitions: Sepsis-3, the SEP-1 management bundle, and the CDC Adult Sepsis Event. Performance varied substantially by definition, with expected calibration error differing across them (0.05 for Sepsis-3, 0.06 for Adult Sepsis Event, 0.07 for SEP-1) and decision-curve analysis showing higher standardised net benefit under Sepsis-3 and SEP-1. Cohen et al. established definitional sensitivity retrospectively, in MIMIC-III, at $n \approx 2,000$, with research models; Dutta et al. reproduce it prospectively, in a live deployment, at $n \approx 200,000$, with a commercial model. The finding is robust to dataset, scale, model class and setting.

The critical gap in this line of work is what makes it the starting point for this study rather than its conclusion. Both papers demonstrate *that* the label matters; neither separates the part of the label effect that is guaranteed by the definitions from the part discovered in the data, and neither attaches an interval to the comparison between label variance and model variance. A spread of point estimates is not an estimand. Cohen et al.’s “0–6% versus 1–5%” is a comparison of two ranges, each computed without uncertainty, on a cohort of roughly two thousand admissions. Whether the two sources of variance are genuinely of comparable size is therefore not settled by it, and Sections 5.6 and 5.10 of this thesis are constructed to settle it on a cohort thirty times larger with intervals attached to both quantities.

Figure 2.2 illustrates the mechanism: the same patient population, subjected to different but equally valid Sepsis-3 interpretations, produces overlapping but

distinct cohorts.

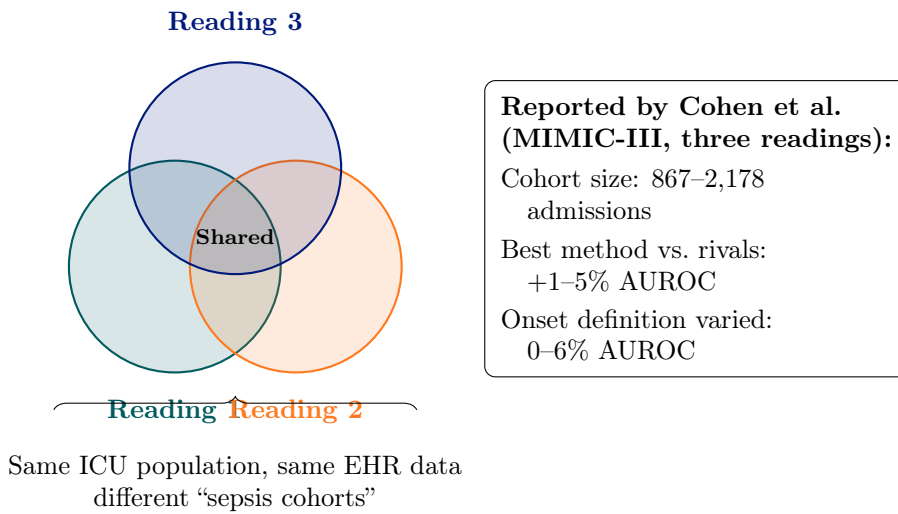


Figure 2.2: Schematic of label variance. Three valid Sepsis-3 interpretations applied to the same ICU population produce overlapping but distinct cohorts. The boxed quantities are those *reported by Cohen et al.* [6] on MIMIC-III; they are shown to convey the scale of the effect that motivates this study and are not predictions, targets, or results of the present work. The corresponding measurements on MIMIC-IV are reported in Chapter 5.

2.4 The Label Is Constituted by Clinical Action

Sepsis-3 defines sepsis as organ dysfunction in a patient with suspected infection [1], and the operational criteria derived by Seymour et al. [26] render “suspected infection” through two clinician actions: the drawing of a culture and the administration of an antibiotic. The consequence is a labelling rule that reads two clocks, one of which is a record of what the treating team did.

Kamran et al. [7] pursued this to its conclusion and produced the deepest threat to the enterprise. Clinicians frequently recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the moment the criteria are met is therefore partly trained to predict *when the clinician will act*, and the physiological record available at prediction time is itself shaped by that developing suspicion: tests are ordered, monitoring escalates, and observations are recorded that would not otherwise exist. Re-evaluating the Epic Sepsis Model on 77,582 hospitalisations, the authors report an AUROC of 0.62 when predictions issued after clinical recognition are retained, falling to 0.47, worse than chance, once they are excluded.

For a hazard model the implication is exact. A coefficient estimated on the interval up to Sepsis-3 onset is in part a coefficient for clinician behaviour. Lactate is drawn because someone is worried; the predictor and the outcome share a common cause in the clinician’s suspicion, and no amount of proportional-hazards testing detects this, because the model is correctly estimating the hazard of a contaminated event. The remedy is definitional rather than statistical: evaluation must be anchored to a prediction time that precedes clinical action.

Lauritsen et al. [10] generalise the problem beyond the label to the surrounding design. They formalise *framing* (the observation window, the prediction window, the window shift, and the event that triggers a prediction) enumerate eight possible framing structures, and evaluate four on the same underlying data. Framing determines both discrimination and calibration, and, critically, can produce opposing interpretations of the same physiological variable: their worked example is SpO₂, whose apparent relationship to sepsis risk reverses direction with the framing. A model with strong discrimination *and* strong calibration may nonetheless be clinically unusable.

Taken together these two results close a trap around the argument for interpretable models. If framing can flip the sign of a coefficient, then the coefficient reports a property of the study design as much as a property of the patient. An uninterpretable model that is wrong is a nuisance; an *interpretable* model that is confidently wrong about the direction of a physiological effect is a hazard, because clinicians act on directions. H2 in this study is the direct test of whether that hazard is realised on MIMIC-IV.

Where this study departs from Kamran et al. is in the scope of the remedy. Treatment-anchored evaluation restricts *when* performance is measured, but it leaves the anchoring inside the label untouched: a Sepsis-3 case is still a case only because someone ordered a culture and an antibiotic. A comparison confined to Sepsis-3 readings (or to SEP-1 and the CDC Adult Sepsis Event, both of which are defined on antibiotic days) cannot separate definitional from empirical label variance, because every member of that family keys on treatment timing. Breaking the circularity requires a label built from physiology alone, which is why this study constructs one (Section 3.6) rather than confining itself to the consensus family.

2.5 The Generalisation Gradient

Assembling the validation evidence into one frame exposes a consistent structure that no single study makes visible. Table 2.2 collects it.

Study	Model	Transition	Degradation
Nemati et al. [18]	Weibull-Cox (AISE)	Emory $\rightarrow$ MIMIC-III; AUROC 0.83–0.85	Reported indistinguishable
Reyna et al. [17]	853 Challenge entries	Two seen hospital systems $\rightarrow$ one unseen; rank correlation $\rho = 0.949$ between seen systems	$\rho = -0.033$ and 0.013 to the unseen system
Moor et al. [21]	Deep temporal	Four international ICU databases, 136,478 admissions; internal AUC 0.846	External AUC 0.761 (-0.085); 0.807 with 10% fine-tuning
Wang et al. [4]	91 models (meta-level)	Internal $\rightarrow$ external, full-window	AUROC 0.811 $\rightarrow$ 0.783 (n.s.); Utility 0.381 $\rightarrow$ -0.164
Wong et al. [27]	Epic Sepsis Model v1	Proprietary $\rightarrow$ Michigan Medicine, 38,455 hospitalisations	AUROC 0.63 ; PPV 12%
Ostermayer et al. [28]	Epic Sepsis Model v1	Proprietary $\rightarrow$ two county emergency departments	Sensitivity 14.7% ; PPV 7.6%
Wong et al. [5]	Epic Sepsis Model v2	Prospective, four US health systems, 227,091 encounters	AUROC 0.82–0.92 but PPV 0.13–0.26 ; 21–35 alerts per case
Guo et al. [8]	Feed-forward network	MIMIC-IV, within database, across admission eras	Sepsis worst-degraded of four tasks

Table 2.2: The published external-validation record for real-time sepsis prediction. Nemati et al.’s internal-external equivalence is the exception and is instructive: their external set shares an era, a country and a care model with the development set, which is a mild shift by the standards of Moor et al.’s Netherlands-and-Switzerland comparison. The apparent counter-example is better read as evidence that the size of the external penalty scales with the distance of the shift.

Three regularities follow, and each dictates a design decision taken in Chapter 3.

The external penalty for a well-built model is roughly 0.085 AUROC. This is Moor et al.’s $0.846 \rightarrow 0.761$ across four databases spanning the United States, the Netherlands and Switzerland, and it is consistent in direction, if not in significance, with Wang et al.’s meta-level $0.811 \rightarrow 0.783$. The number is specific enough to pre-specify, which is what H5 does; and a study observing a substantially *smaller* drop should first suspect that its external cohort is insufficiently distant rather than conclude that its model is unusually robust.

Rank order does not transfer. Reyna et al. report that team Utility Scores were strongly rank-correlated between the two hospital systems participants had trained on ($\rho = 0.949$) but essentially uncorrelated with performance on the third, unseen system ($\rho = -0.033$ and 0.013). The ranking of algorithms carried almost no information out of distribution. For a study that intends to compare its model against published competitors this is foundational: a benchmark is informative only when run on the same data, under the same label, with the same split, and even then it licenses no inference about which model would win at a different hospital. Every comparator in this study is therefore refitted and rescored inside the same pipeline, and no performance figure is compared against a published one except to bound an expectation.

The split design is not neutral, and sepsis is the worst case. Guo et al. [8] partitioned MIMIC-IV into admission eras (2008–2010, 2011–2013, 2014–2016, 2017–2019) and evaluated four prediction tasks under temporal shift: mortality, long length of stay, sepsis, and invasive ventilation. Sepsis was the worst-affected. Domain-generalisation and unsupervised domain-adaptation algorithms failed to beat plain empirical risk minimisation (AUROC differences -0.003 to 0.050), so there is at present no algorithmic fix. The consequence is unavoidable and task-specific: a random train/test split on MIMIC-IV is optimistically biased for sepsis, and more so than for the neighbouring ICU outcomes a reader might use to calibrate their scepticism. This study’s primary validation is therefore a temporal split; the random split is retained only to quantify the optimism, which is H4.

A qualification the field does not usually state: Dutta et al. [9] did *not* detect temporal drift over their eighteen-month prospective period. Shift over eighteen months inside one health system is a different phenomenon from shift across a decade of MIMIC-IV, and the two results are compatible. What follows is that “temporal validation” is not a single practice, and the era gap matters as much as the fact of splitting by time.

2.6 Discrimination and Clinical Value Tell Opposite Stories

The most consequential comparison in this literature sits inside a single paper. Wang et al. [4] report, across ninety-one models, that the median AUROC moved from 0.811 internally to 0.783 externally, a difference they report as not statistically significant, while the median Utility Score moved from +0.381 to −0.164. The same models, the same transition, and two metrics that disagree about whether anything happened. On the Challenge’s utility calculus, a negative score means that deploying the median externally validated sepsis model is worse than deploying nothing.

The mechanism is not mysterious. AUROC is a ranking statistic, invariant to monotone transformation of the score, so it cannot see miscalibration; and under low prevalence it is dominated by the enormous population of easily classified true negatives. What changes across sites is not primarily the *ordering* of patients but the *location* of the score distribution relative to the alerting threshold, which is exactly what determines the false-alarm rate, and therefore the utility. Van Calster et al. [29] name calibration the Achilles heel of predictive analytics for this reason: any model driving a threshold-based action is a calibration-dependent instrument whatever its AUROC.

Two deployment results make the same point visible in a second way. Wong et al. [5], validating the Epic Sepsis Model v2 prospectively across 227,091 encounters at four health systems, report encounter-level AUROC between 0.82 and 0.92, an enormous improvement on version 1’s 0.63 [27], alongside positive predictive value of only 0.13 to 0.26, requiring clinicians to investigate 21 to 35 alerts to find one case. Discrimination near ceiling; usability still contested. Ostermayer et al. [28], validating version 1 independently across two county emergency departments, report sensitivity of 14.7% and PPV of 7.6%, worse than Wong et al.’s original finding, in a different setting and case mix.

The v2 result deserves a qualification, because it cuts against the sceptical reading. Version 1 was a logistic regression and version 2 is a gradient-boosted ensemble, so the improvement is evidence against the strongest form of Christodoulou et al.’s claim [23]. But version 2 also changed from globally trained to locally retrainable, and Moor et al. have already shown that local fine-tuning recovers roughly half of the external penalty ($0.761 \rightarrow 0.807$). Model class and localisation are therefore confounded in the published comparison, and the evidence does not permit them to be separated. What is not ambiguous is that PPV remained low and alert burden high at AUROC 0.92.

Two benchmarks carry forward from this section into the evaluation plan. Moor et al. report their deployed system raising **1.4 false alerts per true alert** while

detecting 80% of septic patients a median of 3.7 hours (95% CI 3.0–4.3) before onset. That is the reference burden against which any alert volume should be read, and the 3.7-hour figure is the honest lead time of a good modern system, not the 28 hours of the original TREWScore headline [19]. And Seymour et al. [3], across 49,331 patients at 149 New York hospitals, estimate an odds ratio of 1.04 per additional hour to antibiotic administration. A model delivering Moor et al.’s 3.7-hour lead time with perfect clinician response would therefore purchase an improvement of order $1.04^{3.7} \approx 1.16$ on the odds scale: real but modest, available only if the alert is acted on, only if the patient would not otherwise have been recognised, and only after the false alerts have been paid for.

The consequence for this study’s evaluation plan is direct. A study reporting AUROC, sensitivity, specificity and lead time, and concluding that its model generalises, would have failed to measure the thing that fails. Calibration, AUPRC, the Utility Score, alert burden and decision-curve analysis are not enhancements to that list; they are the list. This is why the Utility Score is the primary metric here and AUROC is descriptive, and, given that the Challenge metric is threshold-dependent, why it is reported across a swept grid of operating points rather than at a single cut-off (Section 5.4).

2.7 The Survival-Methods Dispute Has No Clean Resolution

A study proposing a survival model for ICU event data inherits a dispute that the applied literature usually passes over. It is worth stating in full, because the specification adopted in Chapter 3 is a position in it rather than a default.

1. **Proportional hazards frequently fails in sepsis cohorts.** Kasal et al. [30] compared a standard Cox model against Gray’s flexible survival model in a large severe-sepsis cohort and found hazards for death after sepsis frequently non-proportional. This is an under-cited caution: the assumption is not merely untested in most MIMIC work, it is known to fail in this disease.
2. **Independent censoring also fails.** Resche-Rigon et al. [31] argue that discharge alive is not a non-informative censoring event in ICU cohorts: patients discharged alive are systematically unlike those who remain, so the assumption underlying Kaplan–Meier estimation is violated. The standard remedy is the Fine–Gray subdistribution hazard model [32].
3. **The remedy is itself contested.** Schoenfeld [33] argues that survival methods, explicitly including competing-risks methods, are inappropriate for

ICU outcome studies, and that studies seeking prognostic covariates should use logistic regression instead.

4. **Landmarking does not restore proportional hazards.** The landmark approach, refitting at each prediction time on the subset still at risk, is the pragmatic route to a continuously updated score [34], and is what most proposals in this space gesture toward. It does not in general satisfy the proportional-hazards assumption. Landmarking is a different model with its own approximation error, not an escape.
5. **Joint models restore rigour at a prohibitive cost.** Modelling the longitudinal biomarker trajectories and the event process through shared latent structure outperforms landmarking when the longitudinal sub-model is correctly specified [35], but the approach is computationally expensive, requires a common baseline time, and scales poorly to the heterogeneous, irregularly sampled, high-dimensional streams characteristic of an ICU record.

There is no assumption-free route. What there is, and what this study adopts, is a formulation that dissolves several of these problems rather than testing them: the **discrete-time hazard model**, fitted as a pooled logistic regression over person-hour records with time-varying covariates and a flexible baseline hazard. D’Agostino et al. [36] established that pooled logistic regression is mathematically equivalent to time-dependent Cox regression when the intervals are short and per-interval event rates are low, and Ngwa et al. [37] confirmed the equivalence by simulation. One-hour ICU intervals with hourly sepsis probability below 1% satisfy both conditions comfortably.

Its properties answer the dispute point by point. It is native to the data’s structure, because ICU data is already binned hourly by convention [17], [18]: a discrete-time hazard model *is* an hourly risk score, and no reconciliation between the model’s time scale and the deployment’s is required. It does not require proportional hazards, so item (1) is dissolved rather than tested. It accommodates competing risks directly through a multinomial cause-specific formulation over the person-hour [31], [38], so that discharge alive and death without sepsis are modelled as what they are, which answers item (2) without incurring item (3). It answers Schoenfeld’s objection in item (3) on its own terms, being a logistic regression, while retaining the time-to-event structure he would discard. Its coefficients are log-odds of onset in the next hour given survival sepsis-free to this hour: arguably a more clinically legible quantity than a hazard ratio. And it emits exactly the object the Challenge utility metric consumes, an hourly risk score, so it is directly comparable with the machine-learning literature. Heyard et al. [38], [39] supply the corresponding development and validation methodology for discrete time-to-event data with competing risks.

The Cox model is retained in this study, but as a deliberately weak comparator rather than as the primary specification: its role is to quantify what the field's default choice costs, and its Schoenfeld test is reported as a finding about the default rather than as a check on the model actually used.

2.8 Which Rule-Based Comparator Is Worth Beating

Most MIMIC-IV sepsis models benchmark against SIRS, qSOFA and NEWS. Two results constrain how that benchmark should be read.

The Surviving Sepsis Campaign guidelines [40] issue a strong recommendation *against* using qSOFA as a sole screening tool, on the grounds of poor sensitivity. Demonstrating superiority over a score the governing clinical guideline has already told practitioners not to use for screening is a weak claim, and a study whose headline is that it beats qSOFA has chosen an opponent that has already conceded. NEWS2 is therefore the minimum meaningful rule-based comparator, and this study reports SIRS and qSOFA for completeness while treating NEWS2 as the benchmark.

Mellhammar et al. [41] supply the second constraint, and it is a direct precedent for the enterprise rather than a background caution. They derived a novel score using LASSO on prospectively collected emergency department data and validated it against NEWS2 and RETTS; the derived score did not outperform NEWS2 (AUC 0.73 versus 0.69, $p = 0.32$), and the authors conclude that even with a statistical approach they could not construct better risk stratification scores for sepsis than NEWS2. Deriving a statistical score and benchmarking it against rule-based scores has been tried, and it did not work. This is a valuable negative result and it shapes what this study claims: the contribution here is not a better score, and the comparison against NEWS2 is used to locate the model within a family of estimators rather than to argue for deployment.

2.9 Equity and Subgroup Validity

Wang, Li, Naidech and Luo [11] provide the only source located by this search that examines subgroup validity in a MIMIC sepsis cohort, and it is important to characterise it precisely, because its scope is narrower than the citation is often taken to imply.

The study [11] analysed 11,791 MIMIC-III critical-care patients, of whom 5,783 met Sepsis-3 criteria, and reports two findings. First, six competing sepsis identification criteria selected systematically different sub-populations with respect to

race, marital status, insurance type and language: who counts as septic in MIMIC is not demographically neutral. Second, **mortality** prediction performance degraded significantly when a model trained on the full population was applied to Asian patients, Hispanic patients and Spanish-speaking patients, with pairwise testing detecting significant discrepancies between Asian and White patients and between English- and Spanish-speaking patients.

The second finding concerns **mortality prediction among patients already identified as septic**, not sepsis onset prediction. It is therefore not a result about the task studied here, and it cannot serve as the prior estimate against which an onset model's subgroup discrimination is compared. What it does establish is the conjunction that motivates this study's equity analysis: the label is differentially sensitive by subgroup, *and* model performance is differentially poor for overlapping subgroups. Subgroup analysis is therefore not a robustness check to be appended in a sensitivity section; it is a validity requirement, and it is entangled with the label problem of Section 2.3.

The gap is that the two halves have never been measured together on the same subgroups in the same cohort. Wang et al. report both, but for different targets (label composition for sepsis identification, discrimination for mortality), so the two cannot be placed side by side. Studies examining label sensitivity for onset prediction do not stratify demographically at all. This study reports, for each subgroup, both the variation in who is labelled septic under each label variant and the variation in onset-model discrimination, under common false-discovery control, so that the relative precision of the two analyses is visible rather than assumed.

One methodological caution follows from the same source and governs how the results in Section 5.9 must be read. MIMIC's demographic fields are administrative records, not measured population attributes. The race field mixes broad categories with more granular ones drawn from the same population, and includes values that record the absence of an answer rather than an answer. Subgroup contrasts computed on such a field compare administrative strata, and the interpretation has to be stated in those terms.

2.10 Does Interpretability Survive the Label Problem?

The case for a statistical specification in this domain is epistemic rather than performance-based: coefficients are interpretable, assumptions are checkable, and the model can be transparently benchmarked. Christodoulou et al. [23] supply the supporting premise that little discriminative performance is sacrificed by the choice.

Cohen et al. [6] and Lauritsen et al. [10] attack that case at its root. If the identified cohort varies by a factor of 2.5 with the reading of the onset rule, and if problem framing can reverse the sign of a covariate’s estimated association, then a coefficient reported without an accompanying label-sensitivity analysis is not an interpretable quantity. It is a number whose provenance is partly biological and partly bureaucratic, and the reader is given no means of decomposing it.

This does not defeat the case for interpretable models. It relocates it. The interpretability of a statistical model is a *conditional* property: it holds given a stable and explicitly specified estimand, and the field has not supplied one. The discipline required to supply one is precisely the discipline a classical specification is better equipped to impose. A coefficient shown to be stable across three defensible readings of the Sepsis-3 onset rule is a genuinely interpretable quantity; one shown to *flip* across those readings is a genuinely important negative result. Either way the analysis is informative, and it is an analysis this search did not find performed for a survival model on MIMIC-IV.

That observation is the seed of the research question in Section 1.3. The field has established that the label matters (Section 2.3), that it is constituted by clinical action (Section 2.4), that the split design is not neutral (Section 2.5), and that the metric conventionally reported cannot see the failure (Section 2.6). Each of these has been demonstrated separately, in different databases, under different models, by different groups. What has not been done is to hold one database and one pipeline fixed and ask how much of the reported performance each of these analyst decisions accounts for, relative to the choice of model that the literature spends most of its effort on. That question is answerable, it is answerable with the intervals the constituent studies lack, and it is what this thesis does.

2.11 Summary: Design Decisions and Their Evidence

Table 2.3 maps each major design decision onto the evidence that warrants it and the section in which that evidence is evaluated.

Table 2.3: Design decisions and their evidence warrants.

Decision	Reason	Source
Three label variants	Label variance is of the same order as model variance, on the only comparison available	Cohen et al. [6]

Decision	Reason	Source
Label as variable	Variance is a finding, not a nuisance; confirmed prospectively at scale	Cohen; Dutta et al. [9]
Discrete-time model	Cox-equivalent at short intervals; no PH assumption; answers Schoenfeld’s objection	D’Agostino [36]; Ngwa [37]; Schoenfeld [33]
Competing risks	Discharge alive is informative, not censoring	Resche-Rigon [31]; Heyard [38]
Cox as weak comparator	PH known to fail in sepsis; quantify what the field’s default costs	Kasal et al. [30]
No case-control matching	Control alignment inflates apparent performance	Moor et al. [24]
Treatment-anchored eval	Models learn clinician suspicion, not deterioration	Kamran et al. [7]
Treatment-independent label	Sepsis-3, SEP-1 and CDC ASE all key on treatment timing, so a within-family comparison cannot separate definitional from empirical label variance	Kamran [7]; Dutta [9]
Time-based split primary	Sepsis is the worst-affected MIMIC-IV task under temporal shift; no algorithmic fix	Guo et al. [8]
Comparators refitted in-pipeline	Published rankings carry no out-of-distribution information	Reyna et al. [17]
Utility Score primary	AUROC survives the transition that inverts clinical value	Wang et al. [4]
Utility swept, not fixed	A threshold-dependent metric at one cut-off measures calibration, not value	Reyna [17]; Wong [5]
Calibration co-primary	Threshold-driven instruments are calibration-dependent whatever their AUROC	Van Calster et al. [29]
Alert burden reported	1.4 false alerts per true alert is the reference; worse is disqualifying, not a limitation	Moor et al. [21]
NEWS2 as benchmark	qSOFA is guideline-deprecated for screening; a LASSO-derived score has already failed against NEWS2	Evans [40]; Mellhammar [41]
GBT comparator	The statistics-versus-ML equivalence is contested precisely on temporal tasks	Christodoulou [23]; Fagerström [22]

Decision		Reason	Source
Subgroup sensitivity	label	Label composition varies demographically; onset-model subgroup discrimination is unreported	Wang et al. [11]
TRIPOD+AI PROBAST+AI	/	The reporting standard covers regression and machine learning under one checklist	Collins [12]; Moons [13]

Chapter 3

Research Methodology

3.1 Research Design

This is a retrospective, observational prediction-modelling study on de-identified critical-care data. It follows a pre-registered protocol (`00_preregistration.md`, locked before any of the modelling reported here) and reports according to TRI-POD+AI [12] and PROBAST+AI [13] guidelines. The central design innovation is treating the label definition, validation split, and evaluation metric as primary variables rather than fixed background choices.

All code is implemented in R (v4.x) and Python (v3.11), with a single configuration module (`config.R`) as the sole source of truth for all paths, clinical identifiers, and thresholds.

Literature search protocol. The design decisions set out in this chapter are warranted by evidence identified through a documented search, and two of the contributions claimed in Section 1.4 are claims about what that search did *not* find. The protocol (databases, date ranges, search strings, inclusion and exclusion criteria, verification standard, and yield) is specified in Table 2.1 and the identification-to-inclusion counts in Figure 2.1. It is a structured narrative search rather than a systematic review: searches followed a written protocol and every included source was verified against a primary publisher, PubMed, PMC or repository record, but no title-and-abstract screening pass was performed over a full de-duplicated yield and no second reviewer screened independently. Every claim of the form “no identified study does X ” in this thesis is therefore stated as “to the best of our knowledge” and is bounded by that coverage, which is recorded as a limitation in Section 6.7.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The primary dataset is MIMIC-IV v3.1 [16], [42], [43], a large de-identified critical-care database from Beth Israel Deaconess Medical Center (BIDMC), Boston. It was accessed under a PhysioNet credentialing agreement. The study uses eight clinical tables (Table 3.1).

Table	Exists	Size (MB)
admissions	yes	19.9
patients	yes	2.8
icustays	yes	3.3
chartevents	yes	3,502.6
labevents	yes	2,593.0
prescriptions	yes	606.7
microbiologyevents	yes	117.2
inputevents	yes	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in this study.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses eICU-CRD v2.0 [44], a multi-centre database from over 200 US ICUs. It uses different electronic health record systems and covers a broader range of hospitals and patient populations than the single-centre MIMIC-IV, making it a meaningful external test.

Transporting the label, not just the model

External validation of a sepsis model requires transporting two things: the model and the *label*. The model transports trivially, it is a frozen set of coefficients applied to harmonised features. The label does not, because Sepsis-3 onset is defined on documentation artefacts (an antibiotic order, a microbiology culture) whose recording conventions differ between databases far more than physiology does. Three properties of eICU-CRD govern how the label is constructed here.

Drugs are recorded across two tables. eICU-CRD has no item dictionary; drugs are free-text `drugname` strings, and they are split between `medication` (physician orders) and `infusionDrug` (continuous infusions). Vasopressors and much of the intravenous antibiotic exposure appear only in the latter. Both tables are read and the antibiotic exposure is their union; vasopressor exposure is likewise

taken from `infusionDrug` so that the SOFA cardiovascular component is not fixed at its no-pressor value.

Microbiology coverage is the binding constraint. The Sepsis-3 suspicion anchor requires a culture, and `microLab` covers 1.55% of the analysis cohort (2,824 of 181,589 unit stays). In the remaining stays the conjunction cannot fire under *any* variant, whatever the patient’s physiology. Coverage of the culture limb is an upper bound on the conjunction rather than a measure of it: both limbs are documented for only 232 stays (0.128%), and the counts narrow further once the window and the organ-dysfunction criterion are applied (Section 5.5.1 states each stage separately). Including those person-hours in the denominator would not estimate a low sepsis incidence; it would report the coverage of one source table as though it were an epidemiological quantity. The primary external analysis is therefore restricted to the microbiology-covered sub-cohort, in which an onset is observable in principle, and the reported external event rate is compared directly against MIMIC-IV’s on that basis.

The organ-dysfunction criterion is a rise, not a level. Onset is gated on a SOFA increase of at least two points over a baseline, evaluated inside each variant’s own window and against each variant’s own baseline (admission for Variants A and B, rolling 24-hour for Variant C), exactly as in the development cohort. An absolute SOFA threshold would be a different criterion, and because missing components are scored at assumed-normal values it would reject stays for incomplete documentation rather than for intact organ function, which in a database with sparser laboratory coverage than MIMIC-IV is a systematic bias towards the null.

Each of the three label variants is applied to eICU-CRD with its own antibiotic–culture windows, so the external label sets differ from one another as they do internally. This is verified rather than assumed: a label-invariant comparator such as NEWS2 can only score an identical AUROC across variants if the underlying label sets are identical, so the pipeline asserts at run time that the three external onset sets are distinct and halts if they are not.

A secondary **antibiotic-only** arm relaxes the culture requirement across the full cohort and is reported separately. It measures how much of the event count the culture requirement removes. Its target is not Sepsis-3, so it is never pooled with the primary arm and never substitutes for it.

3.3 Ethical Considerations

This study uses two publicly available, de-identified critical-care databases. Neither contains protected health information as defined under HIPAA, and no re-identification was attempted.

MIMIC-IV v3.1 is distributed by the MIT Laboratory for Computational

Physiology via PhysioNet [16]. Access requires completion of the Collaborative Institutional Training Initiative (CITI) “Data or Specimens Only Research” course and signing a PhysioNet Credentialed Data Use Agreement (DUA). The DUA prohibits re-identification and restricts data sharing to credentialed collaborators. The principal investigator (the author) completed CITI training and executed the DUA before any data access; credentials are on file with PhysioNet.

eICU-CRD v2.0 [44] is distributed under the same PhysioNet credentialing framework and DUA terms.

Both databases were collected under institutional review board (IRB) approvals held by the original data custodians (Beth Israel Deaconess Medical Center for MIMIC-IV; Philips Healthcare for eICU-CRD), with a waiver of informed consent for the de-identified research dataset. No additional IRB approval was required for secondary analysis of these de-identified records, consistent with the US Common Rule (45 CFR 46.104(d)(4)) and PhysioNet’s data access policies.

3.4 Cohort Construction

The base cohort is built by applying three inclusion criteria uniformly across all label variants:

- Age ≥ 18 at ICU admission.
- First ICU stay per patient only (to prevent within-patient correlation).
- ICU length of stay ≥ 4 hours (to ensure enough data for hourly features and at least one prediction opportunity).

The resulting cohort contains 65,078 **stays** in MIMIC-IV. Figure 3.1 shows the patient flow through each exclusion step.

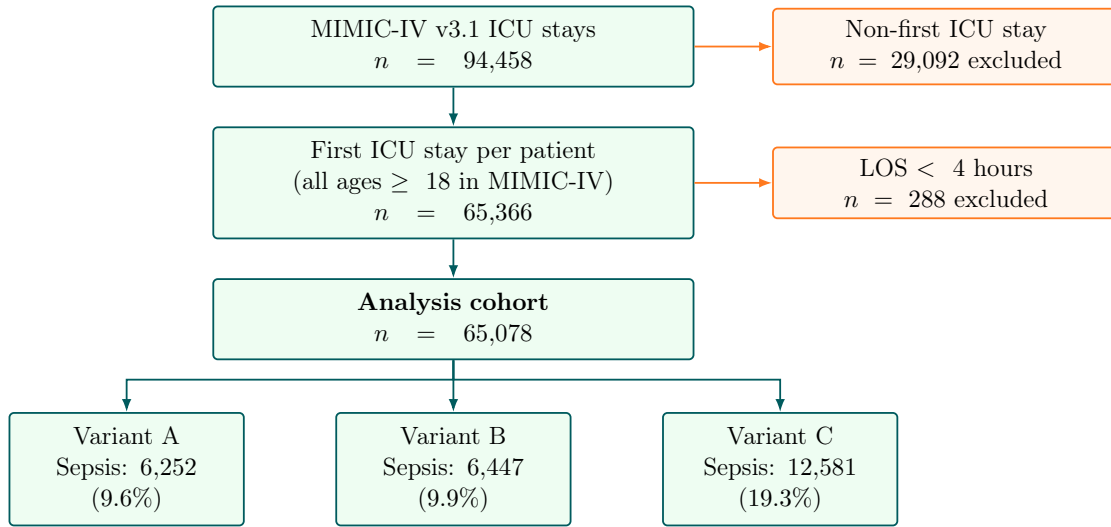


Figure 3.1: CONSORT-style patient flow diagram. All 65,078 stays enter all three label variants; only the sepsis outcome differs. Every count is generated from `02_cohort_flow`, written by script `02_cohort_labels.Rmd`. Sepsis counts are **cohort-wide**, at any hour of the stay; the subset falling inside the 72-hour modelling horizon, which is what the models are trained and evaluated on, is reported in the results chapter.

3.5 Outcome Definition: Three Sepsis-3 Label Variants

Following Cohen et al. [6] and Johnson et al. [15], the Sepsis-3 onset label is defined three different ways. All three are reasonable readings of the same consensus definition [1]. They differ in the antibiotic-culture time windows and the SOFA baseline (Table 3.2).

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline
A	Narrow	24 h	24 h	24 h	12 h	Admission
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed. Baseline = SOFA reference value.

In all three variants, a patient has sepsis if there is suspected infection (antibiotic order paired with a culture within the specified windows) and the SOFA

score increases by at least 2 points. Surveillance cultures (e.g. MRSA screens) are excluded.

Variant B (Seymour-Standard) is the **primary analysis**. Variants A and C are pre-specified sensitivity analyses for the variance decomposition.

3.6 Beyond Sepsis-3: A Treatment-Independent Label

Variants A, B and C are three readings of one definition, and all three fire on the same trigger: a culture drawn near an antibiotic order. That shared anchor is a record of a clinician’s decision to treat, not a measurement of the patient. Two consequences follow, and they must be separated.

The first is **definitional**. Sepsis-3 [1] makes onset a function of when treatment was given, so any variation in the antibiotic–culture windows necessarily moves the onsets. Part of the label-variant spread reported in Chapter 5 is therefore guaranteed by the construction of the definition and would appear even if the underlying physiology were identical across patients. The second is **empirical**: whether a label that owes nothing to treatment timing picks out different patients, at different hours, with different predictability. Only the second is a claim about the world, and it cannot be settled inside a family of labels that all share the anchor.

Sepsis-3 is a conjunction: *suspected infection* **and** *acute organ dysfunction*. Under Protocol Amendment 2 the conjunction is split and each limb is labelled on its own (Table 3.3).

ID	Name	Suspicion anchor	Organ dysf.	Uses treatment timing?
B	Seymour-Std	required	required	membership and timing
E	Anchor-Only	required	not required	yes
D	Deterioration	not required	required	no
F	Sepsis-3 Early	required	required	membership only

Table 3.3: The two limbs of the Sepsis-3 conjunction, labelled separately (Protocol Amendment 2), and the standard onset timing applied to the intact conjunction (Variant F, Protocol Amendment 4). Variant E is Variant B’s suspicion anchor with the SOFA criterion removed; Variant D is acute organ dysfunction with the infection criterion removed; Variant F has Variant B’s membership exactly and differs only in when the onset is placed.

Variant E (Anchor-Only) takes onset to be the suspicion time itself, using Variant B's windows, with no organ-dysfunction requirement. Its label is purely a record of clinician behaviour. Predicting it well means predicting who gets cultured and started on antibiotics, which is the quantity a treatment-anchored sepsis label partly encodes. The share of Variant B's above-chance discrimination that Variant E reproduces,

$$\phi_{\text{anchor}} = \frac{\text{AUROC}_E - 0.5}{\text{AUROC}_B - 0.5}, \quad (3.1)$$

is reported as the **anchor-attributable fraction**.

Variant D (Deterioration) takes onset to be the first hour after hour 1 at which a *physiology-only* SOFA score rises at least 2 points above the patient's admission baseline (the mean of each component over hours 0–3). "Physiology-only" means the vasopressor and catecholamine terms are withheld, so the cardiovascular component is scored from mean arterial pressure alone. No antibiotic order, culture draw or drug administration enters the definition at any point: Variant D is the treatment-independent labelling that the pre-registered family lacked. Both variants are fitted with the identical model specification, feature set and temporal split, so the label is the only thing that changes.

The PhysioNet/CinC 2019 operationalisation [17] was considered as the *treatment-independent* alternative label and rejected for that purpose. It sets onset to $\min(t_{\text{susp}}, t_{\text{SOFA}})$ with the same culture-plus-antibiotic trigger, so membership is still anchored to treatment timing and it would not have broken the circularity. The same objection applies to SEP-1 and to the CDC Adult Sepsis Event [9], both of which are defined on antibiotic days.

Two limits on interpretation are stated here and carried through the results. Variant E is not a sepsis model and its AUROC is not a sepsis result. Variant D is a threshold function of recorded physiology, and five of the six SOFA components are themselves model inputs, so its discrimination is partly self-fulfilling and its *level* is not comparable with Variant B's. What is comparable, and what carries the empirical content, is whether Variant D selects different patients at different hours: reported as onset agreement (Cohen's κ) and median onset shift in Section 5.6.

3.6.1 Standard Onset Timing (Variant F, Protocol Amendment 4)

That rejection settles the circularity question and no more. The min rule does not break the anchor, membership still requires the antibiotic-culture pair, but it does change something the pre-registered family holds fixed and never examines: *when* the onset is placed. Anchored *membership* and anchored *timing* are separable properties, and only the second is what the treatment-anchored evaluation window

tests.

Variants A–C assign $\text{onset} = t_{\text{susp}}$ whenever the SOFA criterion is satisfied anywhere in the evaluation window, so the SOFA criterion gates membership and contributes no timing information. Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [17] instead assign

$$t_{\text{onset}} = \min(t_{\text{susp}}, t_{\text{SOFA}}), \quad (3.2)$$

with t_{SOFA} the first hour in the evaluation window at which the ≥ 2 -point rise over the admission baseline is already present. Equation (3.2) is the branch that permits an onset to precede the first antibiotic or culture; dropping it makes the pre-treatment window empty as a matter of arithmetic (Proposition 5.1 in Chapter 5).

Variant F was added for that reason and only that reason. The pre-registered onset rule was found to make H3 unfalsifiable, so a variant carrying the standard timing was needed for the question to be answerable at all; the amendment was fixed before Variant F was fitted, and nothing about its performance entered the decision to add it.

Variant F (Sepsis-3 Early-Onset) restores it and changes nothing else. The suspicion windows, SOFA threshold, evaluation window and baseline are Variant B’s, so the two variants label exactly the same stays cohort-wide; only the timestamp moves, and only ever earlier. Variant F therefore isolates the timing branch as cleanly as Variants D and E isolate the two limbs.

Three properties are worth stating in advance. First, the SOFA score used for t_{SOFA} is the *full* score, vasopressor components included, as Seymour et al. specify; vasopressor administration is itself a treatment, so an onset placed by the cardiovascular component is free of the antibiotic-and-culture anchor that defines the pre-treatment window but is not free of treatment in general. Variant D remains the fully treatment-independent comparison.

Second, F and B label the same stays septic, membership is identical at the level of the stay, so the informative agreement statistic for F is the onset *shift*, not the overlap. Agreement is nonetheless not perfect once restricted to the modelled horizon, which leads to the third property.

Third, moving an onset earlier can move it *into* the 72-hour prediction window. A stay whose t_{susp} falls after hour 72 contributes no labelled event to any model here, but if its SOFA rise occurred inside the window, Variant F places the onset there and the stay becomes a modellable event. This is a one-directional effect: $F \supseteq B$ within the horizon, and the extra stays are reported in Section 5.6.3. It means the choice of onset rule determines not only *when* events are recorded but *how many* are visible to a real-time model at all.

Variant F is post-hoc, is never pooled with A–C, and takes no part in H1 or the variance decomposition. Its contrast against B joins exploratory family E3, which

enlarges that family and makes every Benjamini–Hochberg adjustment within it stricter than before the amendment.

Variants D, E and F are post-hoc and are never pooled with A–C. They take no part in H1 or in the variance decomposition table, which are computed from exactly the same inputs as before the amendments; their contrasts form their own exploratory family (E3), corrected by Benjamini–Hochberg.

3.7 Prediction Framing

Following Lauritsen et al. [10] and Reyna et al. [17]:

- **Observation window:** Expanding from ICU admission, uses all data available up to the current hour.
- **Estimand:** the *cause-specific discrete-time hazard* of sepsis onset in the current hour, conditional on the stay being at risk at the start of it, $\Pr(\text{onset at hour } t \mid \text{at risk at } t, \mathbf{x}_t)$, with discharge and death as competing causes.
- **Prediction frequency:** Every hour.
- **Start time:** 1 hour after ICU admission.
- **End time:** 72 hours after ICU admission (right-censored).
- **Alerting window:** 6 hours. An alert counts as timely if it is issued in the six hours up to onset; this governs the Utility Score and the per-patient detection metrics, not the quantity the models estimate.

What is estimated and what is scored, stated separately. These are two different objects in this design and conflating them would misdescribe every discrimination number in Chapter 5. The models estimate a per-hour hazard, and the ranking metrics (AUROC, AUPRC and the calibration curves) are computed on that hazard against an indicator that takes the value one at the onset hour and zero at every other at-risk hour. They therefore measure how well a model separates the hour in which onset is recorded from the hours that precede it in the same and other stays; they are not the AUROC of a “sepsis within the next six hours” classifier, and they should not be read as one. The six-hour window enters through the clinically oriented metrics instead: the Utility Score pays a reward that decays linearly in lead time over that window (Section 3.12), episode sensitivity asks whether an alert fired inside it, and the lead-time distribution is reported against it directly.

The choice follows from the competing-risks formulation rather than sitting beside it. A cumulative six-hour risk is a functional of the hazard together with the competing-cause hazards over the same window, so estimating the hazard is the more primitive operation and leaves the horizon a reporting decision rather than a modelling one. The cost is that a single number cannot be quoted as “six-hour discrimination”, and the corresponding limitation is recorded in Section 6.7.

The alerting window and the utility scoring rule match the PhysioNet/CinC 2019 Challenge [17]; the estimand does not, the Challenge having scored a binary label defined over the window itself.

3.8 Person-Hour Table and Competing Events

Each patient stay is expanded into a table with one row per person-hour at risk. Following Heyard et al. [38], each hour is assigned one of four outcomes:

- 0: Still at risk (censored at this hour)
- 1: Sepsis onset (the event of interest)
- 2: Discharged alive without sepsis (competing event)
- 3: Died without sepsis (competing event)

Treating discharge and death as simple censoring would overstate sepsis incidence and bias model estimates. The multinomial competing-risks formulation avoids this problem.

3.9 Feature Construction

Features are extracted at each person-hour from chartevents, labevents, inpuvents, and outputevents. The feature set includes:

1. **Vital signs**, 6 (last value carried forward per hour): heart rate, respiratory rate, SpO₂, MAP (invasive or noninvasive), temperature (°C), GCS total.
2. **Laboratory values**, 5 (last value carried forward): lactate, creatinine, total bilirubin, platelet count, WBC.
3. **Vasopressor indicators**, 2: any vasopressor (yes/no); any norepinephrine or epinephrine (yes/no).
4. **Six-hour trends**, 3: rolling slopes for heart rate, respiratory rate and MAP.

5. **Missingness indicators**, 6: whether each of lactate, creatinine, bilirubin, platelets, white cells and a blood gas (P/F ratio) was ordered in the current hour, one flag per test.

Temperature in Fahrenheit is converted to Celsius. FiO₂ values above 1.0 (percentage format) are divided by 100.

Two boundaries of this list are worth stating because they are easy to misread. The P/F ratio enters as an *order indicator* only: it is used in the SOFA respiratory component that constructs the label, and its measurement flag is a covariate, but the ratio itself is not a model input, because it is recorded for a minority of person-hours and its forward-filled value carries more information about ventilation practice than about the current hour. And there is no lactate slope: rolling slopes are computed for the three variables sampled densely enough to support one.

The declared set therefore holds **22 covariates**, and the fitted model carries **29 terms per outcome** once the six-column natural-spline baseline in elapsed hours and the intercept are counted. The vector is defined once, in `config.R`, and every stage that fits or transports a model reads it from there; `require_features()` raises an error rather than proceeding if any declared predictor is absent from a panel.

3.10 Model Classes

3.10.1 Primary Model: Dynamic Discrete-Time Competing-Risks Hazard Model

The primary model is a multinomial pooled logistic regression over person-hours, following D’Agostino et al. [36] and Heyard et al. [38]:

$$\lambda_r(t \mid \mathbf{X}_{it}) = \frac{\exp(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it})} \quad (3.3)$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$. The baseline hazard $\alpha_r(t)$ uses a restricted natural cubic spline with 5 knots, placed at quantiles of training-set hour within each variant.

This model predicts the risk of sepsis every hour. Under short intervals and low per-interval event probabilities (both conditions met for hourly ICU data with onset probability below 1%), it is mathematically equivalent to time-dependent Cox regression, without requiring the proportional-hazards assumption.

Identification: the specification is penalised, and must be. The unpenalised version of Equation (3.3) is **completely separated** on this design. Driven

to a tight tolerance by iteratively reweighted least squares, the whole coefficient vector diverges together, $|\beta|$ of order 10^{15} under *every* label variant, which is the signature of separation along a direction in covariate space rather than of one badly behaved predictor; the design matrix is full rank, so this is not aliasing. A separated likelihood has no finite maximiser, so the fitted coefficients are not identified and the point at which a quasi-Newton optimiser stops is arbitrary.

This is a trap worth naming, because the standard diagnostic does not catch it: `nnet::multinom` returns `convergence = 0` for every variant here. That flag reports only that BFGS's relative-change stopping rule fired; it says nothing about whether an optimum exists. A ridge penalty of $\lambda = 10^{-4}$ is therefore applied, which makes the penalised log-likelihood strictly concave and the maximiser unique. The value is **not tuned**: across a 20-point ridge path the temporal-test AUROC moves by less than 0.02 for every variant, and that insensitivity, not an optimisation, is the justification for the constant. Every fit is checked after estimation and flagged if $\max |\beta|$ exceeds 20.

Penalisation is not a refinement here; it is a precondition. Coefficient comparisons across label variants (H2, Section 5.10.4) are only meaningful against a unique maximiser, and an unpenalised fit can land in a degenerate region under one label and a benign region under another, manufacturing an apparent label effect where none exists.

Uncertainty for this model comes from the bootstrap, with a cluster-robust covariance alongside it. Person-hours are nested within stays, so a cluster-robust sandwich estimator clustered on stay is the natural choice for coefficient standard errors. It is not available off the shelf: `sandwich` supplies a *bread* method for `multinom` objects but no score-function method, so the whole sandwich family fails on this class. The obstacle is narrow rather than fundamental, and the missing piece is supplied here. For row i and non-reference category k the score with respect to the coefficient on covariate j is $x_{ij}(y_{ik} - p_{ik})$, a residual matrix the fitted object already stores; providing it, and fitting with the Hessian retained, yields a clustered covariance for the primary model.

Two qualifications attach to it and neither is cosmetic. Because the specification is ridge-penalised out of necessity (Section 3.10.1), what the estimator returns is the covariance of the *penalised* estimator: the bread is the inverse penalised Hessian, so the two factors are mutually consistent, but this is a ridge-regularised sandwich and is described as one throughout. And it gives standard errors on *coefficients*, not on H2's count of unstable covariates, which is a threshold rule applied across three separate fits and admits no analytic variance of its own. H2 therefore remains a descriptive count (Section 5.10.4).

What the covariance does permit is a diagnostic on that count, added as **Proto-**

col Amendment 7 (post-hoc, §21 of 00_preregistration.md). A rule that flags a coefficient for doubling in magnitude cannot distinguish a coefficient that genuinely moves from one estimated too imprecisely for a doubling to mean anything, so for each flagged covariate the difference between its two extreme estimates is referred to the pooled standard error of those two. The ratio is reported alongside the pre-registered count and never in place of it (Section 5.10.4). It is deliberately not a test: the three fits share stays, so the differenced estimates are correlated and a pooled standard error assuming independence is conservative by an unquantified amount, and the penalisation qualification above applies to it as well. No p -value is computed for it and it enters no corrected family.

Every interval on a performance quantity in this thesis continues to come from the stay-level cluster bootstrap of Section 3.14, which resamples whole stays and so respects the same clustering. The analytic covariance is reported for the coefficients and does not enter any hypothesis verdict.

3.10.2 Comparator: Gradient-Boosted Trees (GBT)

An XGBoost model is fitted under the same conditions for each label variant and split. Settings: logistic objective, area under the precision-recall curve as the evaluation metric, maximum depth 6, learning rate 0.05, row and column subsampling at 0.8, minimum child weight 5, and positive-class weighting by the observed class ratio. None of these is tuned; they are the library's conventional starting values, fixed once and applied identically to every variant, split and feature arm. Tuning the comparator would confound the model-class contrast with a difference in search budget, which is the same objection that keeps the ablated arm on the full arm's hyperparameters (Section 3.14.5).

The number of boosting rounds is the one quantity that is chosen rather than fixed, and it is chosen by early stopping, which needs a set the model is not fitted on. That set is carved out of the *training* rows: 15 per cent of training stays are held out, the model is fitted on the remainder, and boosting stops after 30 rounds without improvement on the holdout. The split is on the stay rather than the person-hour, because person-hours are roughly forty-five correlated rows per stay and a row-level split would leave the same patient on both sides of it. The test set is never used to select the number of rounds (Protocol Amendment 8).

3.10.3 Comparator: Static Cox Model

A standard Cox proportional-hazards model using only the first observation per stay, included as a deliberately weak baseline to show the cost of ignoring time-varying data. Schoenfeld residual testing is applied to verify the expected violation of proportional hazards.

3.10.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same features. NEWS2 is the headline comparator following Evans et al. [40].

3.11 Validation Architecture

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [8]	Honest internal estimate
Internal (secondary)	Random 75/25 patient-level split (seed = 42)	Measure split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0 [44], no retraining	Generalisability (H5)

Table 3.4: Validation architecture. All splits are patient-level: no patient appears in both training and test sets.

3.12 Evaluation Metrics

3.12.1 Primary Metrics

PhysioNet Utility Score [17] is the primary metric. It rewards early true alerts and penalises late and false ones. AUROC is reported only for comparison, because it hides the clinical failure mode [4].

Each stay contributes a raw utility. A septic stay first alerted at hour t within the reward window $[t_{\text{onset}} - 6, t_{\text{onset}}]$ earns $(t_{\text{onset}} - t)/6$, so a full-horizon warning earns 1 and an alert that merely coincides with onset earns nothing; a septic stay alerted only in $(t_{\text{onset}}, t_{\text{onset}} + 3]$ earns -0.05 per late alert hour; a septic stay never alerted in either window earns -1 ; and every alert hour outside a reward window, on any stay, costs -0.05 .

The raw total U is then referenced to the two strategies that bound it. The *inaction* strategy never alerts and therefore earns $U_{\text{inact}} = -1$ per septic stay; the *optimal* strategy earns the full reward on every septic stay and raises no false alarm, giving $U_{\text{opt}} = +1$ per septic stay. The reported score is

$$\tilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}}, \quad (3.4)$$

so that $\tilde{U} = 1$ is a perfect forecaster, $\tilde{U} = 0$ is *exactly* the no-alert strategy, and $\tilde{U} < 0$ is worse than issuing no alerts at all. The inaction baseline has to

appear in both numerator and denominator: normalising by the optimum alone would score a model that raises no alert at -1 rather than at 0 , contradicting the meaning of the metric. Equation (3.4) is verified against both bounds at run time, a threshold above every predicted probability is required to return exactly zero, and no operating point may exceed one.

Operating points. A score of this kind is a property of a model *and* a threshold. At an hourly event rate below 0.5 per cent no model’s predicted probability approaches the conventional 0.5 cut-off, so that cut-off measures the no-alert strategy for a well-calibrated model and an arbitrary point for a poorly calibrated one. The Utility Score is therefore reported twice: at the 0.5 cut-off, for comparability with the literature, and at its maximum over a threshold grid. The grid is defined on the *alert rate*, the fraction of person-hours alerted, rather than on the probability scale, spanning 10^{-5} to 0.5 over sixty logarithmically spaced points, so that it adapts to each model’s calibration and covers every operating point a ward could contemplate. The maximum is taken over thresholds that raise at least one alert, and the alert burden, alert rate and per-stay detection rate that attain it are reported alongside, because a utility maximum bought at an unusable alarm volume is not a deployable operating point.

Calibration is assessed via calibration intercept and slope, following Heyard et al. [39], and, following Van Calster et al. [29], is additionally reported as a flexible calibration curve (restricted cubic spline of the outcome on the logit of predicted risk) summarised by the Integrated Calibration Index (ICI), E50, E90, E_{max}, and the scaled Brier score. Predicted probabilities are floored at 10^{-6} rather than at the conventional 10^{-3} , because at an hourly event rate below 0.5 per cent the conventional guard overwrites the majority of the distribution being measured.

3.12.2 Secondary Metrics

AUPRC (important at low event rates), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. [21]), lead time (benchmark: 3.7 hours from Moor et al.), and decision-curve net benefit [45]. Net benefit is evaluated on a threshold grid running from one tenth to ten times the observed event prevalence rather than on the conventional 0.01–0.20 grid, which at this prevalence lies entirely above the range of thresholds any deployment would use.

3.12.3 Deployment-Facing Metrics

AUROC, AUPRC and the Utility Score characterise a model but not the alert load a ward would carry. The following are therefore reported alongside them, at a threshold set to the largest cut-off achieving a fixed per-hour sensitivity of 80 per cent rather than at an arbitrary 0.5 cut-off:

- **Positive predictive value** and its reciprocal, the number needed to evaluate (NNE), at that fixed sensitivity.
- **False alerts per patient-day**, which normalises alarm volume by exposure time rather than by the number of true alerts.
- **Per-patient (event-level) detection**: the fraction of septic stays alerted within the actionable window [onset – 6 h, onset], the fraction alerted at any point before onset, patient-level PPV, and the fraction of all stays alerted. Per-hour and per-patient views of the same operating point are reported side by side, because the choice of denominator moves the headline PPV by more than an order of magnitude.
- **False-alarm episodes per patient-day**, counting a contiguous run of alert hours once rather than once per hour.
- **Lead time** from first alert to onset: median, interquartile range, and the fraction of septic stays alerted at least 3 and 6 hours ahead.

The same quantities are computed on eICU-CRD with thresholds re-derived on the external cohort.

3.13 Sample Size

Sample size is calculated using the Riley et al. [46] method, targeting a shrinkage factor ≥ 0.9 with 29 parameters and an anticipated C -statistic of 0.846 from Moor et al. [21]. Each label variant is evaluated separately; an underpowered variant is a finding, not a failure.

The parameter count, and the criterion’s fit to this model. 29 is the number of terms the model estimates for one outcome: 22 covariates, the six-column spline baseline and an intercept (Section 3.9). The mapping is imperfect and the imperfection runs against the study, so it is stated rather than left implicit. Riley et al.’s criteria are derived for a *binary* outcome, whereas the primary model is a four-category multinomial estimating those 29 terms for *each* of three non-reference outcomes, three times that count in all. Taking the per-outcome count treats each cause-specific sub-model as the unit, which is the closest defensible reading of a binary criterion in this setting but still understates the dimensionality of the whole fit. The minimum sample size scales approximately linearly in the parameter count, so the margins reported below absorb even the all-outcomes reading with a factor of ten to spare, and the adequacy verdict does not depend on which reading is taken.

The unit the criterion applies to. Riley et al.’s criteria are derived for a sample of *independent* observations, and person-hours are not independent, they are nested within ICU stays at roughly 45 correlated rows per stay, and one stay per patient. The adequacy calculation is therefore parameterised on the **stay**: the outcome proportion supplied to the criterion is the fraction of training stays labelled septic under that variant, and the resulting minimum is compared against the number of training stays. The person-hour count is reported alongside as the size of the design matrix and is never used as a sample size. This is the same unit that governs every interval in this thesis, for the same reason: the cluster bootstrap of Section 3.14 resamples stays, not rows. Reporting adequacy against 2.3 million person-hours would overstate the available information by roughly the mean number of at-risk hours per stay, and would do so in the direction that makes the study look better powered than it is.

3.14 Statistical Inference and Multiplicity

3.14.1 Uncertainty: the Stay-Level Cluster Bootstrap

Person-hours are nested within ICU stays: a single stay contributes on average about 45 highly correlated hourly rows. Treating those rows as independent would produce standard errors several times too small. Every interval in this thesis therefore comes from a **stay-level cluster bootstrap**, in which whole ICU stays are drawn with replacement and all person-hours belonging to a drawn stay are carried with it. Intervals are 95% percentile intervals from $B = 2,000$ replicates for internal and subgroup analyses and $B = 1,000$ for eICU-CRD, which contains 7.9 million person-hours. The seed is fixed at 42.

Contrasts between quantities measured on the *same* person-hours are evaluated within replicate, so the interval accounts for their correlation. This applies to the label-variant comparison (all three variants are built on the same test stays) and to the model-class comparison (all models score the same hours). Contrasts across *different* evaluation sets (temporal versus random split, and MIMIC-IV versus eICU-CRD) use independent bootstrap streams and propagate both variances.

For subgroup analyses the bootstrap is stratified: stays are resampled within each subgroup so the observed subgroup sizes are held fixed. The subgroups are the object of study, not a random draw from a larger frame.

P-values are obtained by recentring the bootstrap distribution on the null boundary and reading off the tail area, using the $(1 + \text{count})/(B + 1)$ convention so that no p-value is reported as exactly zero. The smallest reportable p-value is therefore $1/(B + 1)$.

3.14.2 Confirmatory and Exploratory Families

The six pre-registered hypotheses form a single **confirmatory** family and are corrected by Holm–Bonferroni at a family-wise error rate of 0.05. The family size is fixed at $m = 6$: the number of hypotheses that were pre-registered, not the number that turned out to be testable. H2 and H3 each reserve a slot without contributing a p-value (Section 5.10 explains why neither admits one), which is deliberately conservative. A hypothesis becoming untestable after the data were seen must not be allowed to relax the correction applied to the others.

All subgroup analyses are **exploratory**. The pre-registration fixed their *scope* (that race, language and insurance would be examined), but not the subgroup levels, the reference level, or the direction of any comparison. They are corrected by Benjamini–Hochberg at $q = 0.05$ within three families: E1, all subgroup AUROC contrasts against the reference level across every equity variable and both models; E2, all subgroup label-sensitivity contrasts; and E3, the alternative-label contrasts of Protocol Amendment 2, in which the AUROC under each limb of the Sepsis-3 conjunction (Section 3.6) is compared with Variant B for both models. The anchor-attributable fraction of Equation 3.1 is a ratio with no meaningful null at zero and is reported as an interval only, outside the corrected family. The correction denominator is the number of contrasts *computed*, not the smaller number selected for presentation in Chapter 5. Bonferroni-adjusted p-values are also reported so that a reader preferring family-wise control over false-discovery control can substitute them.

The two procedures differ by design. A confirmatory claim must survive family-wise control, because one false confirmation invalidates the claim. An exploratory scan exists to generate hypotheses and needs only its false-discovery proportion bounded; applying family-wise control to it would suppress every signal the scan is meant to surface.

Each subgroup is contrasted against the largest level of its own variable. The reference is chosen by sample size alone, never by outcome.

Two event floors, and what each decides (Protocol Amendment 6).

Two distinct thresholds govern the subgroup analysis and the distinction is load-bearing. `MIN_SUBGROUP_EVENTS = 5` decides whether a subgroup AUROC is *computed*: below it, together with a 50 person-hour minimum, an AUROC is not estimable and the subgroup is listed as such rather than shown as a point estimate. `MIN_SUBGROUP_EVENTS_INTERPRET = 100` decides whether a computed AUROC is *interpreted*, and whether its contrast enters family E1. Strata below the interpretation floor are still estimated and still tabulated, flagged as such, because the figure showing their intervals reaching the chance line is itself the argument for the floor; they are excluded from every statement about spread and from the corrected

family. The constant is set equal to the external-validation floor deliberately: it answers the same question on the same warrant, so the two cannot drift apart.

This is **Protocol Amendment 6**, adopted post-hoc on 6 August 2026 and recorded as §20 of the pre-registration. It must be read against the rule stated two paragraphs above, namely that the correction denominator is the number of contrasts computed rather than the number selected for presentation. Restricting E1 from thirty computed contrasts to 6 is on its face the manoeuvre that rule prohibits. Three facts bear on whether it is admissible, and §20.4 of the pre-registration sets out the full argument for a reader who wishes to reject it. The criterion is outcome-blind, depending only on a stratum’s event count. The direction is exculpatory: the reduction removed the family’s only surviving discovery, a language contrast resting on twelve events, rather than creating one; a family narrowed to manufacture significance narrows *around* its significant results. And the verdict is invariant to the choice: no contrast in E1 survives Benjamini–Hochberg under either denominator, the smallest adjusted value being $q = 0.066$ at $m = 6$. Section 5.9.1 reports the consequence.

The floor applies to E1 and *not* to E2, and the asymmetry is deliberate rather than an oversight. E1’s estimand is an AUROC, which requires enough events to order cases against non-cases and is not usefully estimable on a handful; E2’s is a stay-level proportion, which is estimable at any stratum size and simply carries a wide interval when the stratum is small. E2 therefore retains every stratum, including one insurance category resting on seven stays whose interval spans nearly sixty percentage points and is reported with that interval visible. Retaining it makes the correction *stricter*, not more permissive, because it enlarges the denominator over which false-discovery control is applied; dropping it would be the manoeuvre that needs defending.

3.14.3 Analysis Status Tagging

Every result in Chapter 5 carries an explicit status tag rather than relying on a single blanket statement in this chapter:

- **Confirmatory:** estimand, threshold and direction were fixed in `00_preregistration.md` before any modelling.
- **Exploratory:** pre-registered in scope but not in specific contrasts. Generates hypotheses; does not test them.
- **Descriptive,** no inferential claim is attached.

The tags are held in a machine-readable analysis register (`12_analysis_register.parquet`) produced by the pipeline, and are reproduced in Table 5.32.

3.14.4 Status of the Inference Plan Itself

This inference and multiplicity plan is **Protocol Amendment 1**, dated 30 July 2026 and recorded as §13–15 of `00_preregistration.md`. It was written *after* the analyses of Chapter 5 had been run and inspected, and it is therefore itself post-hoc. It is reported as such rather than presented as pre-specification.

The amendment changes no hypothesis, estimand, threshold, direction, subgroup or model. It attaches uncertainty to contrasts that were already pre-specified and imposes the multiplicity control the original protocol omitted. Where the correction changes a verdict recorded in the original protocol, the corrected verdict is the one reported, and the change is stated explicitly.

3.14.5 Feature Ablation: Action-Derived Predictors (Protocol Amendment 5)

The central argument of this thesis is that the Sepsis-3 label is constituted by clinician action rather than merely correlated with it. The same objection applies, with equal force, to part of the *feature* set, and it would be inconsistent to press it against the outcome while leaving the predictors unexamined.

Two kinds of predictor record clinician behaviour rather than patient physiology. **Vasopressor exposure** is a treatment: it is something done to the patient, not something measured in them. The **per-test order indicators** (whether lactate, creatinine, bilirubin, platelets, white cells or a blood gas was obtained in a given hour) record a decision to look. Their predictive value is well established and is usually read as evidence that measurement patterns encode clinical suspicion, which is precisely the problem: a model that leans on them is partly learning when clinicians became worried.

Protocol Amendment 5 (post-hoc, dated 6 August 2026 and recorded as §19 of `00_preregistration.md`) therefore adds an ablated arm. The primary model and the gradient-boosted comparator are re-fitted on the same training rows, the same splits and the same hyperparameters, with the 8 action-derived features removed and every physiological measurement retained.

Three properties make the contrast interpretable:

- **The ablation is deliberately conservative.** Forward-filled laboratory *values* are retained even though a value exists only because someone ordered the test. Dropping them as well would remove most of the laboratory signal and confound the ablation with a loss of physiological information. What is removed is the set of features that encode *only* clinician behaviour.
- **Nothing is re-tuned.** The gradient-boosted arm keeps the hyperparameters of the full fit. Re-tuning would confound the feature set with the search

budget.

- **Both arms are scored by the same metric code on the same test rows**, so the difference is attributable to the feature set alone.

The arm is exploratory. It enters no hypothesis, no confirmatory family and no variance decomposition; it is reported as estimation, and its two readings, the change in discrimination and the change in cross-label coefficient stability, are given in Section 5.7.

3.15 Pre-Registration and Reporting

The full protocol was locked before the cohort extraction reported here. All six hypotheses, label variants, model classes, splits, metrics, and decision thresholds are specified in `00_preregistration.md`. Any analysis not specified there is labelled as exploratory, and, following Section 3.14, each individual result carries its status tag in the results table itself rather than inheriting it from this statement alone. Reporting follows TRIPOD+AI [12]; code availability and the constraints the data use agreement places on it are set out in Section 4.15.2.

Chapter 4

System Design and Pipeline Architecture

The pipeline consists of twelve numbered, single-purpose scripts in `sepsis/project/`, preceded by the locked pre-registration document `00_preregistration.md`. Each script reads from upstream outputs and writes its own results to `sepsis/project/output/`. No script modifies the original source data. All random seeds are fixed for reproducibility. Figure 4.1 shows the data flow, and Appendix B tabulates each stage’s inputs and outputs in full.

This chapter describes what each stage does. It opens with why the architecture takes this shape, because the structure is not incidental: a study whose headline claim is that analyst decisions move results is obliged to make its own analyst decisions inspectable, and several of the choices below exist to make a specific class of silent error impossible rather than merely unlikely.

4.1 Design Principles

Four constraints shaped the architecture. Each is stated here with the decision it forced and what that decision rules out.

The source data does not fit in memory, and must not be modified. MIMIC-IV’s `chartevents` table is 3.5 GB compressed and `labevents` 2.6 GB; expanded, either exceeds the working memory of the machine this study ran on. Both are therefore read through a DuckDB streaming layer that pushes the cohort filter and the item-identifier filter down into the scan, so only rows belonging to the analysis cohort are ever materialised. The alternative, extracting a reduced copy once and working from it, was rejected because it makes the extraction step invisible to everything downstream: a defect in the filter becomes indistinguishable

from a property of the data. Reading from the original tables on every run keeps the extraction inside the reproducible chain.

The corollary is that the source directories are treated as strictly read-only. No stage writes to `data/`, and every stage writes only to `output/`. Reproduction is therefore a matter of deleting `output/` and re-running, with no possibility that a previous run has left the inputs in a state the next run depends on.

Every clinical constant must have exactly one definition. The study varies label definitions deliberately, which means the same quantity (an antibiotic–culture window, a SOFA threshold, an evaluation horizon) is read by several stages and must mean the same thing in all of them. A constant duplicated across two scripts is a constant that can silently diverge, and in a study about definitional sensitivity such a divergence would be indistinguishable from a finding. `config.R` is therefore the single source of truth for every path, identifier, window, threshold, seed, family size, event floor and covariate vector in the pipeline, and no other file hard-codes any of them. The covariate vector is the case that matters most and was the last to be brought inside the rule: it is read by the two fitting stages and again by the external-validation stage, which reconstructs a stored model’s column order from it, so a divergence between copies would have produced a permuted design matrix and predictions that are wrong while remaining perfectly well-formed. Where a stored model records its own feature names, disagreeing with the declaration is an error rather than a warning. The same discipline extends to the thesis: `thesis_constants.R` writes the reported numbers into the document from the result tables, so a value cannot disagree between the code and the text either.

An interface between stages must fail loudly or not at all. The pipeline’s most dangerous failure mode is not a crash but a stage that quietly does less than it was asked to. Selecting predictors by intersecting a declared feature list with the columns actually present, for instance, silently drops any predictor whose name does not match: the model still fits, the metrics still compute, and nothing in the output indicates that a variable is missing. Interfaces are therefore declared rather than inferred: `require_features()` raises an error when a declared predictor is absent instead of proceeding without it, and the fit diagnostics record convergence status alongside the coefficient magnitudes that reveal an unidentified likelihood, because a convergence flag alone does not distinguish the two.

The same principle governs the run-time invariants. Where a computation has a property that must hold by construction, the pipeline asserts it and halts: a threshold above every predicted probability must score exactly zero on the Utility Score, because it *is* the no-alert strategy; no operating point may exceed the optimal strategy’s score of one; and the three label variants must produce three

distinct sets of external onsets, since a label that was never varied would otherwise present as a label that has no effect. These are stops, not warnings. A warning in a sixty-minute batch run is a line in a log nobody reads.

Expensive fits must be refreshable without disturbing the others. Fitting is the dominant cost, and gradient-boosted tree fitting is not bit-stable across runs, so an incidental refit changes numbers throughout the document without any change of substance. Two environment variables scope a run accordingly: `V2_VARIANTS` restricts the per-variant loops to a named subset, and `V2_ARMS` restricts the fitting arm (Section 3.14.5). Adding an alternative label or an ablated specification therefore does not re-execute the fits that every headline number rests on, and the outputs of the additional arms are written to separate files so they cannot overwrite a pre-registered result.

Why twelve scripts rather than one program. The stage boundaries follow the analysis, not the code: each corresponds to a step that has to be separately inspectable because a claim in Chapter 5 rests on it. The inference layer is the clearest case: Stage 12 reads only the stored predictions of earlier stages, so intervals and multiplicity corrections can be recomputed without refitting anything, and the correction applied to a result is verifiable independently of the model that produced it. The cost of this granularity is intermediate storage, which is paid in Parquet; the benefit is that any single stage can be re-run and checked against the rest.

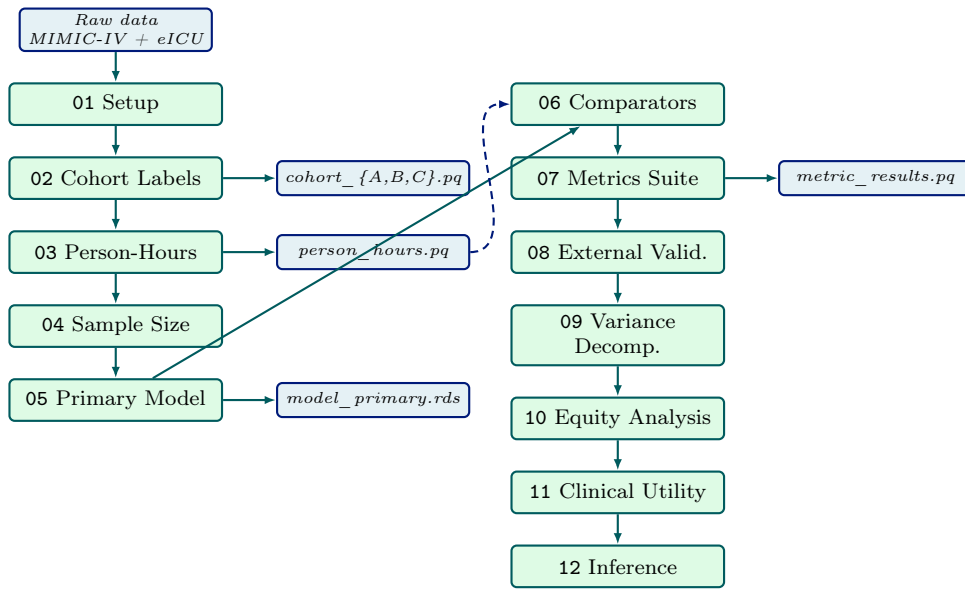


Figure 4.1: Pipeline data flow. Each box is a single script; italic nodes are output files. Raw MIMIC-IV and eICU data are read-only.

4.2 Configuration and Utilities

`config.R` is the single source of truth for all pipeline settings: data paths, cohort criteria, antibiotic name lists, MIMIC-IV item identifiers, label variant definitions, split boundaries, prediction framing constants, equity variable names, and Utility Score parameters. No other script hard-codes a clinical constant or file path.

`utils.R` provides shared helper functions: a DuckDB-backed streaming reader for large CSVs, Parquet reader/writer wrappers, datetime parsing, last-observation-carried-forward (LOCF) imputation, and Schoenfeld test wrappers.

The full pipeline can be reproduced on any MIMIC-IV installation by setting a single environment variable (`MIMIC_RESEARCH_ROOT`).

4.3 Stage 01: Setup Validation

`01_setup.Rmd` checks that all required MIMIC-IV source tables exist and reports their sizes. The pipeline stops if any table is missing.

4.4 Stage 02: Cohort Extraction and Labelling

`02_cohort_labels.Rmd` builds all cohorts in a single pass:

1. **Base cohort:** Adult first ICU stays with $\text{LOS} \geq 4$ h (65,078 stays). Equity variables (race, language, insurance) are attached.
2. **Suspected infection:** Qualifying antibiotic-culture pairs within each variant's time windows. Surveillance cultures are excluded.
3. **SOFA organ dysfunction:** SOFA components are computed from `chartevents`, `labevents`, `inputevents`, and `outputevents`. A SOFA rise ≥ 2 within each variant's windows defines organ dysfunction.
4. **Onset time:** For the pre-registered variants A–C, the suspicion time t_{susp} , provided the SOFA rise occurs somewhere in the evaluation window. The SOFA criterion therefore gates *membership* and contributes no timing information: an implementation choice with consequences developed in Section 3.6.1.
5. **Alternative labels (Protocol Amendments 2 and 4):** The Sepsis-3 conjunction is also split into its two limbs (Section 3.6). Variant E keeps the suspicion anchor and drops the SOFA criterion; Variant D keeps organ dysfunction and drops the anchor, so its onsets are derived from physiology in Stage 03 rather than here. Variant F (Amendment 4) keeps the conjunction intact and re-times the onset to $\min(t_{\text{susp}}, t_{\text{SOFA}})$, also in Stage 03 because it needs the hourly SOFA trajectory. All three are held outside the pre-registered variant set in `config.R` and their outputs are written to separate files so they cannot enter H1 or the decomposition table.

A single environment variable, `V2_VARIANTS`, restricts the per-variant loops in Stages 02–07 to a named subset. Adding the alternative labels therefore did not require refitting variants A–C: XGBoost is not bit-stable across runs, so an incidental refit would have moved every GBT-dependent number in this thesis without any change of substance.

4.5 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each stay into hourly rows. Large source tables (`chartevents`: 3.5 GB; `labevents`: 2.6 GB) are processed via DuckDB streaming, pre-filtered to cohort stays, to avoid loading everything into memory. For each person-hour:

- Vitals and labs are LOCF-imputed from the most recent measurement.
- Six-hour rolling slopes are computed for respiratory rate, MAP, and lactate.

- Vasopressor indicators are set from concurrent infusions.
- Missingness flags are set for each lab test.
- The competing-event code (0/1/2/3) is assigned based on the hour's relationship to onset, discharge, or death.
- The temporal split group is assigned from the stay's admission year.
- Treatment anchor timestamps (first antibiotic, first culture) are recorded.

This is the pipeline's most expensive stage after model fitting, and the only one that reads the full-size source tables; it is also the stage whose outputs every later stage depends on, which is why the streaming and read-only disciplines of Section 4.1 matter most here.

4.6 Stage 04: Sample Size Calculation

`04_sample_size.Rmd` applies the Riley et al. [46] criteria to the training portion (2008–2016) of each label variant. The criteria assume independent observations, so the calculation is parameterised on the *stay* rather than the person-hour: the outcome proportion supplied is the fraction of training stays labelled septic, and the resulting minimum is compared against the number of training stays. The per-person-hour parameterisation is computed alongside and retained as a labelled reference, so that the two cannot be substituted for one another.

4.7 Stage 05: Primary Model

`05_primary_model.Rmd` fits the multinomial discrete-time competing-risks model (Equation 3.3) for each label variant on the temporal training split. Baseline hazard splines use 5 knots, placed at quantiles of the training hour distribution. A random patient-level 75/25 split is also fitted to support H4.

The fit carries two guards that are part of the specification rather than defensive extras. The declared predictor set is required rather than intersected, so a name mismatch stops the stage instead of producing a model with fewer terms than specified. And because the unpenalised likelihood for this design is completely separated (its maximum lies at infinity, while the optimiser's stopping rule reports success regardless (Section 3.10.1)) the fit is ridge-penalised and the stage logs the largest fitted coefficient alongside the convergence flag, since only the former distinguishes an identified solution from an arbitrary stopping point. No analytic

variance is available for the fitted object, so all uncertainty for this model comes from the stay-level cluster bootstrap of Stage 12.

Both stages 05 and 06 run two **arms**: the pre-registered specification and the ablated one of Protocol Amendment 5 (Section 3.14.5), which drops the action-derived features. The ablated arm writes to its own files and never overwrites a pre-registered fit, so it can be added or refreshed without re-executing the fits every headline number rests on.

4.8 Stage 06: Comparator Models

`06_comparators.Rmd` fits the XGBoost GBT model, the static Cox model (with Schoenfeld testing), and computes NEWS2, qSOFA, and SIRS at each person-hour. GBT models are fitted under both temporal and random splits. GBT feature importance is exported for each variant.

The gradient-boosted comparator is additionally fitted on the ablated feature set with identical hyperparameters. NEWS2, qSOFA and SIRS never used the ablated features, and the Cox comparator's role here is the proportional-hazards check rather than discrimination, so neither is ablated.

4.9 Stage 07: Metrics Suite

`07_metrics_suite.Rmd` evaluates all models on all splits (temporal, random, treatment-anchored) using the full metric set: AUROC, AUPRC, calibration intercept and slope, PhysioNet Utility Score, alert burden, lead time, and decision-curve net benefit. The ablated arm is scored by the same code on the same test rows, and the paired difference is written to `07_ablation_metrics.csv`.

The Utility Score is computed at the default 0.5 cut-off and swept across a sixty-point grid of alert rates, in a single pass: the per-stay reward is monotone in a falling threshold, so the whole grid costs one sort plus one traversal of the reward-window rows. Two invariants are asserted at run time rather than assumed: a threshold above every predicted probability must return exactly zero, since it *is* the no-alert strategy, and no operating point may exceed the optimal strategy's score of one. A stand-alone Python implementation of the same scoring rule, `utility_score.py`, is kept in step with the notebook so that prediction tables outside the pipeline can be scored and the two implementations cross-checked.

4.10 Stage 08: External Validation on eICU-CRD

`08_external_validation.Rmd` applies the frozen MIMIC-IV models (primary, GBT and NEWS2) to eICU-CRD without any retraining. It builds the external person-hour panel once from physiology alone, then derives onsets *per label variant* from that panel using each variant’s own antibiotic–culture windows and SOFA-delta criterion (Section 3.2.2). Antibiotic and vasopressor exposure is taken from `medication` and `infusionDrug` together.

Two arms are reported. The primary arm applies the Sepsis-3 conjunction to the microbiology-covered sub-cohort; the secondary arm applies an antibiotic-only anchor to the full cohort and is labelled as a different target. AUROC and AUPRC are reported for each variant and model in both, alongside a labelling-input coverage table and an explicit reconciliation of the eICU-CRD and MIMIC-IV event rates.

The stage asserts at run time that the three variants produce distinct external onset sets and halts otherwise, so that a label-invariant result can never be produced by a label that was never varied.

4.11 Stage 09: Variance Decomposition

`09_variance_decomposition.Rmd` computes the pre-registered variance decomposition by varying one factor at a time while holding all others fixed:

- **Label spread (H1):** AUROC range across Variants A/B/C.
- **Model-class spread:** AUROC range across all models.
- **Split optimism (H4):** Random vs. temporal split comparison.
- **Metric divergence (H5):** AUROC vs. Utility Score decline.

It then runs the **label-anchor attribution** of Protocol Amendment 2: performance under each limb of the Sepsis-3 conjunction, the anchor-attributable fraction of Equation 3.1, and onset agreement (Cohen’s κ , median onset shift) between every label variant and Variant B.

4.12 Stage 10: Equity Analysis

`10_equity_analysis.Rmd` evaluates model performance and label sensitivity within each stratum of the equity variables declared in `config.R` (race, ethnicity, language, insurance). MIMIC-IV v3.1 records race and ethnicity in a single

field, so three variables carry data and are reported: **race**, **language**, and **insurance**. Levels are restricted to the eight largest per variable, and are strata of an administrative field rather than populations (Section 5.9.1). Discrimination is estimated for every stratum with enough rows to support an estimate, but only strata carrying at least `MIN_SUBGROUP_EVENTS_INTERPRET` sepsis onsets enter the exploratory contrast family, so the multiplicity correction is applied over the analyses the study is willing to interpret rather than over every stratum that happens to exist. Label sensitivity, being a stay-level proportion, is reported for all strata.

4.13 Stage 11: Clinical Utility Metrics

`11_clinical_metrics.Rmd` computes flexible calibration curves, decision-curve net benefit, operating points at a fixed per-hour sensitivity, per-patient (event-level) detection and alarm-episode counts, and lead time to onset, for both MIMIC-IV and eICU-CRD.

4.14 Stage 12: Inference and Multiplicity

`12_inference.Rmd` is the inference layer. It reads only the Parquet outputs of the earlier stages, so it can be re-run without refitting anything, and it produces three things: stay-level cluster-bootstrap confidence intervals for every headline and subgroup estimate; the Holm-corrected confirmatory tests of H1–H6 and the Benjamini–Hochberg-corrected exploratory families (E1 and E2 subgroup analyses, E3 alternative-label contrasts); and the machine-readable analysis register that supplies the status tags used throughout Chapter 5. The bootstrap resamples ICU stays, never person-hours, and evaluates AUROC and AUPRC as weighted rank statistics on pre-sorted data so that 2,000 replicates over half a million person-hours remain tractable. Setting `V2_BOOT_B` overrides the replicate count for a fast smoke test; reported results always use the values fixed in `config.R`. The statistical rationale is given in Section 3.14.

4.15 Reproducibility

The pipeline is orchestrated by `run_pipeline.R` (or the equivalent `run_pipeline.sh`), which executes all stages sequentially or by phase, renders each stage to PDF in `output/pdf/`, and writes a per-stage and a combined run log to `output/logs/`. All intermediate outputs are Parquet (compressed, columnar). Small result tables (<1,000 rows) are additionally written as CSV for readability.

The end-to-end run reported in this thesis completed in 60 minutes on an Apple silicon workstation; the dominant cost is stage 05, which fits the multinomial model on 2.3 million person-hours for each label variant.

4.15.1 Fixed Random Seed

All random operations use a single fixed seed (42) to ensure bitwise reproducibility. Every Rmd script sets `set.seed(42)` in its setup chunk before any computation. The same value is used for the patient-level random split (`RANDOM_SPLIT_SEED` in `config.R`), the `pmsampsize` sample-size calculation, and all model-fitting calls. No seed is drawn from wall-clock time or left to the system default.

The full pipeline can be reproduced on any MIMIC-IV v3.1 installation by setting the environment variable `MIMIC_RESEARCH_ROOT` to the directory containing the `data/` folder.

4.15.2 Code Availability

The complete pipeline (the twelve numbered scripts, `config.R`, `utils.R`, `clinical_scores.R`, `thesis_constants.R`, `utility_score.py`, the pre-registration document and the orchestration scripts) is held in a public version-controlled repository at <https://github.com/ahadarif007/RESEARCH>, released under the Apache License 2.0. Appendix A describes the layout, the reproduction procedure, and the boundary of that licence, which covers the source written for this study and neither the databases nor the institutional template.

The repository contains **code only**. No MIMIC-IV or eICU-CRD record, derived person-hour table, or fitted model object is included: the `data/`, `processed_data/` and `output/` directories are excluded from version control by design, because the PhysioNet Credentialed Data Use Agreement (Section 3.3) prohibits redistribution of the underlying data. A reader wishing to reproduce these results must obtain MIMIC-IV v3.1 and eICU-CRD v2.0 through their own PhysioNet credentialing, then re-run the pipeline; with the fixed seed above it reproduces the reported estimates exactly.

Chapter 5

Results

This chapter reports the results in two parts. The first and larger part presents the measurements: cohort description and label variance, sample size adequacy, internal temporal performance and calibration, clinical utility (net benefit, operating points, event-level detection and lead time), external validation on eICU-CRD, label-anchor attribution, the action-derived feature ablation, model diagnostics, and the equity analysis. The second part, Section 5.10, assembles those measurements into the variance decomposition and adjudicates the six pre-registered hypotheses; Section 5.11 then records the status of every analysis reported here.

The verdicts come last because each is computed from quantities the earlier sections establish, and two of them cannot be read without those quantities in hand: H5 is a contrast between the internal and external estimates, and the decomposition is an assembly of contrasts drawn from four different sections. A reader who wants the answers first may begin at Section 5.10, but the evidence for them lies above it rather than below. Chapter 6 interprets them, states how far each estimate can be pushed, and sets out the limitations.

Every result below carries an explicit status tag ([C] confirmatory, [E] exploratory, [D] descriptive) defined in Section 3.14 and enumerated in full in Table 5.32. Confirmatory results test a hypothesis whose estimand, threshold and direction were fixed before any modelling; exploratory results were pre-registered in scope only and generate hypotheses rather than testing them. All intervals are 95% percentile intervals from a stay-level cluster bootstrap. The six pre-registered hypotheses are corrected by Holm–Bonferroni at a family-wise error rate of 0.05; the exploratory subgroup families are corrected by Benjamini–Hochberg at $q = 0.05$.

5.1 Cohort Description and Label Variance [D]

Table 5.1 summarises the three cohorts. The base extraction identifies 65,078 adult first ICU stays. The database and patients are the same across all three variants; only the sepsis definition differs.

Var	Name	Stays	Onsets (cohort)	%	Deaths	Onset h (IQR)
A	Narrow	65,078	6,252	9.6	5,934	10[4–26]
B	Seymour-Std	65,078	6,447	9.9	5,892	11[4–30]
C	Liberal	65,078	12,581	19.3	4,956	10[4–30]

Table 5.1: Cohort description by label variant. The Liberal variant identifies about twice as many sepsis cases as the Narrow or Seymour-Standard variants, mainly because it accepts wider antibiotic-culture windows and uses a rolling SOFA baseline. Onsets are counted **cohort-wide, at any hour of the stay**. Table 5.15 reports the subset falling inside the 72-hour modelling horizon (5,662, 5,887 and 11,531), which is the count the models are trained and evaluated on; the two columns are different quantities, not conflicting ones.

The twofold difference in sepsis prevalence between Variant A (9.6%) and Variant C (19.3%) confirms the label-sensitivity finding from Johnson et al. [15] at MIMIC-IV scale.

5.2 Sample Size Adequacy [C]

Riley et al.’s criteria [46] are derived for a sample of *independent* observations. The modelling unit here is the person-hour, and person-hours are not independent: they are nested within ICU stays, one per patient. The adequacy assessment is therefore stated on the unit that carries the information, the stay, and the person-hour count is reported alongside it as the size of the design matrix, not as a sample size. Table 5.2 keeps the two apart.

Var	Name	Train stays	Sepsis stays	Stay prev.	Riley min- N (stays)	EPV
A	Narrow	52,946	4,647	0.0878	1,729	160.2
B	Seymour-Std	52,946	4,847	0.0916	1,666	167.1
C	Liberal	52,946	9,403	0.1776	991	324.2

Table 5.2: Sample size adequacy on the independent unit. Every row is a count of *stays*: the training split contains 52,946 stays, of which the given number are labelled septic under that variant. Riley min- N is the minimum number of independent observations required at that variant’s stay-level prevalence, an anticipated C -statistic of 0.846 [21], and 29 candidate predictors. EPV is events per candidate predictor. All three variants exceed the requirement by a wide margin ($30.62\times$, $31.78\times$ and $53.43\times$ respectively).

The corresponding design matrices contain 2,304,923, 2,301,842 and 2,120,376 training person-hours, at per-hour event rates of 0.00202, 0.00211 and 0.00443. Those figures govern the computational cost of the fit and the precision of an hourly hazard estimate conditional on the clustering; they do not multiply the effective sample size, and reading 2.3 million rows as 2.3 million observations would overstate the available information by roughly the average number of at-risk hours per stay. The relevant denominator for every inferential statement in this thesis is the stay, which is why all intervals reported in later sections come from a stay-level cluster bootstrap rather than a row-level one.

Sample size is not the binding constraint on this study, and stating the criterion on the right unit is what makes that visible: the requirement is 1,666 independent observations under the primary label against 52,946 available, a margin of $31.78\times$. Precision binds elsewhere, where the events are, in the external cohort (Section 6.3) and in the smaller demographic strata (Section 5.9), not in the development sample.

5.3 Model Performance: Internal Temporal Evaluation [C]

Table 5.3 reports performance on the temporal test set (2017–2019) under the unrestricted evaluation window.

Var	Model	AUROC	AUPRC	Utility (0.5)	Utility (best)	Alert burden
A	Primary	0.7713	0.0093	0.000	0.026	∞
A	GBT	0.7455	0.0088	-1.614	0.038	192.1
A	NEWS2	0.6413	0.0032	-0.509	-0.001	236.4
A	Cox	0.6589	0.0035	-4.790	0.000	356.4
A	qSOFA	0.6012	0.0027	-2.293	-0.263	343.1
A	SIRS	0.6157	0.0032	-3.209	-0.035	346.9
B	Primary	0.7590	0.0096	0.000	0.025	∞
B	GBT	0.7424	0.0087	-1.781	0.031	196.5
B	NEWS2	0.6415	0.0032	-0.491	0.000	246.7
B	Cox	0.6444	0.0033	-4.720	0.000	361.6
B	qSOFA	0.6091	0.0028	-2.193	-0.252	316.8
B	SIRS	0.6114	0.0030	-3.086	-0.035	351.7
C	Primary	0.7695	0.0201	0.000	0.095	∞
C	GBT	0.7550	0.0188	-0.557	0.111	89.2
C	NEWS2	0.6483	0.0074	-0.099	0.001	102.2
C	Cox	0.6733	0.0085	-1.613	0.003	148.7
C	qSOFA	0.6012	0.0060	-0.705	-0.048	150.9
C	SIRS	0.6242	0.0070	-1.065	0.000	148.4

Table 5.3: Internal temporal test performance (2017–2019, all variants use temporal split). N test rows: 548,827 (B). **Utility (0.5)** is the normalised PhysioNet Utility Score at the default cut-off; **Utility (best)** is its maximum over the threshold grid of Section 3.12. Zero is the no-alert strategy under both. Alert burden is at the 0.5 cut-off (∞ = no true alerts there); the benchmark is 1.4 [21]. Per-threshold curves are in Figure 5.1 and the attaining operating points in Table 5.5. Confidence intervals for Variant B are in Table 5.4. **[C] confirmatory.**

Model	AUROC	95% CI	AUPRC (95% CI)
Primary	0.7590	0.7434–0.7749	[0.0082, 0.0123]
GBT	0.7424	0.7265–0.7583	[0.0075, 0.0110]
NEWS2	0.6415	0.6248–0.6589	[0.0029, 0.0037]
Cox	0.6444	0.6271–0.6614	[0.0030, 0.0038]
qSOFA	0.6091	0.5934–0.6254	[0.0026, 0.0033]
SIRS	0.6114	0.5950–0.6281	[0.0028, 0.0039]

Table 5.4: Variant B temporal test performance with 95% stay-level cluster-bootstrap intervals ($B = 2,000$ replicates over 548,827 test person-hours carrying 1,040 sepsis onsets). All six models are scored on the same replicates, so differences between rows may be read as paired contrasts. The primary model and GBT overlap almost completely; both are separated from the four remaining models, which overlap one another. **[C] confirmatory.**

5.3.1 Key Finding: No Model Reaches a Deployable Operating Point

At the default 0.5 cut-off no model does better than issuing no alerts. That statement needs one immediate qualification, because the cut-off is doing much of the work. The primary model scores 0.000 under Variant B not because it alerts badly but because it never alerts: its largest predicted hourly probability is far below 0.5, so at that cut-off it *is* the no-alert strategy and scores exactly the no-alert strategy’s zero. The models that do cross 0.5 (the rule-based scores, whose outputs are bounded integer counts rescaled onto the unit interval, and the Cox linear predictor) cross it so often that they score deeply negative: -4.720 for Cox and -2.193 for qSOFA under Variant B. A single arbitrary operating point therefore separates the models by their calibration rather than by their clinical value, and no conclusion about clinical failure can rest on it.

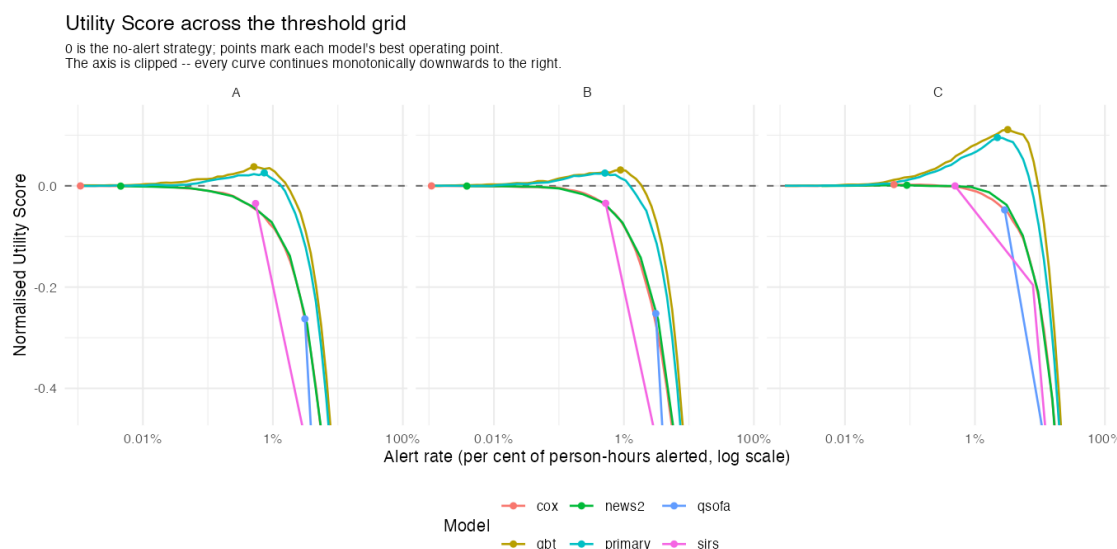


Figure 5.1: Normalised Utility Score against alert rate for every model and pre-registered label variant on the temporal test set. The dashed line at zero is the no-alert strategy: a curve below it describes a configuration that would be better switched off, and points mark each model’s best operating point. Only the primary model and GBT rise above the line at any alert rate, peaking between 0.5 and 3 per cent of person-hours alerted, two orders of magnitude away from the 0.5 probability cut-off. The vertical axis is clipped: every curve continues monotonically downwards to the right, reaching -5 or below at a 50 per cent alert rate.

Sweeping the threshold answers the question the cut-off cannot. Figure 5.1 traces the whole grid, and Table 5.5 reports each model’s best attainable score together with the operating point that attains it.

Var	Model	Max utility	Alert rate (% p-h)	Stays detected (%)	Alert burden
A	Primary	0.026	0.737	15.6	44.9
A	GBT	0.038	0.511	13.0	40.2
A	NEWS2	−0.001	0.005	0.0	∞
A	Cox	0.000	0.001	0.0	∞
A	qSOFA	−0.263	3.114	13.9	254.1
A	SIRS	−0.035	0.542	4.1	122.9
B	Primary	0.025	0.511	11.2	47.3
B	GBT	0.031	0.885	17.0	51.8
B	NEWS2	0.000	0.004	0.0	∞
B	Cox	0.000	0.001	0.0	∞
B	qSOFA	−0.252	3.095	13.5	248.8
B	SIRS	−0.035	0.523	3.5	135.8
C	Primary	0.095	2.213	28.5	29.3
C	GBT	0.111	3.194	35.5	34.0
C	NEWS2	0.001	0.090	0.8	49.2
C	Cox	0.003	0.057	0.6	46.7
C	qSOFA	−0.048	2.868	13.0	111.2
C	SIRS	0.000	0.493	3.4	59.7

Table 5.5: Maximum achievable Utility Score and the operating point that attains it, over the sixty-point threshold grid, on the temporal test set. “Alert rate” is the percentage of test person-hours alerted; “stays detected” is the percentage of septic stays alerted within the reward window [onset − 6, onset]; alert burden is false alerts per true alert at that same point, against the benchmark of 1.4 [21]. An ∞ burden with a max at or below zero means the model’s best strategy is to raise no alert. **[E] exploratory:** the threshold is chosen on the test set, so these are upper bounds on achievable utility, not estimates of deployed performance.

Four things follow, and they are more specific than a blanket verdict of failure.

Tuning matters, and it moves the sign. The two fitted models cross into positive utility once the threshold is chosen sensibly: the primary model reaches 0.025 under Variant B and 0.095 under Variant C, and GBT reaches 0.031 and 0.111. A claim that these models are worse than doing nothing would be an artefact of the operating point, and is not made here.

The margin is nonetheless very small. On a scale whose upper bound is 1, the best a model achieves under the primary label is 0.031. The optimum is a shallow peak a few per cent above the no-alert strategy, not a clinically meaningful gain, and it is measured with the threshold selected on the same test set, so even that margin is optimistic.

Four of the six models cannot be tuned above zero at all. Under Variants A and B, NEWS2, Cox, qSOFA and SIRS have no threshold whatever at which alerting beats not alerting; their maxima are 0.000, 0.000, -0.252 and -0.035 . For NEWS2 and Cox the maximum is attained where the model issues almost no alerts, which is the sweep’s way of saying that the best available strategy is silence. The rule-based scores in routine ward use are in this group.

The alarm volume is the binding constraint. Every maximum that is positive at all is attained at between 29.3 and 51.8 false alerts per true alert, twenty to thirty-seven times the benchmark of 1.4 [21], and the two fitted models detect only 11.2 to 35.5 per cent of septic stays in the actionable window there. The Utility Score’s own optimum is a configuration that misses most septic patients and alarms constantly on the rest. That, rather than the sign of a score at an arbitrary cut-off, is what the evidence supports.

This is consistent with Wang et al. [4]: the internal-to-external utility collapse they found across ninety-one models already has its internal counterpart on MIMIC-IV under an honest time-based split, in the form of an optimum that sits barely above zero before the model leaves its development cohort. The failure is only visible because the Utility Score is reported; AUROC alone would hide it, and Table 5.3 shows discrimination in the high 0.7s for the same configurations.

One further qualification is developed in Section 5.4. The Utility Score and decision-curve net benefit disagree here because they price false alarms differently, at a threshold matched to the event prevalence the primary model does achieve positive net benefit over both default strategies (Table 5.7), and that disagreement is itself evidence for H5 rather than a reason to prefer one number. The conclusion that survives both metrics is the alarm-volume one: no configuration tested reaches an alert burden a ward could absorb.

5.3.2 Calibration [C]

Calibration intercept and slope are two summary numbers describing a whole curve, so both the curve itself and a scalar summary of its distance from the diagonal are reported. Following Van Calster et al. [29], the flexible calibration curve is a restricted cubic spline of the outcome on the logit of predicted risk, and it is summarised by the Integrated Calibration Index (ICI, the mean absolute distance between predicted risk and that curve) and the scaled Brier score, $1 - \text{Brier}/\text{Brier}_{\text{null}}$, where the null model predicts the prevalence for every person-hour.

Var	Model	Intercept	Slope	ICI	$E_{\max}$	Scaled Brier
A	Primary	−0.077	1.013	0.0002	0.019	0.0037
A	GBT	−6.049	1.143	0.3789	0.893	−89.1792
A	Cox	−6.418	1.034	0.4986	0.978	−140.9298
B	Primary	−0.106	1.055	0.0002	0.033	0.0038
B	GBT	−6.059	1.306	0.3993	0.864	−93.5482
B	Cox	−6.399	0.957	0.5016	0.975	−138.8289
C	Primary	−0.035	1.084	0.0003	0.006	0.0082
C	GBT	−5.295	1.253	0.3971	0.836	−42.3814
C	Cox	−5.564	1.168	0.4933	0.984	−60.3299

Table 5.6: Calibration on the temporal test set. Perfect calibration is intercept = 0, slope = 1, ICI = 0. A scaled Brier score at or below zero means the model is no better than predicting the prevalence for everyone. NEWS2, qSOFA and SIRS are omitted: their score-to-probability lookups take only 21, 4 and 5 distinct values respectively, too few to support a flexible calibration curve.

Two results stand out. First, only the primary model is calibrated at all, and it is calibrated under *every* label variant: its intercept sits within 0.11 of zero, its slope within 0.08 of one (1.013, 1.055, 1.084), and its largest ICI across the three variants is 0.0003, under a fifth of the 0.2 per cent event rate. Its scaled Brier score is positive under all three, though barely (0.0037 to 0.0082), meaning it improves on the null model by less than one per cent. Calibration this close to the diagonal at this event rate is the result that makes the model’s outputs usable as probabilities at all, and it is what distinguishes it from every comparator here.

Second, GBT and Cox are not merely imperfectly calibrated; they are miscalibrated by roughly three orders of magnitude. Both carry intercepts near −6, and their scaled Brier scores run into the tens and hundreds below zero. A scaled Brier of −93.5482 means the GBT model’s squared error is some ninety-eight times worse than simply predicting the prevalence for every person-hour. Figure 5.2 shows why: GBT emits predicted risks between 0.2 and 0.9 for person-hours whose observed event rate never exceeds 0.04. This matters directly for Section 5.4: a model whose probabilities are inflated by two orders of magnitude cannot be operated at a threshold chosen from clinical reasoning about risk, because its outputs do not mean what they appear to mean. Its useful operating points exist, GBT’s best lies near 0.8157 under Variant B (Table 5.5), but they can only be located empirically, by sweeping, and they carry no interpretation as a risk a clinician has agreed to act on.

The consistency across variants is itself worth noting. Calibration is the one respect in which the three label definitions make no difference at all: slopes of 1.013,

1.055 and 1.084 and $E_{\max}$ values of 0.019, 0.033 and 0.006 are indistinguishable for practical purposes. Whatever the label changes about which patients are septic, it does not change how faithfully the fitted hazard tracks the observed rate.

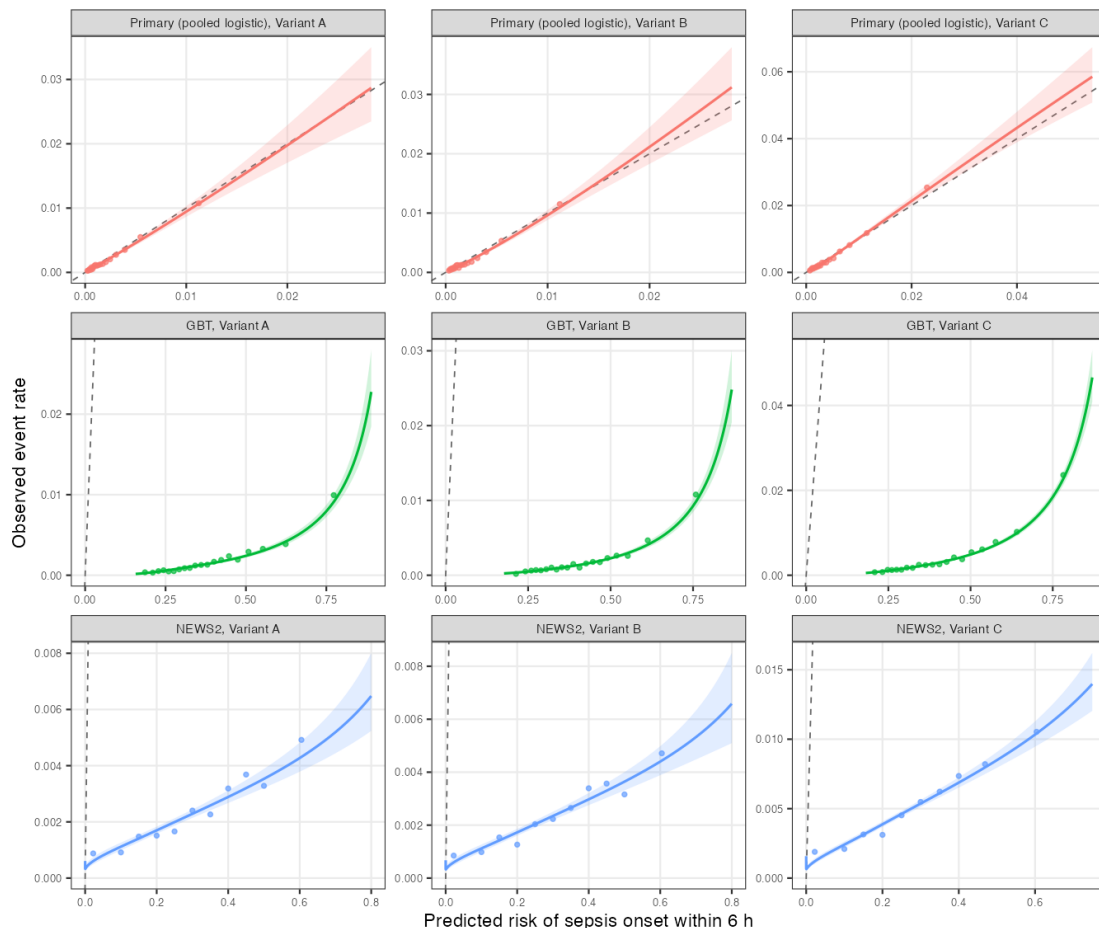


Figure 5.2: Flexible calibration curves on the temporal test set. Dashed line = perfect calibration; points = twenty equal-sized bins of predicted risk; shaded band = 95% confidence interval. Each panel has its own scale because the predicted-risk ranges differ by orders of magnitude. The primary model tracks the diagonal under all three variants; GBT sits far to the right of it in every variant.

A Correction to the Clamped Estimates

The calibration intercepts and slopes above supersede those produced by the metric-suite notebook, which computed them after clamping predicted probabilities to $[0.001, 0.999]$. That guard is standard for models whose predictions approach 0 or 1, but at a 0.2 per cent hourly event rate it is actively harmful: 46.8%

(Variant A), 39.6% (Variant B) and 8.8% (Variant C) of the primary model’s predicted probabilities fall *below* the clamp floor, so the clamp was overwriting a large share of the quantity being measured. No predicted probability exceeds 0.2342 under any variant, so the ceiling never binds; it is only the floor that does damage. The values in Table 5.6 use a floor of 10^{-6} , below the smallest non-zero prediction in the test set. Under the conventional clamp the primary model’s Variant A slope reads 1.137 against 1.013 on the 10^{-6} floor: the clamp makes a well-calibrated model look materially worse than it is. Both figures are given rather than only the second, because the gap between them is exactly the kind of scale-dependent analyst decision this study set out to measure: a calibration guard chosen without reference to the event rate is itself a source of reported variance.

5.4 Clinical Utility: Net Benefit, Operating Points, and Event-Level Detection

AUROC, AUPRC and the Utility Score describe a model; they do not describe what a ward would experience if the model were switched on. This section reports the four quantities that do: net benefit across the range of thresholds a clinician might plausibly accept, the positive predictive value and false-alarm rate at a fixed sensitivity, detection expressed per patient rather than per person-hour, and the lead time gained when detection occurs.

One framing point governs everything below. Under the discrete-time competing-risks specification, $\text{event} = 1$ marks the sepsis onset hour itself, so each septic stay contributes exactly one event hour out of a mean of 45. Per-hour sensitivity therefore demands that the alert fire *in the onset hour*; per-patient sensitivity asks only whether the patient was alerted at all before onset. The two are different questions, and reporting only the first, as the metric suite in Section 5.3 does, understates detection while hiding the alarm volume that buys it.

5.4.1 Decision-Curve Analysis [C]

Decision-curve analysis, pre-specified in Chapter 3, requires a choice of threshold grid, and at this prevalence that choice determines the entire result. A threshold probability of t encodes a willingness to accept $(1 - t)/t$ false alarms per true detection. The hourly event rate here is 0.19–0.42 per cent, so the conventional grid of 0.01–0.20 sits between five and one hundred times the prevalence: at $t = 0.01$ the analyst is asserting a tolerance of only 99 false alarms per detection, far below what an hourly sepsis alert must absorb, and every curve falls trivially below “treat none”. Net benefit is therefore evaluated on a grid running from one tenth to ten

times the prevalence, following Vickers and Elkin [45]. The first run of the metric suite used the conventional grid and produced no usable curve; that null result is a property of the grid, not of the models.

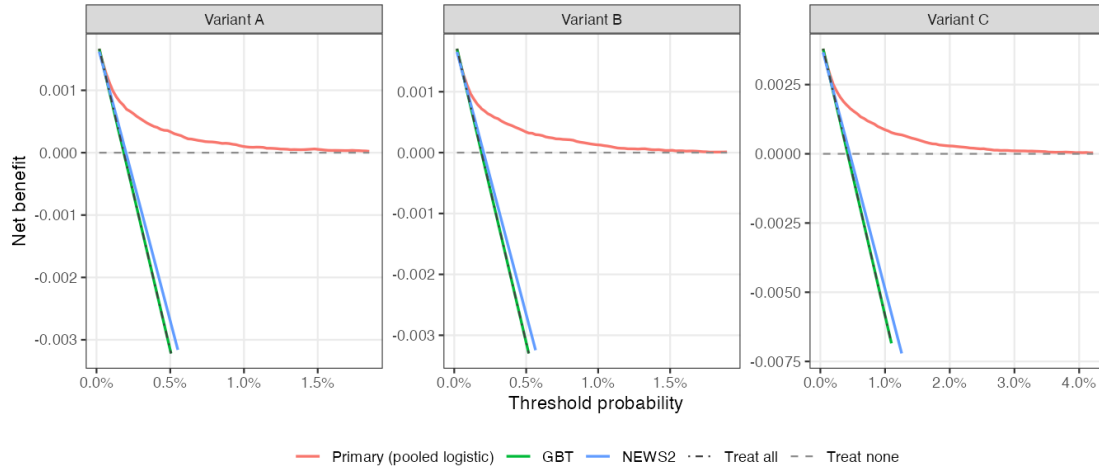


Figure 5.3: Decision curves on the temporal test set. Net benefit against threshold probability, with both default strategies shown. The GBT curve lies exactly on top of “treat all” in every variant. Curves are clipped below $-1.75\times$ prevalence so the near-zero region stays legible; a line leaving the bottom of a panel is worse than treating nobody from that threshold onward.

Var	Model	Net benefit	Std NB	Alert rate	Alert-rate span
A	Primary	7.53×10^{-4}	0.407	0.305	0.987
A	GBT	0	0.000	1.000	0.000
A	Cox	0	0.000	1.000	0.000
A	NEWS2	1.33×10^{-4}	0.072	0.894	0.000
A	qSOFA	2.70×10^{-4}	0.146	0.656	0.000
A	SIRS	2.51×10^{-4}	0.136	0.676	0.000
B	Primary	7.19×10^{-4}	0.380	0.318	0.995
B	GBT	0	0.000	1.000	0.000
B	Cox	0	0.000	1.000	0.000
B	NEWS2	1.31×10^{-4}	0.069	0.894	0.000
B	qSOFA	2.84×10^{-4}	0.150	0.656	0.000
B	SIRS	2.83×10^{-4}	0.149	0.676	0.000
C	Primary	1.73×10^{-3}	0.411	0.289	0.997
C	GBT	0	0.000	1.000	0.000
C	Cox	0	0.000	1.000	0.000
C	NEWS2	2.98×10^{-4}	0.071	0.888	0.000
C	qSOFA	6.45×10^{-4}	0.153	0.644	0.000
C	SIRS	6.91×10^{-4}	0.164	0.665	0.000

Table 5.7: Net benefit at a threshold probability equal to the event prevalence. At that threshold both defaults have net benefit exactly zero and the maximum attainable net benefit is the prevalence itself, so standardised net benefit (Std NB = net benefit / prevalence) is the fraction of the attainable benefit captured. “Alert-rate span” is the change in alert rate across the whole threshold grid; a span of zero means the score cannot be tuned anywhere in the plausible range.

Read on the correct threshold scale, decision-curve analysis gives the most favourable result in this thesis, and it is still a weak one. At a threshold equal to the prevalence, the primary model captures 40.7% (Variant A), 38.0% (Variant B) and 41.1% (Variant C) of the attainable net benefit. The three variants agree closely, spanning barely three percentage points, so unlike discrimination, where the label moves the answer, the share of attainable benefit captured is insensitive to which reading of Sepsis-3 is used. This is genuine, if modest, value over both defaults, and it is the same qualitative verdict the swept Utility Score reaches in Table 5.5: a small positive margin for the fitted models once the threshold is set on the right scale, and nothing at all for the rule-based comparators. The two metrics still differ in magnitude, because they price errors differently: net benefit weights a false alarm by $t/(1-t)$, which at $t = 0.00189$ is almost nothing, whereas the Utility Score imposes a fixed -0.05 penalty per false-alert hour regardless of threshold. That is why net benefit is the more generous of the two here, and why

the gap between them widens as the alert rate rises. Which pricing is right is a clinical judgement, not a statistical one, and the divergence is itself an instance of H5.

The alert-rate span column carries the more damaging finding. For every model except the primary one it is exactly zero: across the entire plausible threshold range, the alert rate does not move. GBT and Cox alert on *100% of person-hours* at every threshold in this range, so their decision curves are not merely poor but identical to treating everyone: the green GBT line in Figure 5.3 sits exactly on the treat-all reference. This is the direct consequence of the calibration failure in Table 5.6: with predicted probabilities inflated to the 0.2–0.9 range, no threshold below 4 per cent excludes anything. NEWS2, qSOFA and SIRS are non-tunable for the opposite reason (their integer scores map to a handful of fixed probabilities, all above the grid), and alert on a constant 58–86 per cent of person-hours. The primary model is the only one of the six that can be operated as a threshold-based alert at all.

5.4.2 Operating Points at Fixed Sensitivity [E]

The default 0.5 cut-off used in Section 5.3 is not how an alerting system is commissioned; a sensitivity target is chosen and the threshold follows. Table 5.8 sets the threshold to the largest cut-off achieving 80 per cent per-hour sensitivity and reads off the resulting operating characteristics.

Var	Model	Spec	PPV	NNE	Alerts/pt-day	FA/pt-day	FP:TP
A	Primary	0.562	0.0034	296	10.52	10.48	295
A	GBT	0.540	0.0032	311	11.05	11.02	310
A	NEWS2	0.301	0.0023	442	16.79	16.75	441
A	Cox	0.404	0.0025	403	14.32	14.28	402
B	Primary	0.522	0.0032	315	11.49	11.45	314
B	GBT	0.511	0.0031	322	11.74	11.70	321
B	NEWS2	0.301	0.0023	430	16.79	16.75	429
B	Cox	0.380	0.0024	409	14.88	14.85	408
C	Primary	0.565	0.0077	129	10.48	10.40	128
C	GBT	0.541	0.0073	136	11.04	10.96	135
C	NEWS2	0.315	0.0052	191	16.47	16.38	190
C	Cox	0.423	0.0058	171	13.87	13.78	170

Table 5.8: Operating characteristics at a fixed per-hour sensitivity of 80%. NNE = number needed to evaluate (1/PPV), the alerted person-hours required per true event hour. FA/pt-day = false-alert hours per patient-day. qSOFA and SIRS are reported separately in the text because they degenerate at this target.

At 80 per cent sensitivity, PPV never exceeds 0.0077. The best case, the primary model under the Liberal label, still requires 129 alerted person-hours per true event hour; under the Seymour-Standard label the same model requires 315. Every configuration generates between 10.40 and 16.75 false-alert hours per patient-day. At the best operating point that is 10.40 false alert-hours per occupied bed per day: a twelve-bed unit would field roughly 125 of them daily, every day, before a single true alert is counted.

The ordering here is also the one that matters clinically, and it is not the AUROC ordering. The primary model and GBT are separated by less than 0.02 AUROC (H6), but at a fixed sensitivity target the primary model needs fewer alerts per true event under all three labels. Discrimination summarised over all thresholds and discrimination at the threshold anyone would actually deploy are not the same comparison.

Setting the threshold this way also exposes the coarse rule-based scores. qSOFA cannot be tuned to 80 per cent sensitivity under the Liberal label at all: with only 4 distinct score values, the lowest cut-off reaching the target is the one that alerts on every person-hour (specificity 0.000, 23.90 false-alert hours per patient-day: an alert on every hour of every stay). It reaches the target under Variants A and B only at specificity 0.344 and 15.72 false-alert hours per patient-day. Their apparent competitiveness on AUROC in Table 5.3 does not survive being asked to hit a sensitivity target.

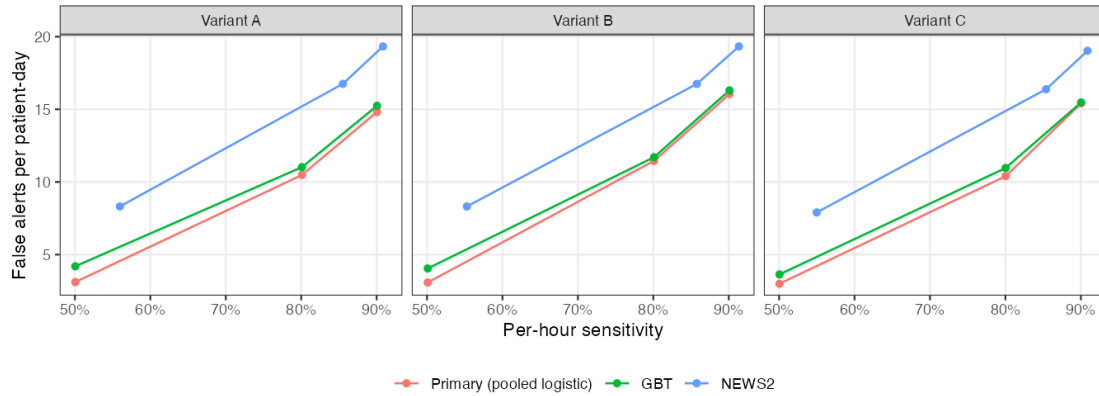


Figure 5.4: False alerts per patient-day against per-hour sensitivity, at sensitivity targets of 50%, 80% and 90%. The relationship is close to linear across the whole range: there is no operating point at which sensitivity is obtained cheaply.

5.4.3 Per-Patient (Event-Level) Detection [C]

Table 5.9 re-expresses the same operating points per patient. Episode sensitivity within the actionable window counts a septic stay as detected if any alert fired between six hours before onset and onset itself. A false-alarm episode is a contiguous run of alert hours that does not overlap the window from six hours before to three hours after onset, so a five-hour alert run counts once rather than five times.

Var	Model	Ep. sens (6 h)	Ep. sens (any pre)	Patient PPV	% stays alerted	FA ep./pt-day
A	Primary	0.880	1.000	0.084	100.0	0.92
A	GBT	0.914	0.989	0.085	97.0	1.37
A	NEWS2	0.968	0.987	0.084	98.2	2.02
A	Cox	0.895	0.924	0.083	93.6	1.94
B	Primary	0.880	0.996	0.085	100.0	1.03
B	GBT	0.904	0.987	0.088	95.8	1.19
B	NEWS2	0.967	0.991	0.086	98.3	2.02
B	Cox	0.902	0.923	0.084	93.8	1.89
C	Primary	0.864	1.000	0.175	100.0	0.95
C	GBT	0.902	0.988	0.181	95.7	1.03
C	NEWS2	0.964	0.988	0.177	98.1	2.03
C	Cox	0.896	0.925	0.177	91.9	1.77

Table 5.9: Per-patient detection at the 80% per-hour sensitivity operating point, on 12,132 test stays of which 1,015 (A), 1,040 (B) and 2,128 (C) are septic. Ep. sens (6 h): fraction of septic stays alerted within $[\text{onset} - 6 \text{ h}, \text{onset}]$. Ep. sens (any pre): alerted at any point before onset. FA ep./pt-day: false-alarm episodes, not alert-hours, per patient-day.

Read alone, the first two columns look like a success: 0.880 of septic patients are alerted inside the actionable six-hour window under the primary label, and 0.996 are alerted at some point before onset. This is the number a paper reporting only per-patient detection would headline.

The fifth column shows what it costs. To achieve it, the models alert on 93.8–100.0 per cent of *all* stays. Patient-level PPV is consequently 0.085 under Variant B and 0.175 under Variant C, which are, to three decimal places, the septic fractions of the cohort themselves (1,040/12,132 and 2,128/12,132). At the patient level, being alerted carries no information about whether a patient will become septic, because essentially every patient is alerted. The primary model under Variants A and C alerts on 100.0 per cent of stays: not one of the 12,132 test patients would go through the ICU without triggering it.

Per-Hour and Per-Patient, Side by Side

Var	Model	Sens (hr)	Sens (pt)	PPV (hr)	PPV (pt)	FA hr/pt-day	FA ep./pt-day
A	Primary	0.801	1.000	0.0034	0.084	10.48	0.92
A	GBT	0.801	0.989	0.0032	0.085	11.02	1.37
A	NEWS2	0.855	0.987	0.0023	0.084	16.75	2.02
B	Primary	0.801	0.996	0.0032	0.085	11.45	1.03
B	GBT	0.801	0.987	0.0031	0.088	11.70	1.19
B	NEWS2	0.858	0.991	0.0023	0.086	16.75	2.02
C	Primary	0.800	1.000	0.0077	0.175	10.40	0.95
C	GBT	0.800	0.988	0.0073	0.181	10.96	1.03
C	NEWS2	0.854	0.988	0.0052	0.177	16.38	2.03

Table 5.10: The same operating point viewed per person-hour and per patient. Both views are correct; across these rows they differ by more than twentyfold in PPV and by roughly tenfold in alarm count. A study reporting only one column of each pair can make the same model look either useless or usable.

The gap between the two views is the methodological point of this section. Per-hour PPV of 0.0032 and per-patient PPV of 0.085 describe one model at one threshold on one test set. The choice of denominator moves the headline number by a factor of twenty-six, which is larger than the spread this thesis attributes to label definition (H1) or model class (H6). Denominator choice belongs on the list of analyst decisions that require pre-specification.

5.4.4 Lead Time to Onset [C]

Lead time is the interval between the first alert issued at or before onset and the onset hour, over septic stays that were alerted at all.

Var	Model	Median (h)	IQR (h)	≥ 3 h	≥ 6 h
A	Primary	8.0	3–19	0.850	0.593
A	GBT	8.0	3–19	0.832	0.591
A	NEWS2	8.0	4–20	0.868	0.618
A	Cox	8.0	3–19	0.749	0.539
B	Primary	8.0	3–21	0.827	0.585
B	GBT	8.0	3–21	0.819	0.579
B	NEWS2	8.0	4–21	0.866	0.605
B	Cox	7.0	3–21	0.744	0.531
C	Primary	7.0	3–21	0.822	0.574
C	GBT	7.0	3–21	0.822	0.577
C	NEWS2	8.0	4–21	0.865	0.605
C	Cox	7.0	3–21	0.722	0.511

Table 5.11: Lead time at the 80% sensitivity operating point. The ≥ 3 h and ≥ 6 h columns are fractions of *all* septic stays, not only detected ones.

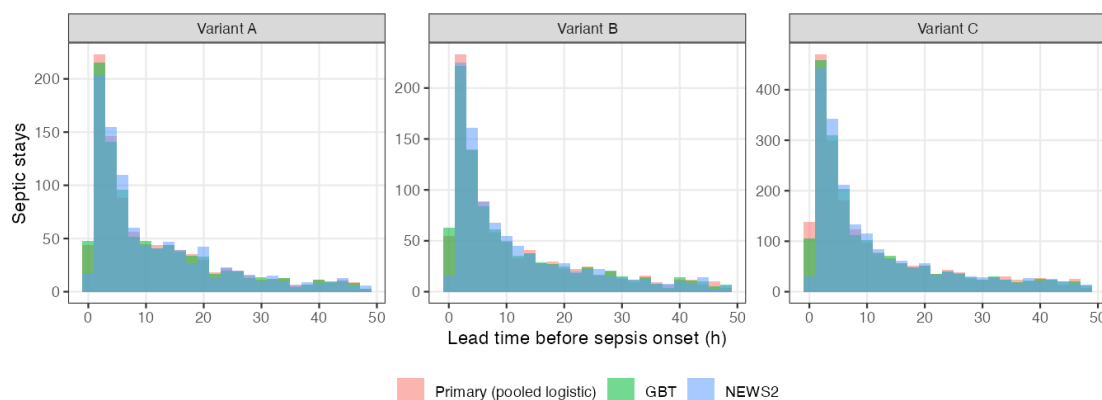


Figure 5.5: Distribution of lead time before onset at the 80% sensitivity operating point, truncated at 48 h. The mode sits in the first four hours with a long right tail.

Median lead time is 7.0–8.0 hours for the primary model, with 0.585 of septic stays alerted at least six hours ahead under the primary label.

These numbers are reported for completeness and should not be compared with the 4–12 hour horizons in the literature [19], [21], because they are not measuring the same thing. Lead time is measured over stays that were alerted, and at these thresholds almost every stay is alerted; an alarm that fires on every patient will, mechanically, fire early on the septic ones. The right-skewed distribution in Figure 5.5, with a long tail past 24 hours, is a signature of alerts unrelated to the

impending onset. Lead time is a meaningful metric only once the false-alarm rate is controlled, and it is not controlled here.

5.5 External Validation on eICU-CRD [C]

External validation asks whether a frozen model ranks person-hours as well in a different health system as it does in the one it was fitted on. Answering it requires transporting the label as well as the model, and in this pair of databases the label is by far the harder of the two. This section therefore reports the labelling inputs first, then discrimination, then the operating characteristics a ward would experience, because each is only readable in the light of the one before it.

5.5.1 What the Sepsis-3 Anchor Can Reach in eICU-CRD

Sepsis-3 onset is anchored on the co-occurrence of an antibiotic order and a microbiology culture. In eICU-CRD the culture limb is documented for 1.55% of the analysis cohort (2,824 of 181,589 unit stays), against near-universal coverage in MIMIC-IV, while antibiotic exposure, pooled across the `medication` and `infusionDrug` tables, reaches 30.44% of stays. The asymmetry is decisive: in a stay with no culture record the Sepsis-3 conjunction cannot fire under *any* variant, whatever the patient's physiology.

Coverage of a single limb, however, is an upper bound on what the conjunction can reach rather than a description of it, and the two quantities differ by an order of magnitude. Requiring *both* limbs to be documented at all, at any offset and before any window is imposed, leaves 232 stays, 0.128% of the cohort. Requiring them to co-occur inside a variant's own antibiotic-culture window, which is what suspicion of infection means, leaves 186, 196 and 211 stays under Variants A, B and C. Adding the organ-dysfunction criterion leaves 27, 31 and 73 labelled stays.

One further step separates the labelled stays from the events the external AU-ROC is computed on, and it accounts for the only discrepancy between the two: under Variant B, 31 stays are labelled but 29 events are scored. An onset falling after a stay's last observed hour cannot be predicted from inside that stay's panel, so it is treated as right-censoring rather than as an event, the same horizon rule the development cohort applies at 72 hours. Two of Variant B's onsets fall that way; none of Variant A's or Variant C's do, which is why only Variant B moves between the two tables. No labelled onset hour is lost for any other reason: every one that falls inside its panel carries a complete covariate vector and receives a prediction from all three models.

The four coverage counts answer four different questions and are stated together here because a single percentage cannot carry the constraint: 1.55% is the sub-

cohort in which an onset is *observable*, and it is the denominator of the external analysis; 0.128% is the ceiling on how many stays the conjunction could ever label; and the labelled counts are what the AUROC estimates actually rest on.

This has a direct consequence for how an external event rate may be read. Applying the conjunction across all 7,875,057 person-hours would place the overwhelming majority of the denominator in stays where the numerator is structurally zero, and the resulting rate would describe the completeness of one source table rather than the incidence of sepsis. The primary external analysis is therefore confined to the 163,458 person-hours in which an onset is observable in principle.

Var	Cohort	Person-hours	Events	Events / 1,000 person-h
A	MIMIC-IV temporal test	548,854	1,015	1.849
A	eICU, microbiology-covered	162,070	27	0.167
A	eICU, antibiotic-only anchor	7,656,521	5,589	0.730
B	MIMIC-IV temporal test	548,827	1,040	1.895
B	eICU, microbiology-covered	161,992	29	0.179
B	eICU, antibiotic-only anchor	7,627,689	6,140	0.805
C	MIMIC-IV temporal test	504,828	2,128	4.215
C	eICU, microbiology-covered	159,580	73	0.458
C	eICU, antibiotic-only anchor	7,450,975	9,886	1.327

Table 5.12: Event-rate reconciliation. The rate is stated for the development cohort and for both external anchors so that the comparison is made explicitly rather than inferred. The antibiotic-only anchor is not Sepsis-3 and its rate is not interchangeable with the other two.

On the restricted cohort the two databases fall within one order of magnitude of each other: 0.179 events per thousand person-hours against MIMIC-IV’s 1.895 under the primary label, a ratio of 0.094. Relaxing the culture requirement closes most of the remaining gap (0.805 per thousand, ratio 0.425), which locates the residual difference precisely: it is the culture limb, not the antibiotic limb and not the patients. eICU-CRD documents antibiotic exposure for 30.44% of stays and cultures for 1.55%, so the conjunction is throttled by one of its two inputs.

A gap of this size is a substantive finding about how differently the two systems document infection, and it is the reason the restricted cohort is the right denominator. A gap of two or more orders of magnitude would not be a finding at all: no plausible difference in case mix separates two large adult ICU populations by that much, and such a gap would indicate that the labelling inputs, not the patients, differ.

5.5.2 Discrimination Under Transport

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary	162,070	27	0.6913	0.7713
A	GBT	162,070	27	0.7106	0.7455
A	NEWS2	162,070	27	0.5866	0.6413
B	Primary	161,992	29	0.6960	0.7590
B	GBT	161,992	29	0.6815	0.7424
B	NEWS2	161,992	29	0.5259	0.6415
C	Primary	159,580	73	0.7370	0.7695
C	GBT	159,580	73	0.6877	0.7550
C	NEWS2	159,580	73	0.5074	0.6483

Table 5.13: External validation on eICU-CRD (frozen models, no retraining), Sepsis-3 anchor on the microbiology-covered sub-cohort. Each variant carries its own external label set, so the row counts and event counts differ across variants as they do internally.

The point estimate falls under every label. The primary model falls from 0.7713 to 0.6913 (Narrow), from 0.7590 to 0.6960 (Seymour-Standard) and from 0.7695 to 0.7370 (Liberal). Only the Seymour-Standard contrast carries an interval, because only it is the tested contrast of H5, and that interval covers zero (Section 5.10.7); the Narrow and Liberal figures are point estimates on 27 and 73 events with no interval attached, so “under every label” describes the direction of three estimates and not three established degradations. The Seymour-Standard drop of 0.0630 is about three-quarters of the 0.085 that Moor et al. [21] report as typical, so the pre-registered expectation is approximately met on the point estimate. Section 6.3 sets out why it is not established at conventional confidence.

The degradation is ordered by label, and in the same direction as the internal results: the narrowest label transports worst and the most liberal transports best. That the ordering survives transport is a point in its own right: the label determines not only which cohort exists but how well what is learned from it travels.

NEWS2 degrades further and faster, to 0.5259 under the primary label: effectively chance. On this arm, then, the fitted models retain more of their advantage over a bedside score than the score retains of its own internal performance. That is the one comparison in this section that favours the fitted model, and it should not be carried further than the arm it is computed on: Section 6.3 reports the same three comparators under the antibiotic-only anchor, where the event count is two orders of magnitude larger and the ordering between the boosted model and

NEWS2 does not hold.

Two further properties of the table are diagnostic rather than substantive, and both are worth stating because their violation would be invisible otherwise. First, the event counts differ across variants. They must: the variants differ in their antibiotic–culture windows, so they select different stays, and an external analysis reporting one event count for all three would be reporting a single label applied three times. Second, NEWS2’s AUROC differs across variants. Its *predictor* is label-invariant, the score is a function of physiology alone and is computed once, but its AUROC is not, because AUROC is a property of the predictor *and* the outcome it is ranked against. An identical NEWS2 AUROC across variants is therefore a signature of an unvaried label, and the pipeline halts on it rather than reporting it.

Every external event count above falls below the interpretability floor adopted in this study. Section 6.3 states what that permits the external estimates to support, and reports the antibiotic-only sensitivity arm that bounds them.

5.5.3 External Clinical Utility on eICU-CRD [C]

Discrimination is a ranking property and survives transport in the attenuated form reported above. The quantities a ward would actually experience are reported here, on the same sub-cohort and under the same event floor, and they attenuate further. Everything in this subsection inherits the coverage limitation of Section 5.5.1: the denominator holds only person-hours in which an onset is observable at all.

Var	Model	Spec	PPV	NNE	FA/pt-day	Patient PPV
A	Primary	0.459	25.10×10^{-5}	3,984	12.98	0.00956
B	Primary	0.500	29.65×10^{-5}	3,372	11.99	0.01027
C	Primary	0.484	71.65×10^{-5}	1,395	12.38	0.02516
A–C	NEWS2	0.110	16.65×10^{-5}	6,004	21.35	0.00889

Table 5.14: eICU-CRD operating characteristics at the same 80% target sensitivity, thresholds re-derived on the external cohort (161,992 person-hours, 2,824 stays, 29 event hours).

Operating characteristics on eICU-CRD are substantially worse than internal ones at the same target sensitivity. Reaching 80 per cent sensitivity requires 1,395 to 3,984 alerted person-hours per true event hour, at 11.99–12.98 false-alert hours per patient-day, roughly one alert every two hours for every patient in the unit, indefinitely. NEWS2 cannot reach the target without alerting on nearly every person-hour (specificity 0.110, 21.35 false-alert hours per patient-day). Patient-level PPV of 0.01027 is again the cohort septic fraction (29/2,824): at this thresh-

old the model alerts on essentially every stay, so identifying a patient as alerted conveys nothing beyond the base rate.

How much of this is attributable to the model and how much to the label is not separable at this event count. PPV, number-needed-to-evaluate and false-alarm rate all scale directly with the event rate, so any residual under-recovery of onsets depresses them mechanically, independently of how well the model ranks. The internal analysis already establishes the substantive point, that a stable AUROC and a usable alert are different claims, and establishes it on a cohort where the label is fully observed (Section 5.4.2). The external operating characteristics are therefore reported as an upper bound on transferability and are not used to support any conclusion of their own.

5.6 Label-Anchor Attribution: Definitional or Empirical? [E]

Everything to this point varies the label *within* Sepsis-3. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor, so H1 cannot distinguish two very different claims: that the reading of a definition matters (definitional), or that the choice of construct matters (empirical). This section separates them by splitting the Sepsis-3 conjunction into its two limbs and labelling each on its own (Section 3.6). It is post-hoc, under Protocol Amendment 2, and its contrasts form exploratory family E3.

5.6.1 The Anchor Determines Every Onset Time

The first result is structural rather than statistical, and it can be stated exactly. Under variants A, B and C the onset time *is* the suspicion time t_{susp} ; the SOFA criterion decides only *whether* a stay is labelled, never *when*. Variant E, which removes the SOFA criterion, therefore turns out to be a strict superset of Variant B: every one of B's 5,887 labelled stays is also labelled by E, and in 100.0% of **them the onset hour is identical**. Adding the organ-dysfunction requirement discards 11,364 of E's 17,251 labelled stays, two thirds, and moves not one onset by a single hour.

Var	Label	Onsets (≤ 72 h)	%	κ vs B	Sens. to B	Exact onset match
A	Narrow	5,662	8.7	0.267	0.325	97.5%
B	Seymour-Std	5,887	9.0	n/a	n/a	n/a
C	Liberal	11,531	17.7	0.344	0.625	98.6%
E	Anchor-Only	17,251	26.5	0.432	1.000	100.0%
D	Deterioration	27,209	41.8	0.057	0.555	6.5%
F	Sepsis-3 Early	6,074	9.3	0.983	1.000	73.7%

Table 5.15: Onset agreement with Variant B across all five comparison variants, on the 65,078 shared stays. Counts are restricted to onsets inside the 72 h modelling horizon so that Variant D, whose onsets exist only within the panel, is comparable. “Sens. to B” is the fraction of B’s labelled stays that the variant also labels; “exact onset match” is computed among stays labelled by both.

Table 5.15 separates the definitional component from the empirical one, and the three blocks answer different questions.

Within Sepsis-3 (top block), when two variants agree that a stay is septic they agree on the hour almost always: 97.5% for A, 98.6% for C. The pre-registered label variance is therefore a disagreement about *which patients*, essentially never about *when*, because the “when” is fixed by the treatment anchor in all three. That disagreement is nonetheless substantial: A and B share only 0.325 of B’s septic stays ($\kappa = 0.267$) despite near-identical prevalence.

Variant D breaks the pattern completely. A label built from physiology alone agrees with Sepsis-3 at $\kappa = 0.057$, barely distinguishable from chance, and matches B’s exact onset hour in only 6.5% of the stays both label, with a median absolute shift of 9.0 hours (IQR -11 to $+7$). **This is the empirical component, and it is large.** A treatment-independent label is not a re-reading of Sepsis-3; it is a different construct, identifying different patients at different times.

Variant F isolates the third possibility: same construct, same patients, different clock. It labels every stay B labels ($\kappa = 0.983$, the shortfall being the 187 stays discussed above whose onset only enters the horizon under the earlier rule) yet agrees with B on the exact hour in just 73.7% of them. **A quarter of Sepsis-3 onsets are placed at a different hour by two defensible readings of the same definition, on the same patients.** Neither the pre-registered variants nor Variant D could expose that, because the first hold timing fixed and the second changes everything at once.

5.6.2 Discrimination Under Each Limb

Model	AUROC				AUPRC			
	B	E	D	F	B	E	D	F
Primary	0.7590	0.7697	0.8493	0.7951	0.0096	0.0321	0.0767	0.0131
GBT	0.7424	0.7665	0.9147	0.7458	0.0087	0.0334	0.1362	0.0090
NEWS2	0.6415	0.6650	0.6810	0.6409	0.0032	0.0129	0.0243	0.0034
Cox	0.6444	0.6854	0.7868	0.6528	0.0033	0.0147	0.0382	0.0035
qSOFA	0.6091	0.6243	0.7308	0.6167	0.0028	0.0105	0.0312	0.0031
SIRS	0.6114	0.6407	0.5549	0.5983	0.0030	0.0127	0.0146	0.0029
Events	1,040	3,073	4,774	1,082				

Table 5.16: Discrimination under each limb of the Sepsis-3 conjunction and under standard onset timing (temporal test set, unrestricted window). B = Sepsis-3 (anchor **and** organ dysfunction); E = suspicion anchor only; D = organ dysfunction only, treatment-independent; F = Sepsis-3 with onset at $\min(t_{\text{susp}}, t_{\text{SOFA}})$. AUPRC is reported alongside AUROC because the four labels have very different prevalences.

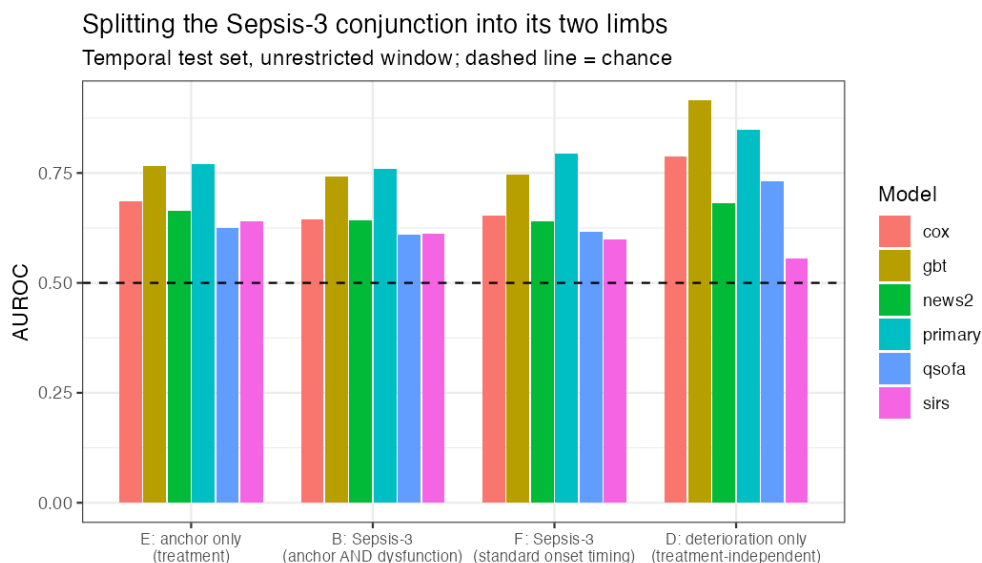


Figure 5.6: Splitting the Sepsis-3 conjunction. The two limbs and the intact conjunction sit close together; only the treatment-independent deterioration label separates. The dashed line is chance.

The anchor-attributable fraction is indistinguishable from one. Equation 3.1 asks what share of Sepsis-3’s above-chance discrimination the treatment

anchor reproduces on its own. The answer, with the ratio bootstrapped end to end, is

$$\phi_{\text{anchor}} = 1.041 \text{ [0.990, 1.097]} \quad (\text{primary}), \quad 1.099 \text{ [1.037, 1.164]} \quad (\text{GBT}).$$

For the primary model the interval **includes 1**, and the underlying contrast is not significant: E against B is $\Delta = 0.0107$ $[-0.0029, 0.0238]$, $p_{\text{BH}} = 0.139$ in family E3. For GBT the interval sits just above 1 ($\Delta = 0.0241$, $p_{\text{BH}} = 0.003$), a small but detectable effect.

The reading is therefore **negative and worth stating as such**: on this cohort the treatment anchor alone is about as predictable as Sepsis-3 itself, neither more nor less. Dropping the organ-dysfunction requirement nearly triples the event count (3,073 against 1,040) and leaves discrimination essentially where it was. The SOFA gate selects which stays are labelled without selecting the separable ones, so “how much of the model’s performance is attributable to the treatment anchor?” has the answer “essentially all of it, but the anchor is not itself the easy part”.

What does separate is the treatment-independent label. Variant D is the one contrast that moves: $\Delta = 0.0904$ $[0.0733, 0.1075]$ for the primary model and 0.1723 for GBT, both significant in family E3. Its AUPRC gap is larger still (0.0767 against 0.0096), which matters more at this prevalence. The caveat of Section 3.6 governs the *level*: D’s label is a threshold function of physiology of which five of six components are model inputs, so part of that separation is self-fulfilling.

And so does moving the clock. Variant F, whose label is Sepsis-3 unmodified in membership, is also significantly more separable than B ($\Delta = 0.0361$ $[0.0257, 0.0461]$, $p_{\text{BH}} = 0.002$). Its AUPRC is *higher* than B’s despite a comparable event count, so this is not a prevalence artefact. Onsets placed at the moment organ dysfunction is first detectable are easier to anticipate than the same onsets placed at the moment a clinician acted, which is what one would expect if the physiological signal leads the clinical response, and is consistent with the pre-treatment result of Section 5.6.3.

Where this leaves the difficulty. Sepsis-3 is not harder to predict than its limbs; it is about as hard as its anchor limb, easier than its physiological limb, and harder than itself under a better clock. The difficulty of sepsis onset prediction on MIMIC-IV therefore does not live in the conjunction as such. It lives in *when* the conjunction is stamped, at the clinician’s action rather than at the patient’s deterioration, and that is a choice the definition leaves open and this literature has largely closed one way.

5.6.3 The Empty Pre-Treatment Window Is a Property of the Onset Rule

Section 5.10.5 proved that the pre-treatment window must be empty under the onset rule variants A–C use. This section reports what is in that window once the rule is changed, and separates three distinct rows.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
E	Anchor-Only	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration	147,299	1,362	0.8498 / 0.9360

Table 5.17: Events in the treatment-anchored (pre-treatment) evaluation window. E’s and B’s zeros are entailed by Proposition 5.1. Variant F holds the Sepsis-3 conjunction and the labelled stays fixed and changes only the onset timestamp to the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$; Variant D removes the infection anchor altogether.

Two of the four rows carry no information. Variant E’s zero is a tautology statable without any data: E’s onset *is* the anchor, so the pre-anchor window is empty by construction. Variant B’s zero is the same tautology one step removed: B is a subset of E with identical onset times, so if E cannot fire before the anchor, neither can B.

Variant F: the same label, timed as the standard specifies. F is the row that answers the question directly, because it changes one thing. The suspicion windows, the SOFA threshold and the evaluation window are Variant B’s; the septic stays are Variant B’s, stay for stay: all 6,447 of them. Only the timestamp differs. On that single change, 1,738 onsets move earlier (27.0%; median -14.0 h, largest -48.0 h), 419 land before the first antibiotic or culture across the cohort, and 71 of those fall in the temporal test set. **Sepsis-3 onsets can and do precede clinical action.** The primary model identifies those hours at 0.8081 AUROC.

This is the number the pre-registered analysis could not produce, and it is independent of the Variant D caveat below: F’s label is Sepsis-3, unmodified in membership, so nothing about it is a different construct.

Onset timing also changes how many events exist at all. Agreement between F and B is 99.7% ($\kappa = 0.983$) rather than exact, and the discrepancy is one-directional and informative. Within the 72-hour prediction horizon F labels 6,074 stays against B’s 5,887, and every one of B’s is among them: $F \supseteq B$. The 187 additional stays are those whose t_{susp} falls after hour 72, so no model here ever

sees a labelled event for them, but whose SOFA rise occurred inside the window. Standard timing makes them modellable; conjunction timing discards them. The onset rule therefore governs not only *when* events are recorded but *how many* a real-time model can be trained and tested on at all, which is a second and quite separate route by which an operationalisation choice propagates into reported performance.

Variant D: a different construct, and a larger count. Variant D's 1,362 pre-treatment onsets are consistent with F and larger, as expected of a label that drops the infection requirement entirely. A model identifies those hours at 0.8498 AUROC (0.9360 for GBT). The caveat of Section 3.6 applies to that *level*, D's label is a threshold function of physiology of which most components are model inputs, but not to the event count.

The form contribution C2 takes. C2 is deliberately *not* the proposition that no patient develops sepsis before clinicians act. That proposition is unsupportable in general and the external literature contradicts it: the Epic Sepsis Model v2 evaluation [5] reports median lead times of 1.4–7.1 hours ahead of clinician recognition across four sites at encounter AUROC 0.80–0.90. C2 is instead a two-part claim, one part analytic and one measured, and the two parts are load-bearing in different ways:

1. **Analytic.** Under a conjunction-based onset rule (the rule used by the pre-registered variants here and, in the author's reading, by a substantial share of the MIMIC-IV sepsis literature) pre-treatment prediction is structurally impossible. Proposition 5.1 is the proof. Any study reporting pre-treatment performance on such a label is reporting an empty set, and any study *not* restricting the window is, to that extent, scoring anticipation of a decision inside a window that decision opened.
2. **Measured.** Restoring the standard timing branch recovers 71 pre-treatment onsets on the identical cohort at 0.8081 AUROC. The prediction problem is real and is measurable; it is the operationalisation, not the data, that conceals it.

Sepsis-3 *membership* remains constituted by clinical action under every timing rule, without a qualifying antibiotic-culture pair there is no label at all, so Kamran et al.'s [7] concern survives at the level of which patients can ever enter the positive class. The remedy Chapter 7 identifies is unchanged in direction and weaker in force: change the label, and at minimum time it as the standard requires.

Variant D’s discrimination is not comparable with Variant B’s, for reasons fixed in advance by its construction; what the comparison does and does not license is set out in Section 6.4.

5.6.4 Exploratory Family E3: Intervals and Correction

Var	Model	AUROC	95% CI	Δ vs B	95% CI (Δ), q_{BH}
E	Primary	0.7697	[0.761, 0.779]	0.0107	[−0.0029, 0.0238], $q = 0.139$
E	GBT	0.7665	[0.757, 0.775]	0.0241	[0.0096, 0.0375], $q = 0.003$
D	Primary	0.8493	[0.844, 0.855]	0.0904	[0.0733, 0.1075], $q = 0.002$
D	GBT	0.9147	[0.911, 0.918]	0.1723	[0.1545, 0.1885], $q = 0.002$
F	Primary	0.7951	[0.780, 0.810]	0.0361	[0.0257, 0.0461], $q = 0.002$
F	GBT	0.7458	[0.730, 0.762]	0.0033	[−0.0087, 0.0148], $q = 0.559$
B	Primary	0.7590	[0.7434, 0.7749]	n/a	n/a
B	GBT	0.7424	[0.7265, 0.7583]	n/a	n/a

Table 5.18: Exploratory family E3: AUROC under each limb of the Sepsis-3 conjunction with 95% stay-level cluster-bootstrap intervals, and the paired difference against Variant B corrected by Benjamini–Hochberg across the family of $m = 6$ contrasts. All variants are labelled on the same test stays, so each difference is evaluated within replicate.

Every contrast is positive, and two of the six do not survive correction. The first is the one that matters most for the anchor-attribution argument: **E against B for the primary model is not significant** ($\Delta = 0.0107$, 95% CI -0.0029 – 0.0238 , $q = 0.139$), so on the interpretable model the treatment anchor alone and the full Sepsis-3 conjunction are statistically indistinguishable. The corresponding GBT contrast is significant but small (0.0241).

The second is F against B for GBT ($\Delta = 0.0033$, 95% CI -0.0087 – 0.0148 , $q = 0.559$), and it is a difference between the two models rather than a difference between the two labels. Re-timing the onset separates the labels for the interpretable model (0.0361 , significant) and not for the boosted one. The onset-timing argument of Section 5.6.3 rests on the event counts F recovers in the pre-treatment window rather than on this contrast, so it is unaffected; what the split does show is that a claim about how much easier the re-timed label is to predict is model-specific and should not be stated in general.

The anchor-attributable fraction reflects this directly: $\phi_{\text{anchor}} = 1.041$ (95% CI 0.990 – 1.097) for the primary model, an interval that **includes 1**, and 1.099 (95% CI 1.037 – 1.164) for GBT, which does not. The defensible reading is the weaker one: removing the organ-dysfunction requirement nearly triples the number of

labelled events and leaves discrimination where it was. The SOFA gate selects *which* stays are labelled without selecting the separable ones.

The two contrasts that do move substantially are D, a different construct entirely, and F for the interpretable model, which is Sepsis-3 with a different clock. The latter is the more surprising of the two, because F’s membership is B’s exactly.

One caution on reading these intervals. They quantify sampling uncertainty only. They say nothing about the construct differences between the labels: prevalence differs by a factor of 3.0 (E) and 4.6 (D) relative to B, and Variant D’s label overlaps the model’s own inputs. The intervals establish which orderings are stable, not that the labels are measuring comparable things.

5.7 Action-Derived Features: An Ablation [E]

The label is not the only place where clinician action enters this study. 8 of the 22 covariates record what clinicians did rather than what was measured in the patient: vasopressor exposure, and the per-test indicators of whether a laboratory value was obtained in a given hour. Section 5.10.4 showed that cross-label coefficient instability is concentrated in exactly these features. This section asks the sharper question that observation raises, whether the models *depend* on them, by removing them and refitting.

The specification is otherwise unchanged: same training rows, same temporal split, same hyperparameters, same metric code, and every physiological measurement retained (Section 3.14.5). The arm is exploratory and enters no hypothesis or decomposition.

Var	Model	AUROC full	AUROC ablated	Δ	Utility _{max} full	Utility _{max} ablated
A	Primary	0.7713	0.7633	−0.0080	0.026	0.013
A	GBT	0.7455	0.7418	−0.0037	0.038	0.037
B	Primary	0.7590	0.7514	−0.0076	0.025	0.014
B	GBT	0.7424	0.7359	−0.0065	0.031	0.032
C	Primary	0.7695	0.7602	−0.0093	0.095	0.092
C	GBT	0.7550	0.7516	−0.0034	0.111	0.095

Table 5.19: Discrimination and best-threshold Utility Score with and without the 8 action-derived features. Both arms are fitted on the same training rows and scored on the same test rows. **[E] exploratory.**

5.7.1 Coefficient Stability After Ablation

The second reading is the one that bears on Section 5.10.4. If the cross-label instability were a property of the fitted model in general, removing eight covariates

would leave the remainder roughly as unstable as before. If it were specific to the action-derived features, the remaining specification should be markedly more stable.

Specification	Covariates	Unstable	% unstable
Full	22	6	n/a
Ablated	14	1	7.1

Table 5.20: Cross-label coefficient instability before and after removing the action-derived features, under the identical stability rule (sign reversal or magnitude change $\geq 2\times$ across Variants A, B and C). **[E] exploratory.**

The second alternative is what the data show, and on both readings at once. Discrimination barely moves: removing 8 of the 22 covariates costs -0.0076 AU-ROC for the primary model and -0.0065 for GBT under the primary label, and no more than -0.0093 for either model under any variant. Stability moves a great deal: the unstable count falls from 6 of 22 to 1 of 14, or 7.1% of the remaining specification. The reduction is not a subset relation, and the difference is worth stating: the single flag that remains in the ablated arm falls on `lactate`, a covariate the full fit does *not* flag. Removing the action-derived features does not simply delete unstable coefficients, it re-estimates the ones that remain, which is what a refit does, and why the ablated count is a property of the ablated specification rather than a subset of the full one.

The two readings agree with the by-kind split of Section 5.10.4, where the action-derived features are 50.0% unstable against 14.3% for the physiological ones. The features that record clinician behaviour contribute almost nothing to discrimination and carry most of the cross-label instability: on this cohort they are close to free to drop, and dropping them buys a markedly more stable model. Section 6.5 states what that does and does not license.

5.8 Model Diagnostics

Two checks on the comparator models are collected here rather than reported beside the results they qualify, because neither feeds a hypothesis and neither is read on its own. The first tests the assumption the static Cox comparator rests on, and so establishes whether that comparator is a fair one; the second describes which predictors the gradient-boosted trees actually used, and so bears on the ablation in Section 5.7 and on the coefficient instability in Section 5.10.4.

5.8.1 Proportional-Hazards Testing [C]

The Schoenfeld residual test on the static Cox model (Variant B) decisively rejects proportional hazards ($\chi^2 = 126.2$, $\text{df} = 12$, $p = 3.6 \times 10^{-21}$). 8 of the 12 covariates violate the assumption individually at $\alpha = 0.05$, led by GCS, bilirubin, respiratory rate and creatinine. These are per-covariate tests reported without multiplicity adjustment; the global test alone is sufficient for the pre-registered check, and the smallest p-values survive Bonferroni correction over the covariate set by many orders of magnitude. The assumption the static Cox comparator rests on therefore fails on these data, so its coefficients cannot be read as time-constant hazard ratios, and the discrete-time formulation is the better-founded choice for the primary model.

5.8.2 GBT Feature Importance [E]

Feature	Gain	Cover	Frequency
Temperature (°C)	0.145	0.144	0.083
Resp. rate slope (6 h)	0.138	0.085	0.075
MAP slope (6 h)	0.087	0.098	0.068
GCS	0.072	0.069	0.055
Heart-rate slope (6 h)	0.083	0.069	0.084
Creatinine	0.058	0.076	0.065
WBC	0.065	0.086	0.078
Heart rate (level)	0.054	0.061	0.085
Lactate	0.044	0.047	0.054
MAP (level)	0.050	0.038	0.084

Table 5.21: Top-10 GBT features by information gain (Variant B). Three of the five most informative predictors are six-hour trend features, and heart rate contributes through both its level and its slope. The direction of change over the past six hours carries as much discriminative signal as the current value.

The prominence of six-hour slope features is consistent with Lauritsen et al.’s [10] concern: the GBT model draws a large share of its signal from *trends* rather than *levels*, meaning its performance depends on the length of the history window used for feature construction: a construction choice that is rarely reported and never varied in the studies this thesis compares against. Heart rate appears in both forms, level and slope, which is what one would expect of the most frequently recorded vital sign in the ICU.

5.9 Equity Analysis [E]

5.9.1 What the Subgroup Strata Are

Two properties of the stratifying variables govern how everything in this section may be read, and both are properties of MIMIC-IV’s administrative record rather than of the analysis.

The strata are documentation categories, not populations. `race` in MIMIC-IV is a single code assigned at registration. Its levels mix broad categories with more granular ones drawn from the same population: “White” and “White – Other European” are separate levels, as are “Asian” and “Asian – Chinese”. A stay carries exactly one code, so the levels partition the cohort, but they do not partition it into disjoint *populations*: the reference level “White” is a residual category comprising stays not assigned a more specific European code, and a contrast against it is a contrast against that residual, not against White patients in general. “Unknown” and “Unable to obtain” record the *absence* of an answer. They are retained because the completeness of the demographic record is itself informative, Section 5.9.3 shows why, but they are not demographic groups and no result concerning them is a result about a population.

An event floor is applied before anything is interpreted. A subgroup AUROC estimated on a few dozen onsets is not a quantity about which anything can be said. The floor adopted here is 100 sepsis onsets, the same threshold and the same warrant as the external-validation floor of Section 6.3 [12], [46]. Strata below it are estimated and tabulated, so that the reader can see which groups exist in the cohort and how few events they carry, but they are **not interpreted**, are excluded from every statement about spread, and are excluded from the exploratory contrast family E1, which therefore contains 6 contrasts, not one per level. Of the racial strata, 2 clear the floor.

5.9.2 Performance Disparities by Subgroup [E]

Racial stratum	Rows	Sepsis	GBT AUROC (95% CI)	Primary AUROC (95% CI)
White	317,688	597	0.739 (0.719–0.761)	0.756 (0.736–0.777)
Unknown	107,158	235	0.773 (0.742–0.803)	0.773 (0.740–0.804)
Black / African American [†]	31,716	58	0.749 (0.676–0.817)	0.759 (0.685–0.825)
Other [†]	22,825	35	0.669 (0.571–0.756)	0.724 (0.631–0.806)
White – Other European [†]	11,768	18	0.672 (0.566–0.780)	0.730 (0.600–0.846)
Asian [†]	6,543	16	0.765 (0.625–0.886)	0.841 (0.726–0.929)
Hispanic / Latino – Puerto Rican [†]	6,462	16	0.624 (0.448–0.792)	0.702 (0.546–0.845)
Asian – Chinese [†]	5,386	13	0.601 (0.398–0.802)	0.701 (0.511–0.869)

Table 5.22: AUROC by racial stratum (Variant B, temporal test) with 95% stay-level cluster-bootstrap intervals, stratified within stratum, $B = 2,000$. Levels are ordered by exposure; interval width grows by roughly an order of magnitude from the largest stratum to the smallest. Strata carrying fewer than 100 sepsis onsets are marked [†]: they are shown so that the composition of the cohort is visible, and are neither interpreted nor entered into family E1 (Section 5.9.1). [E] exploratory.

Read as a list of point estimates the table appears to show a wide spread, and it is tempting to set that spread beside the model-class and label spreads of Section 5.10. It should not be. The three quantities are not commensurable, for three reasons that are independent of each other. They are different estimands: the decomposition rows are contrasts computed on the whole test cohort with one factor varied and everything else held fixed, whereas a subgroup range is the extreme order statistic of a set of separate estimates computed on disjoint slices of that cohort. They rest on different amounts of information: the decomposition rows use every event in the test set, while the subgroup estimates divide those events between strata, the smallest of which carry a few dozen. And the range of a set of noisy estimates is biased upward by construction, it grows with the number of strata and with their imprecision even when every underlying value is identical, so it is not an effect size at all.

What the intervals show is that the apparent spread is largely a function of how few events the small strata contain. Of the racial strata, 2 carry at least 100 onsets; the remaining 6 are estimated on counts between 13 and 58, and their intervals are correspondingly wide.

Table 5.23 therefore reports contrasts rather than ranges, each against the reference level on the same bootstrap replicates.

Variable	Stratum	Sepsis	Model	Δ AUROC vs. reference (95% CI)	p	q_{BH}
Insurance	Private	251	Gbt	+0.050 (+0.012+0.088)	0.011	0.066
Insurance	Private	251	Primary	+0.039 (+0.003+0.077)	0.033	0.099
Race	Unknown	235	Gbt	+0.033 (−0.003+0.069)	0.067	0.134
Race	Unknown	235	Primary	+0.017 (−0.022+0.055)	0.387	0.580
Insurance	Medicaid	183	Primary	+0.008 (−0.036+0.052)	0.683	0.819
Insurance	Medicaid	183	Gbt	−0.003 (−0.049+0.046)	0.888	0.888

Table 5.23: Exploratory family E1 in full: all 6 subgroup AUROC contrasts clearing the 100-event floor, across race, language and insurance and for both fitted models, ordered by adjusted p -value. q_{BH} is the Benjamini–Hochberg adjusted value over the whole family. **No contrast is significant**, on race or on any other variable. Two intervals nonetheless exclude zero, which is the distinction the surrounding text turns on: an unadjusted interval and a family-corrected verdict answer different questions, and only the second is read here. The reference level is the largest stratum of its own variable and, for race, is a residual administrative category rather than a population (Section 5.9.1). **[E] exploratory**.

Applying the floor leaves 6 contrasts in family E1 across all three equity variables and both models, and **none survives false-discovery correction**. The smallest adjusted value in the family belongs to a contrast on insurance rather than on race: Private insurance against the Medicare reference, 0.050 AUROC for the gradient-boosted model, with a bootstrap interval of [0.012, 0.088] and $q_{BH} = 0.066$.

That row is worth pausing on, because it is where an unadjusted interval and a corrected verdict part company. The interval excludes zero; the adjusted p -value does not clear the threshold. Both statements are correct, and the second is the one this study reads, because the contrast was not pre-specified: §9 of the protocol fixed the scope of the equity analysis but neither the levels, nor the reference, nor the direction of any contrast, so an uncorrected interval selected after the fact from six is not evidence of the strength it appears to have. The contrast is reported rather than suppressed, and it is not claimed. Reading it as a finding would be exactly the practice this thesis objects to elsewhere.

One consequence deserves stating plainly, because it bounds the whole analysis. Of the racial strata, only two carry enough events to be interpreted, and one of them is *Unknown*: a missingness category rather than a population (Section 5.9.1). **This study therefore contains no sufficiently powered comparison between two racial strata that both record an actual demographic answer**. That is not a null result about racial disparity in sepsis prediction; it is the absence of the measurement that would be needed to have one.

The correct reading of Table 5.23 is accordingly that this study is **underpowered to detect subgroup disparities**, not that no disparities exist. Even among

the strata clearing the event floor the intervals are wide enough to be compatible with a disparity larger than any effect reported elsewhere in this thesis, so the data are consistent both with a stratum being served materially worse than the reference and with its being served slightly better. Non-significance here is a statement about precision and nothing else.

Wang et al.’s [11] finding of degraded performance for Asian, Hispanic and Spanish-speaking patients is therefore neither confirmed nor contradicted. It could not be: that result concerns mortality prediction among patients already identified as septic, not onset prediction (Section 2.9), and the strata in which it would have to be checked here are precisely the ones that fall below the event floor. The point estimates for the smaller strata do sit below the reference under GBT, in the direction Wang et al. report, but they are not interpreted and no inference is drawn from them.

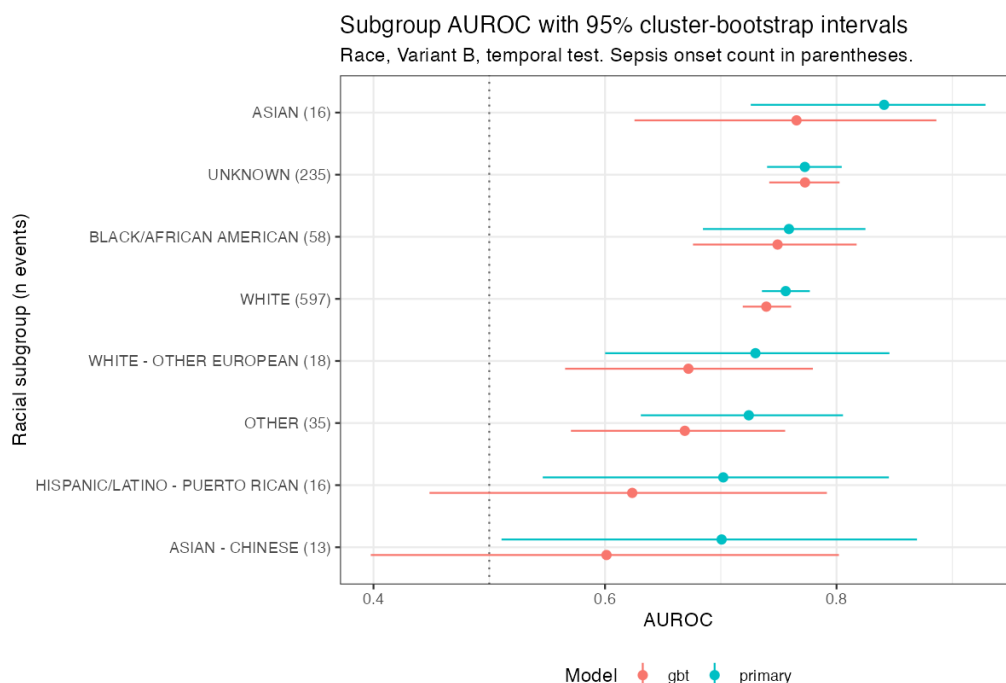


Figure 5.7: AUROC by racial subgroup with 95% stay-level cluster-bootstrap intervals (Variant B, temporal test); sepsis onset count in parentheses. Interval width is governed almost entirely by the stratum event count. The two strata with the most extreme point estimates are also the two least precisely estimated, and several intervals reach the 0.5 chance line. The figure is the argument for the event floor of Section 5.9.1, and is shown in full for that reason.

5.9.3 Label Sensitivity by Subgroup [E]

Racial group	Stays	A (%)	B (%)	C (%)	Range (pp, 95% CI)
White	40,212	9.4	9.6	18.6	9.3 (8.9–9.7)
Unknown	7,264	11.6	12.8	23.9	12.3 (11.3–13.3)
Black / African American	4,929	9.3	9.1	19.0	9.9 (9.0–11.1)
Other	2,254	8.8	9.8	17.5	8.7 (7.1–10.2)
Unable to obtain	1,629	9.7	11.1	22.5	12.8 (10.7–14.9)
White – Other European	1,589	9.4	7.7	16.6	8.9 (7.2–10.7)
Asian	799	10.5	11.5	21.3	10.8 (8.3–13.8)
Asian – Chinese	721	10.1	11.1	19.0	8.9 (6.4–11.9)

Table 5.24: Percentage of stays labelled as septic by racial subgroup and label variant, with a 95% stratified cluster-bootstrap interval on the A–C range. The fraction roughly doubles from Variant A to C across all groups. Per-variant intervals are in `12_subgroup_labelsens_ci.csv`. **[E] exploratory.**

The fraction of patients labelled as septic changes by 8.7–12.8 percentage points across label variants within each racial stratum. Unlike the discrimination analysis, this contrast *does* survive correction, and the reason is arithmetic rather than substantive: it is a stay-level proportion computed on the full 65,078-stay cohort, whereas discrimination is computed on the few hundred onsets falling in the test window. The same strata that cannot support an AUROC support a proportion comfortably. No event floor is therefore applied to this analysis; the construct limitation of Section 5.9.1 applies to it in full.

Racial group	Stays	Δ range vs. White (pp, 95% CI)	p	q_{BH}
Unknown	7,264	+3.01 (+1.93–+4.06)	0.001	0.006
Unable to obtain	1,629	+3.49 (+1.38–+5.55)	0.002	0.009
Black / African American	4,929	+0.62 (–0.35–+1.87)	0.186	0.347
Asian	799	+1.48 (–1.04–+4.50)	0.249	0.407
Other	2,254	–0.63 (–2.26–+0.98)	0.425	0.510
White – Other European	1,589	–0.41 (–2.10–+1.41)	0.635	0.714
Asian – Chinese	721	–0.40 (–3.02–+2.61)	0.675	0.714

Table 5.25: Difference in label sensitivity (A–C range) against the reference group (White). q_{BH} is adjusted over exploratory family E2, all 18 label-sensitivity contrasts across race, language and insurance. Two racial contrasts survive at $q < 0.05$; across the full family, 4 do (adding Spanish-speaking patients, +2.61 pp, $q = 0.006$, and privately insured patients, –1.76 pp, $q = 0.006$). **[E] exploratory.**

2 racial strata are significantly more label-sensitive than the reference after false-discovery correction, and *both of them are missingness categories*: Unknown (+3.01 pp, $q = 0.006$) and Unable to obtain (+3.49 pp, $q = 0.009$). No stratum

recording an actual demographic answer differs from the reference. Across the full family, Spanish-speaking patients are more label-sensitive than English-speaking patients (+2.61 pp, $q = 0.006$), and privately insured patients are *less* label-sensitive than Medicare patients (-1.76 pp, $q = 0.006$).

That composition dictates the reading, and it is narrower than a racial-disparity reading would be. What varies with the label definition is concentrated in the stays whose demographic record is incomplete, together with two strata, non-English language and insurance type, that index access to the care pathway rather than the patient’s physiology. The coherent interpretation is about **documentation**, not about race: a stay whose registration record is incomplete is a stay whose clinical record is also more likely to be incomplete, and because the Sepsis-3 label is constituted by two documented clinician actions (Section 5.10), incomplete documentation propagates directly into unstable labelling. Loosening the definition recovers proportionally more of those stays than of well-documented ones.

This is a finding about the label’s dependence on recording practice, and it is worth stating in that form because it is actionable in a way a racial-disparity finding computed on these strata would not be: it identifies the mechanism, documentation completeness, rather than the demographic correlate. Whether that mechanism is itself differentially distributed across patient populations is a question these strata cannot answer, for the reasons given in Section 5.9.1, and this thesis does not answer it.

Of the two equity analyses, this is the one with the precision to support any claim at all. The discrimination analysis does not, and the pairing is the point: had only the discrimination analysis been reported, its wide intervals would have been read as an absence of disparity; had only the label analysis been reported, its significant contrasts would have been read as a demographic disparity. Reported together, each bounds how the other may be read.

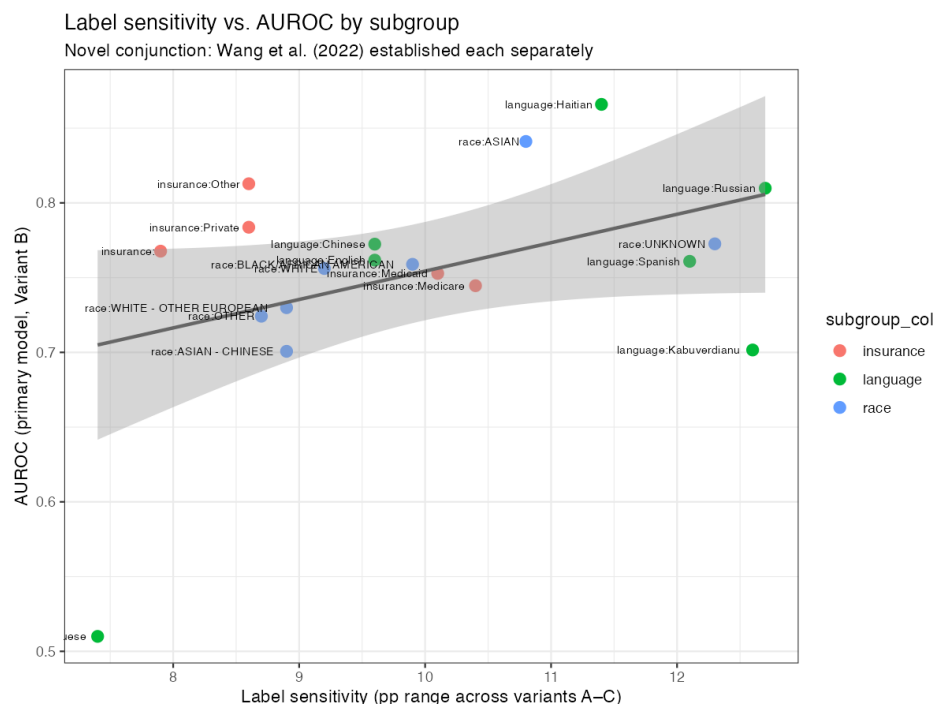


Figure 5.8: Label sensitivity (fraction labelled septic) vs. AUROC by racial subgroup, across all three label variants. Horizontal spread = label sensitivity; vertical spread = performance variation. The two effects appear independent, but no association is tested: the vertical axis carries the uncertainty shown in Figure 5.7, which this plot does not display. **[E] exploratory, visual only.**

To the best of our knowledge, bounded by the narrative search documented in Section 2.1, subgroup model-performance disparity and subgroup label sensitivity have not previously been reported together, on the same strata and in the same cohort, for a sepsis *onset* prediction model. Neither analysis is novel on its own: subgroup performance disparity is reported for sepsis mortality prediction by Wang et al. [11], who also report that competing sepsis criteria select demographically different sub-populations. What the search did not identify is the pairing, and the pairing is what makes the two results separable here: only one of them is precise enough to act on, and that is not knowable from either in isolation.

5.10 Variance Decomposition and Hypothesis Verdicts [C]

This section adjudicates the research question and the six pre-registered hypotheses. It introduces no new measurement: every estimate below is assembled from

quantities the preceding sections establish, which is why it comes last. The label spread is read across the three variants of Section 5.3, the model-class spread across the model rows of the same table, the split contrast against the random-split arm, and the transport contrast against the external estimates of Section 5.5. Assembling them in one place is what makes the four analyst decisions commensurable; each is a contrast in AUROC on the same scale, and each carries an interval from the same stay-level cluster bootstrap.

What a spread is here, and what the choice costs. Each spread is a *range*: the largest AUROC in a set minus the smallest. The label spread is therefore a range over three variants and the model-class spread a range over six model classes, and that asymmetry has to be declared, because Section 5.9 rejects the subgroup range as an effect size on exactly this ground. A range grows with the number of values it is taken over even when the underlying quantities are identical, so a range over six is expected to exceed a range over three by roughly half again. The bootstrap does not remove this: the bias is present in every replicate.

The ordering nonetheless does not rest on the statistic. The model-class range spans a structural gap rather than sampling scatter, between the two fitted models and the four rule-based and static comparators, and *any* single contrast across that gap is larger than the whole label range. The narrowest of them, GBT against the Cox model, is 0.7424 against 0.6444, which exceeds the label range by a factor of eight and involves no order statistic at all. An asymmetry worth about half again cannot account for a difference of 0.1499 against 0.0123.

A variance-components partition would be the more usual instrument, and it is not used here for two reasons. The AUROCs being compared are computed on the same test rows rather than on independent samples, so the design does not supply the error term such a model needs. And §7 of the pre-registration fixed the estimand as the *spread* before any modelling; replacing it now with a variance component would substitute a post-hoc quantity for a pre-registered one, which is the practice this thesis objects to elsewhere. The range is reported as what it is, a descriptive summary carrying a bootstrap interval, rather than as a partition of variance.

Two of the six hypotheses turn out not to admit a test statistic at all, for reasons that are properties of their design rather than of the data. Those are stated where each is reported, and the consequence for the correction (that both still consume a slot in the family, which makes the adjustment on the remaining four stricter) is set out in Section 5.10.2.

5.10.1 The Decomposition Table

Factor (others fixed)	ΔAUROC	95% CI	H
Model class (label B, temporal)	0.1499	0.134–0.171	n/a
Label definition (primary, temporal)	0.0123	0.003–0.032	H1
Split method (primary, label B)	−0.0165	−0.0372–0.0040	H4
External transport (primary, label B)	0.0630	−0.0215–0.1563	H5
Primary – GBT (label B, temporal)	−0.0165	−0.0308–0.0012	H6

Table 5.26: Variance decomposition with 95% stay-level cluster-bootstrap intervals. Model class produces the largest effect and the split method none. External transport is second by point estimate but carries by far the widest interval, because it is the only row not measured on the full test cohort (Section 6.3). Most studies fix three of these factors and compare only models. The first row tests no hypothesis: it is the range over six model classes, and H1 contrasts it against the label range. H6 is a different quantity on the same axis, the contrast between the two *flexible* estimators alone, and is shown below the rule for that reason. **[C] confirmatory.**

The decomposition does not distribute variance evenly across the four analyst decisions, and the imbalance is the result, though not the imbalance the study anticipated. **Model class dominates** (0.1499). **External transport is second** (0.0630), five times the label definition’s contribution. The label itself moves discrimination by 0.0123 and the choice of split by −0.0165, both small, and the split’s interval covers zero comfortably.

The ranking matters more than the individual magnitudes. The two decisions this literature scrutinises most closely, how the label is read and how the data are split, are the two that move discrimination least. *Where the model is deployed* moves it several times more than *how sepsis is defined*, and that ordering is the opposite of the one the study was designed around.

The study was designed on the expectation that these four would be comparable, and that expectation is what fails. Three qualifications keep the result from being read too broadly:

- The model-class spread compares six estimators ranging from a six-item bedside score to a 29-term hazard model, so it is a heterogeneous factor

by construction. Between the two *flexible* estimators the gap is 0.0165, an order of magnitude below the six-class range and of the same order as the label's (H6, non-inferiority established in the direction that matters).

- The label factor is small *in discrimination*, which is not the same as small in consequence. The three variants disagree about which patients are septic at $\kappa = 0.267$ (A vs B), so they define different cohorts, different prevalences and different denominators for every clinical claim downstream. Discrimination is simply insensitive to that.
- The split result is a null with power behind it: both arms are measured on the full MIMIC-IV test cohort, and it contradicts a widely held expectation (Section 5.10.6). The transport row is the opposite case: a substantial point estimate with weak support. It rests on 29 events, because the Sepsis-3 anchor is recoverable in only 1.55% of eICU-CRD stays (Section 5.5.1), so its interval $[-0.0215, 0.1563]$ is wide enough to contain both zero and a degradation twice the size of the label effect. It ranks second on the point estimate; it is not established at the confidence the other three rows carry.

What survives is therefore a sharper claim than the study's premise, pointing in a different direction. Analyst decisions vary widely in how much they move discrimination, and the two the literature scrutinises most, label reading and split design, move it least. The decision that appears to matter is the one this framing treats as a validation step rather than a choice: which population the model is asked to serve. That is a decomposition result, and it could not have been obtained by fixing three factors and varying one.

5.10.2 Corrected Confirmatory Verdicts

H	Contrast θ	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread – model spread	−0.1375	−0.1623–−0.1083	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random – temporal AUROC	−0.0165	−0.0372–0.0040	1.000	NOT confirmed
H5	Internal – external AUROC	0.0630	−0.0215–0.1563	0.551	NOT confirmed
H6	GBT – primary AUROC	−0.0165	−0.0308–−0.0012	1.000	H6 not rejected

Table 5.27: The confirmatory family. One-sided cluster-bootstrap tests against each hypothesis’s pre-registered threshold ($\theta_0 = 0$ for H1 and H5; 0.02 for H4 and H6), Holm–Bonferroni corrected at family size $m = 6$ and FWER 0.05. H2 and H3 reserve a slot without contributing a p-value, which makes the correction on the other four stricter. H3’s count is entailed by the onset rule rather than estimated (Proposition 5.1); its empirical counterpart is Variant F, reported in Section 5.10.5. The smallest reportable p-value is $1/(B + 1)$. **[C] confirmatory.**

None of the four testable hypotheses is confirmed, and for two of them multiplicity control is not what does the work. H1 and H4 return adjusted p-values of 1.000 and 1.000: each would have failed at any correction, and each fails on its uncorrected value too. What overturns them is quantifying uncertainty at all, and the estimate landing on the opposite side of zero from the one predicted.

H6 returns 1.000 as well, and it is not a failure of the same kind. H6 is framed as a null, so a large adjusted value is the outcome that leaves that null standing rather than a signal the study looked for and missed. What it withholds is the two-sided equivalence the pre-registration stated; what it supports is the one-sided non-inferiority of Section 5.10.8.

H5 is the exception and is worth separating. Its estimate does fall on the predicted side of zero (0.0630, in the direction of external degradation), and its uncorrected p-value is the only one in the family below 0.10. It nonetheless fails, on an interval that still covers zero and on an adjusted p-value of 0.551. The reason is resolution rather than direction: the external estimate rests on 29 events (Section 6.3), and an interval that wide cannot separate the pre-registered degradation of 0.085 from no degradation at all. H5 is the one hypothesis in this study that a better-powered external cohort might have confirmed.

The pattern is worth stating plainly before the individual verdicts. H1 and H4 fail *directionally*: the label does not out-vary the model, and the random split does not flatter the model. H5 fails *for want of precision*, not for want of an effect. H6 alone “fails” in the study’s favour: its null is not rejected in the one direction the test addresses, so the boosted trees are not better by more than the margin, which is a positive finding about interpretable models rather than a negative one. A pre-

registered design that returns four non-confirmations is doing what pre-registration is for: the point estimates under the original protocol, without intervals, would have supported three confident claims that the data do not carry.

5.10.3 H1: Label Dominance (Not Confirmed)

H1 [C]: Spread in AUROC across label variants $\geq$ spread across model classes.

Point estimates: label spread = 0.0123; model spread = 0.1499.

Contrast: $\theta = -0.1375$ (95% CI -0.1623 – -0.1083), Holm-adjusted $p = 1.000$. **H1 NOT CONFIRMED, and decisively so, in the direction opposite to the one hypothesised.**

H1 fails, and not marginally. Model class produces an AUROC spread of 0.1499 against the label's 0.0123, more than tenfold, and the interval on the difference, $[-0.1623, -0.1083]$, excludes zero. **Which model class is fitted matters far more for discrimination than which of three defensible Sepsis-3 readings is adopted.** The hypothesis anticipated the reverse.

This has to be read against what the two spreads contain. The model-class spread runs from rule-based scores (qSOFA 0.6091, SIRS 0.6114, NEWS2 0.6415) through a static Cox model (0.6444) to the two flexible estimators (primary 0.7590, GBT 0.7424). It is therefore not a comparison of like with like: it is dominated by the gap between a six-item bedside score and a fitted hazard model with 22 covariates, and a gap of that size is not surprising. The label spread, by contrast, is measured with the model held fixed and is genuinely small: the three variants disagree about *which* patients are septic far more than they disagree about how predictable those patients are.

That distinction is what makes the negative result informative rather than merely deflationary. Label choice moves the cohort substantially: variants A and B agree on only 88.1% of stays ($\kappa = 0.267$) despite near-identical prevalence (9.6% against 9.9%). Yet all three labels turn out to be about equally learnable. **The label determines who is counted; it barely determines how well anyone can be predicted.** A literature that fixes the label and compares only models is therefore not mismeasuring discrimination, but it is silently fixing the denominator of every clinical claim built on top of it.

One caveat applies to both spreads and is developed in Section 5.6. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets: part of the label spread measured here is guaranteed by the construction of the definition rather than discovered in the data. Whether a label owing nothing to treatment

timing behaves differently cannot be answered inside this family, and is taken up separately.

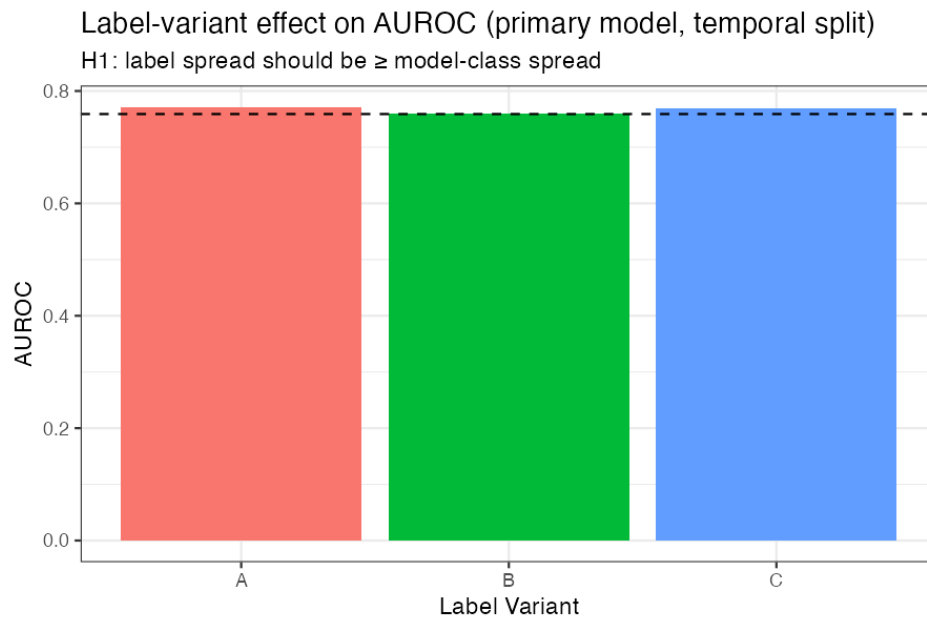


Figure 5.9: AUROC by label variant for each model (temporal split). Both quantities are *ranges*, largest minus smallest: the range within each model group is the label contribution (0.0123, over three variants) and the range across groups is the model-class contribution (0.1499, over six model classes). The model-class contribution is the larger by roughly an order of magnitude, which is the reverse of the ordering H1 predicted; Section 5.10 sets out why the unequal group counts do not account for the gap.

5.10.4 H2: Coefficient Instability (Met, Descriptive)

H2 [C]: At least one feature changes sign or magnitude $\geq 2\times$ across label variants.

Result: of the 22 covariates shared by all three variants, 2 reverse sign, 5 change magnitude by $\geq 2\times$, and 6 do at least one of the two. **H2's criterion is met (descriptive; no test statistic).**

Multiplicity: H2 reserves a slot in the confirmatory family of six but contributes no p-value, which makes the Holm correction on the remaining hypotheses stricter rather than weaker.

H2 is a stability criterion, not a contrast between two estimable quantities, so no p-value is defined for it. The count above is the pre-registered answer and is

reported unmodified. Its magnitude is conditional on the fit being identified: a penalised specification with a unique maximum is a precondition for the question “does this coefficient change sign across labels?” to have an answer at all, and Section 3.10.1 states the penalty used.

The criterion has a weakness that its own form conceals, and which is worth making explicit before the composition of the list is read. A threshold rule applied to point estimates cannot separate a coefficient that genuinely takes different values under different labels from one estimated so imprecisely that a doubling falls within its noise. For a covariate whose standard error is of the order of its estimate, the second is close to certain and says nothing about label sensitivity. The count of 6 is therefore an *upper bound* on the instability present, and how loose a bound was, until now, unmeasured.

Feature	What changed
Bilirubin	Sign reversal across variants
Any vasopressor	Sign reversal <i>and</i> magnitude $\geq 2\times$
Platelets	Magnitude $\geq 2\times$
Creatinine measured	Magnitude $\geq 2\times$
Platelets measured	Magnitude $\geq 2\times$
WBC measured	Magnitude $\geq 2\times$

Table 5.28: Features with sign reversal or magnitude instability across label variants. 2 of the 22 shared covariates reverse sign and 6 are unstable on either criterion. Full per-covariate flags are in `12_h2_coefficient_stability.csv`.

Does the instability exceed its own estimation noise? (Protocol Amendment 7)

The bound above can now be tightened. A cluster-robust covariance for the primary model, clustered on stay, is available (Section 3.10.1), so each flagged covariate carries a standard error under each label. For the two variants at the extremes of a covariate’s estimated range, the difference between them can be referred to the pooled standard error of those two estimates. Table 5.29 reports that ratio.

Covariate	β_A	β_B	β_C	Max diff.	Pooled SE	Ratio	Kind
wbc_measured	0.0970	0.5374	0.8989	0.8020	0.0517	15.51	action
platelets_measured	-0.2564	-0.5614	-0.8173	0.5610	0.0529	10.60	action
creatinine_measured	0.2877	0.2303	0.0343	0.2534	0.1024	2.48	action
platelets	0.0002	0.0001	0.0003	0.0002	0.0002	1.07	physiol.
vaso_anyTRUE	-0.0374	0.0271	0.0025	0.0645	0.0864	0.75	action
bilirubin_total	0.0014	-0.0016	-0.0020	0.0034	0.0056	0.61	physiol.

Table 5.29: Each covariate flagged by H2, with its coefficient under each label variant, the difference between the two extremes, the pooled cluster-robust standard error of those two estimates, and their ratio. 3 of the 6 flagged covariates move by more than roughly twice their pooled standard error; the remaining 3 move by less, the largest such ratio being 1.07. The separation is clean rather than marginal: no covariate falls between 1.07 and 2.48, so the split does not depend on where the cut is placed. “Kind” marks whether the covariate records clinician action or patient physiology (Section 5.10.4). **This is a diagnostic, not a test:** the three fits share stays, so the estimates being differenced are correlated and the pooled standard error assumes they are not; and because the specification is ridge-penalised out of necessity, the covariance is that of the penalised estimator. No p -value is computed and none should be inferred from the ratio.

Two things follow, and they should be kept apart.

First, **H2’s count is the pre-registered answer and stands.** The criterion was fixed before any modelling, it is met, and 6 of 22 covariates satisfy it. This diagnostic is post-hoc, is disclosed as Protocol Amendment 7, and does not replace that count. Restating the headline as the number surviving the diagnostic would substitute a post-hoc estimand for a pre-registered one, which is precisely the practice this thesis objects to elsewhere; the fact that the substitution would be flattering here is a reason for more caution, not less.

Second, **the flagged set is not homogeneous, and the inhomogeneity has a pattern.** The covariates whose movement survives comparison with their own standard errors are 3 of 3 action-derived, with ratios from 2.48 to 15.51, and all 3 are indicators of whether a test was ordered. Those that do not clear the threshold include both physiological covariates in the flagged set and, equally, *both sign reversals*: bilirubin and vasopressor exposure carry the two smallest ratios in the table. The direction changes that a bare count makes most striking are therefore the two the diagnostic supports least, and no claim in this thesis rests on either of them individually. The reading is that the label-sensitivity H2 detects is real but narrower than the raw count suggests, and that it sits where the previous subsection and the ablation both locate it: in the features that record what a clinician did rather than what the patient’s body was doing.

That convergence is worth stating carefully, because it is easy to overstate.

It is not independent corroboration. The ablation’s split by feature kind (Section 5.10.4), the ablation arm of Section 5.7 and this diagnostic are all computed from the same three fitted models and share the same flagged set. They are three readings of one body of evidence, and they agree because they are looking at the same thing from different angles, not because three separate experiments returned the same answer.

The instability is concentrated in the action-derived features

The composition of this list matters more than its length, and it is not arbitrary. The covariates fall into two kinds. **Action-derived** features record clinician behaviour: vasopressor exposure, which is a treatment, and the per-test order indicators, which record a decision to look. **Physiological** features record the patient: vital signs, laboratory values and their rolling slopes. Splitting the instability by kind separates them sharply.

Feature kind	Covariates	Unstable	% unstable
Action-derived	8	4	50.0
Physiological	14	2	14.3

Table 5.30: H2 instability by feature kind. Action-derived covariates are a minority of the specification but a majority of the unstable terms. **[E] exploratory.**

Action-derived covariates are a minority of the specification and a majority of the unstable terms. The reading this licenses is more specific, and more useful, than “the model is unstable across labels”: what is unstable is the part of the model that encodes clinician behaviour, and the part that encodes physiology is comparatively stable.

That distinction matters because it is the same mechanism the thesis identifies in the label, appearing now on the other side of the equation. Changing the label variant changes which treatment episodes count as sepsis. Features built from treatment and from measurement decisions are therefore re-anchored by a label change in a way that a heart rate is not. The sign reversal on *any vasopressor* is the sharpest case: a published association between vasopressor exposure and sepsis hazard, estimated from a model of this kind, cannot be interpreted without knowing the exact label operationalisation, and the same caution does not attach with equal force to the physiological terms.

Two limits on this reading should be stated. With 22 covariates in total the counts are small, and the difference between the two rates is not established at any conventional level of confidence; H2 carries no test statistic (Section 5.10.4) and the split inherits that limitation. And a concentration of instability is not by itself

evidence that the model *depends* on these features for its performance. Section 5.7 addresses the second point directly by removing them and refitting.

The instability is therefore real but narrower than a count alone would suggest: 6 of 22 covariates, most by magnitude rather than direction, and concentrated in features that are not measurements of the patient. That supports Lauritsen et al.’s [10] framing-instability concern for specific effect estimates, and does not support a claim that the fitted model is unstable as a whole.

5.10.5 H3: Treatment Leakage, Definitional and Empirical

H3 [C]: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03 .

Definitional result: 0 sepsis onset events in 179,003 pre-treatment person-hours. AUROC is undefined, so the pre-registered contrast cannot be formed. This zero is *proved* below (Proposition 5.1) to follow from the onset rule used by variants A–C; it is a statement about the implementation, not a measurement.

Empirical result (Protocol Amendment 4): relabelling the same stays with the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset (Variant F) moves 1,738 of 6,447 onsets earlier (27.0%, median -14.0 h) and places 71 onsets before treatment, identified at 0.8081 AUROC (0.8018 GBT). **H3 is not evaluable as pre-registered; the definitional and empirical statements replace it.**

Multiplicity: like H2, H3 reserves a slot in the confirmatory family without contributing a p-value. Variant F’s contrast against B is exploratory (family E3).

The definitional half. When only hours before the first antibiotic or culture are considered, there are zero sepsis onset events for variants A–C and E. This is not a small drop in performance, and it is also not a finding. It is entailed by how the onset timestamp is assigned:

Proposition 5.1. *Let t_{anchor} be the earlier of the first qualifying antibiotic order and the first culture draw of an admission, and let t_{onset} be the onset assigned by variants A, B, C or E. Then $t_{\text{onset}} \geq t_{\text{anchor}}$ for every labelled stay, so the window $[\cdot, t_{\text{anchor}})$ contains no onset.*

Proof. Under E, $t_{\text{onset}} = t_{\text{susp}}$, and t_{susp} is defined as the *later* element of a qualifying antibiotic-culture pair; both elements are at or after t_{anchor} , so $t_{\text{susp}} \geq t_{\text{anchor}}$. Under A, B and C the SOFA criterion is a *membership* gate: a stay is labelled only if the ≥ 2 -point rise occurs somewhere in the evaluation window, but the timestamp written is still t_{susp} (Section 3.6). The inequality

therefore carries over unchanged. $\square$

The exact one-sided 95% upper bound on the pre-treatment onset rate reported in Chapter 5’s inference tables is arithmetically correct but informationally empty: it bounds a quantity the definition fixes at zero. It is retained only to make that point explicit.

The empirical half. The standard operationalisation does not have this property. Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [17] take

$$t_{\text{onset}} = \min(t_{\text{susp}}, t_{\text{SOFA}}),$$

where t_{SOFA} is the first hour in the evaluation window at which the organ-dysfunction rise is already present. When the SOFA rise is detected before the clinician acts, the onset precedes the anchor and Proposition 5.1 does not apply. Variant F implements exactly this and nothing else: identical suspicion windows, identical SOFA criterion, therefore **the same septic stays as Variant B**, all 6,447 of them, with only the timestamp moved.

Under Variant F, 1,738 onsets (27.0%) move earlier, by a median of -14.0 hours, and 71 of them fall in the pre-treatment window of the temporal test set, where variants A–C record none. A model identifies those hours at 0.8081 AUROC. **Sepsis-3 onsets do precede clinical action; the conjunction-timing rule used for the pre-registered variants is what prevents any from being recorded.**

What the two halves together license, and what they do not. Neither half supports the absolute proposition that no patient develops sepsis before clinicians begin treatment. The definitional half cannot support it, because it is a statement about an onset rule rather than about patients; the empirical half contradicts it. External evidence contradicts it too: the Epic Sepsis Model v2 evaluation [5] scores against a composite of clinician recognition and reports encounter-level AUROC of 0.80–0.90 with median lead times of 1.4–7.1 hours ahead of that recognition. Prediction ahead of clinical action is demonstrably possible.

What the two halves do license is narrower and more consequential. Sepsis-3 *membership* is constituted by clinical action (no antibiotic-culture pair, no label, under any timing rule), so Kamran et al.’s [7] argument holds at the level of *which* patients can ever be labelled. And a conjunction-based onset rule, in common with much of the MIMIC-IV literature, discards the timing branch that makes pre-treatment prediction measurable at all. A result reported on such a label is measuring anticipation of the clinician’s decision within a window that decision opened. Section 5.6.3 sets Variant F beside the treatment-independent deterioration label and reports both.

5.10.6 H4: Split Optimism (Not Confirmed)

H4 [C]: Temporal-split AUROC is at least 0.02 lower than random-split AUROC.

Result: primary model, random AUROC 0.7424 against temporal 0.7590. Difference = -0.0165 (95% CI -0.0372 – 0.0040): the temporal split is the *higher* of the two. Test against the pre-registered threshold $\theta_0 = 0.02$: Holm-adjusted $p = 1.000$. **H4 NOT CONFIRMED.**

Model	Temporal	Random	Random – temporal
Primary	0.7590	0.7424	–0.017
GBT	0.7424	0.7132	–0.029
Cox	0.6444	0.6310	–0.013
NEWS2	0.6415	0.6088	–0.033

Table 5.31: Temporal vs. random split comparison (Variant B). Every model scores lower on the random split, not higher. H4 is an AUROC hypothesis; the Utility Score is not shown because the split contrast is defined on discrimination alone.

H4 anticipated that a random split would flatter the model by letting later patient patterns leak into training. It does not, and the direction is consistent across all four models fitted under both splits: each scores *lower* on the random split, by 0.013 to 0.033. The interval on the primary model’s difference, $[-0.0372, 0.0040]$, is centred below zero and excludes the 0.02 threshold the hypothesis required.

The explanation is that the two splits differ in more than optimism. The temporal test set is the 2017–2019 admission cohort, whereas the random test set is a 25% patient-level sample spanning all years. The random arm therefore trains on *less* data from any given era and tests on a more heterogeneous mixture, and that cost evidently outweighs whatever leakage advantage it gains. Guo et al.’s [8] concern is not refuted by this (their argument is about optimism from temporal leakage, which is real), but on this cohort it is not the dominant term, and a study choosing between the two designs cannot assume the random split is the more permissive one.

There is a second reason not to read too much into the size of this contrast. The two test sets are disjoint samples of different composition, so the difference confounds split design with cohort composition; it is not the clean within-sample optimism estimate the hypothesis presumed. This is recorded in the limitations (Section 6.7).

5.10.7 H5: Metric Divergence (Not Confirmed)

H5 [C]: Utility Score declines proportionally more than AUROC from internal to external validation.

Testable arm: AUROC degradation internal $\rightarrow$ external (primary, Variant B) = 0.0630 (95% CI -0.0215 – 0.1563); internal 0.7590, eICU-CRD 0.6960 on 29 external events. Holm-adjusted $p = 0.551$. **The point estimate falls, but the interval does not exclude no change.**

Non-testable arm: the internal Utility Score is 0.000 at the default cut-off and 0.025 at its best threshold, so a proportional decline cannot be formed. **H5 NOT CONFIRMED.**

Neither arm of H5 supports the hypothesis, and the reason differs sharply between them.

The estimand tested is a substitute, and the adjusted p -value belongs to it. H5 as pre-registered compares two *proportional* declines, and that contrast cannot be formed here: a proportional decline requires a baseline to be proportional to, and the internal Utility Score under the primary label is 0.000 at the default cut-off and 0.025 at its best threshold. What occupies H5's slot in the confirmatory family is therefore the AUROC limb alone (internal minus external AUROC, against $\theta_0 = 0$), and the Holm-adjusted value of 0.551 reported above applies to that substituted contrast, not to the metric-divergence estimand the protocol specified. The substitution, its motivation and the constraints placed on it are recorded in §13.6 of `00_preregistration.md` alongside the two hypotheses that are degenerate outright. It should be read with one caution stated in the same place: this substituted contrast produces the smallest raw p -value in the family, and it is still a non-confirmation.

The AUROC arm fails on precision rather than on direction. External discrimination on eICU-CRD is 0.6960 against 0.7590 internally, a degradation of 0.0630: the sign H5 predicted, and roughly three-quarters of the 0.085 anticipated from Moor et al. [21]. But the interval $[-0.0215, 0.1563]$ covers zero, so the estimate is equally consistent with no loss and with a loss twice as large. This is the only hypothesis in the family whose uncorrected p -value falls below 0.10, and the only one a better-powered external cohort might have confirmed.

The precision is bounded by documentation coverage, not by cohort size. eICU-CRD contains 181,589 unit stays, but the Sepsis-3 conjunction is recoverable in 1.55% of them, so the analysis rests on 29 events against an interpretability floor of 100 (Section 6.3). Person-hours from the remaining stays would enlarge the denominator without contributing a single observable onset.

Two independent observations nonetheless point the same way, which is why

the degradation is reported as a direction rather than dismissed as noise. First, the effect is ordered across labels exactly as the internal results are: the Narrow label loses most (0.7713 to 0.6913) and the Liberal label least (0.7695 to 0.7370), which is not a pattern noise produces. Second, the antibiotic-only arm (a different target, but one measured on 6,140 events and therefore well clear of the floor) shows a degradation of the same sign and larger magnitude (0.7590 to 0.6644) with the same ordering across labels. Two arms differing by two orders of magnitude in event count agree on sign, ordering and approximate size.

The honest verdict is therefore neither “discrimination transports” nor “discrimination collapses”, but that this study can establish the direction of external degradation and not its magnitude.

The Utility Score arm fails for a different and more consequential reason: the margin whose collapse Wang et al. [4] documented, a median internal Utility Score of +0.381 falling to −0.164, does not exist here to begin with. The best internal score any configuration reaches under the primary label is 0.031, against internal AUROC in the high 0.7s. A ratio of proportional declines is undefined against a baseline this close to zero, and the 29 external events are in any case too few to estimate an external Utility Score reliably.

The finding that replaces H5 is therefore not “utility degrades faster than AUROC” but something the internal results establish on their own, without needing the external arm at all: **clinical utility never arrives, at any threshold, on the cohort where the label is fully observed**. The gap Wang et al. identified between the two metrics is confirmed, discrimination is a poor proxy for clinical value, while their specific mechanism, external collapse of a healthy internal utility, is not what happens here. What the external arm adds is a qualification in their favour on the other metric: AUROC does appear to decline on transport, so the stability they observed in it (0.811 to 0.783) is not reproduced as stability here either.

There is a methodological point in this that outlives the particular numbers. External validation is conventionally treated as a test of the model, and the label is assumed to travel with it. It does not. Sepsis-3 is defined on clinician actions and on the documentation of those actions, so transporting it requires the destination database to record the same actions in the same retrievable form. Where it does not, the failure presents as an implausibly low event rate rather than as an error, and an AUROC computed on the survivors is still a well-formed number, which is precisely what makes it dangerous. The event-rate reconciliation in Table 5.12 exists so that this check is performed and reported rather than assumed.

5.10.8 H6: Model-Class Null (Not Rejected; Non-Inferiority Established)

H6 [C]: $|\Delta_{\text{AUROC}}(\text{GBT} - \text{primary})| \leq 0.02$.

Result: $\Delta = -0.0165$ (GBT 0.7424, primary 0.7590), 95% CI -0.0308 – -0.0012 . The *upper* limit lies inside the bound $\theta_0 = 0.02$; the lower limit does not. **H6 NOT REJECTED. What this establishes is non-inferiority of the interpretable model (the boosted trees are not better by more than the margin) and not two-sided equivalence.**

This is the strongest positive result in the study, and it is worth stating exactly what it is and is not.

An interpretable multinomial hazard model with 29 terms per outcome (22 covariates, a six-column natural-spline baseline in elapsed hours, and an intercept) is not outperformed by gradient-boosted trees on discrimination under the primary label: 0.7590 against 0.7424. Both models are scored on the same person-hours, so the contrast is paired and the interval is correspondingly tight. The ordering is the same under the other two labels (Table 5.27).

What the interval supports, and what it does not. The pre-registered statement is two-sided, $|\theta| \leq 0.02$, but the test performed against it is one-sided: the bootstrap p -value is computed for $\theta > 0.02$ alone, which is the direction in which the boosted trees would beat the hazard model. That direction is comfortably excluded, the interval’s upper limit being -0.0012 . Establishing two-sided equivalence would require the second of the two one-sided tests as well, and the data do not deliver it: the lower limit -0.0308 falls outside -0.02 , so a difference larger than the margin in the *interpretable* model’s favour remains consistent with these data. The defensible reading is therefore asymmetric: the boosted trees buy nothing here, but the two estimators have not been shown to be interchangeable, and a claim of strict equivalence would overstate the evidence.

One further property of the interval is worth reading off, because it is stronger than non-inferiority and weaker than it might appear. The interval lies entirely below zero, so these data are not consistent with the two estimators discriminating equally well: on the primary label the interpretable model is ahead, not merely not behind. The size of that lead is what keeps the statement modest. It is -0.0165 of AUROC, of the same order as the label spread and far smaller than the model-class range, and the utility comparison below shows it buys nothing at any operating point. The claim this licenses is that the accuracy case for the opaque model fails, not that the transparent one should be preferred on accuracy grounds; those are different claims and only the first is needed. A methodological

caution on the comparator’s fitting procedure, which bears on the direction of this result, is stated in Section 6.7.

The asymmetry does not weaken the methodological conclusion, because the conclusion only ever needed one side of it. The argument for preferring an opaque model over a transparent one is an argument that the opaque model performs better; the interval excludes that in the only direction it is made.

The methodological consequence is direct. On tabular ICU time-series of this kind, the accuracy argument for preferring an opaque model over a transparent one is not supported here. Whatever a boosted ensemble is extracting from these features, a pooled logistic hazard model with the same features and an appropriately flexible baseline extracts as well, and the latter yields per-covariate coefficients, a calibration slope that can be inspected, and a functional form a clinician can be shown.

The clinical-utility comparison tells the same story from the other side, and matters more. At the default cut-off the primary model appears far ahead (0.000 against -1.781) purely because it never crosses 0.5 and GBT does; that contrast measures calibration, not clinical value. Once each model is evaluated at its own best threshold the two are close and GBT is marginally ahead, 0.031 against 0.025. The two metrics therefore order the models oppositely, and both differences are too small to act on: whichever way the comparison is run, neither model is worth choosing over the other on clinical value. Both maxima sit a few per cent above the no-alert strategy, which is the more important observation: **the model-class choice is nearly free, because neither class delivers clinical value worth choosing between.**

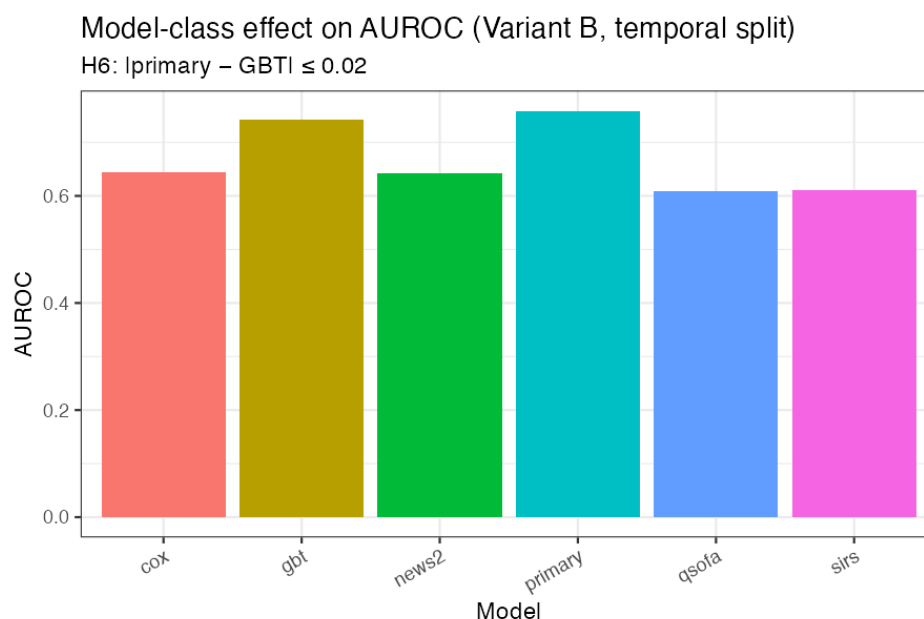


Figure 5.10: AUROC by model class (Variant B, temporal split). The two flexible estimators are indistinguishable from one another and clearly ahead of the rule-based scores and the static Cox model; the best maximum Utility Score among them is 0.031, barely above the no-alert strategy. The model-class range of 0.1499 over six model classes is an order of magnitude larger than the label range of 0.0123 over three variants (Section 5.10).

5.11 Analysis Register [D]

Table 5.32 records the status of every analysis reported in this chapter and the multiplicity correction applied to it. Its confirmatory and exploratory rows are transcribed from `12_analysis_register.parquet`, which the inference stage writes from the same objects it corrects, so the status and correction attached to a hypothesis are the ones the code applied rather than the ones this chapter claims for it. Two rows have no counterpart in that file and are marked accordingly: the antibiotic-only external arm and the by-kind stability split are scored in stages 07 and 08 and enter no corrected family, so the inference stage never sees them. They are recorded here because a register that omitted them would understate how much of this chapter is exploratory.

Table 5.32: Analysis register. [C] confirmatory, [E] exploratory, [D] descriptive. Section numbers refer to 00_preregistration.md. *§9 pre-registered the equity analysis in scope only: neither the subgroup levels, the reference level, nor the direction of any comparison was fixed in advance, so these analyses are exploratory. †A2 = Protocol Amendment 2 (§§16–17), written after notebooks 02–12 had been run; these analyses are post-hoc and change no pre-registered estimand. ‡A4 = Protocol Amendment 4 (§18), which adds the standard onset timing as Variant F; also post-hoc, and it enlarges family E3, so every BH-adjusted p-value in that family is stricter than it would be without the amendment. §A6 = Protocol Amendment 6 (§20), the subgroup interpretability floor; post-hoc, and it *reduces* family E1, which the inference plan’s own denominator rule prohibits on its face; Section 3.14 states the tension and §20.4 of the pre-registration gives the argument. ¶These two analyses are scored outside the inference stage and enter no corrected family, so they have no row in 12_analysis_register; they are listed here so that the exploratory record in this chapter is complete.

Analysis	Status	Pre-reg.	Correction
Cohort description by label variant	[D]	§2	none
Sample size adequacy	[C]	§10	none (single criterion)
Internal temporal performance	[C]	§5–6	none (estimation)
Calibration intercept and slope	[C]	§5	none (estimation)
Decision-curve net benefit	[C]	§6	none (estimation)
Operating points at fixed sensitivity	[E]	n/a	none (descriptive)
Per-patient detection and lead time	[C]	§6	none (estimation)
H1 label dominance	[C]	§7	Holm ($m = 6$)
H2 coefficient instability	[C]	§7	slot reserved; no test statistic
H2 instability vs. estimation noise	[E]	Amd. 7	none; ratio to pooled SE, not a test
H3 treatment leakage	[C]	§7	slot reserved; exact bound
H4 split optimism	[C]	§7	Holm ($m = 6$)
H5 metric divergence	[C]	§7	Holm ($m = 6$)
H6 model-class null	[C]	§7	Holm ($m = 6$)
External validation on eICU-CRD	[C]	§4	none (estimation); feeds H5
antibiotic-only anchor (sensitivity)¶	[E]	n/a	none; different target, not pooled
Action-derived feature ablation	[E]	A5	none (estimation); enters no family
instability by feature kind¶	[E]	n/a	none (descriptive split of H2)

Analysis	Status	Pre-reg.	Correction
Variance decomposition table	[C]	§1	none (assembly of H1–H6)
Proportional-hazards testing	[C]	§8	none (single check)
GBT feature importance	[E]	n/a	none (ranking, no inference)
Subgroup AUROC by demographic group	[E]	§9*	Benjamini–Hochberg, family E1
interpretability floor on E1	[E]	A6 [§]	restricts E1 to 6 contrasts
Subgroup label sensitivity	[E]	§9*	Benjamini–Hochberg, family E2
Label sensitivity vs. AUROC association	[E]	n/a	none (visual, no test)
Alternative-label AUROC vs. Sepsis-3	[E]	A2 [†]	Benjamini–Hochberg, family E3
Anchor-attributable fraction ϕ	[E]	A2 [†]	none (ratio, no null at zero)
Onset agreement between definitions	[D]	A2 [†]	none (descriptive)
Standard onset timing $\min(t_s, t_{\text{SOFA}})$	[E]	A4 [‡]	Benjamini–Hochberg, family E3
Pre-treatment onsets under standard timing	[D]	A4 [‡]	none (event count, not a test)

Chapter 6

Discussion

Chapter 5 reports what was measured. This chapter interprets it: what the six verdicts mean taken together, how far each estimate can be pushed, and what the study cannot support. Section 6.1 states the central finding and Section 6.2 resolves the tension the pre-registration deliberately built between H1 and H6. Sections 6.3 to 6.5 bound the three analyses whose readings are most easily over-extended, the external estimate, the treatment-independent label, and the feature ablation. Section 6.6 draws out what the findings imply for how studies of this kind should be reported, and Section 6.7 sets out the limitations in full.

6.1 The Core Finding

The most important finding is definitional, not numerical: under a conjunction-based onset rule the Sepsis-3 label on MIMIC-IV *cannot* contain an event before clinical action begins (Proposition 5.1), and under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule applied to the identical stays it contains 71 of them at 0.8081 AUROC. A published MIMIC-IV sepsis result is therefore only as informative about physiological deterioration as its onset rule permits, and a result reported without the timing branch is measuring anticipation of clinician behaviour inside a window that behaviour opened. The Utility Score stays close to the no-alert strategy for every model because the temporal split exposes them to a test population where the same treatment-related signals do not rescue performance as consistently as in the training years, and because the score charges for every false-alert hour while crediting only the first correct one. AUROC survives because the ordinal ranking of patients is preserved even as the achievable alert volume becomes unusable.

That divergence is what Wang et al. [4] documented across ninety-one models moving from internal to external validation, and what Section 2.6 identified as the field’s central evaluation failure. The result here locates it one step earlier.

On MIMIC-IV, under an honest time-based split, discrimination and clinical value have already parted company *internally*: there was scarcely any margin left for external validation to destroy.

6.2 H1 and H6 Together

H1 (label dominance) and H6 (model-class null) are in productive tension, as designed, and the resolution is sharper than either hypothesis anticipated, because both resolve in the same direction, which the pre-registration treated as unlikely.

H6 is *not* rejected: the two flexible estimators differ by 0.0165 AUROC, and the interval's upper limit sits inside the 0.02 bound, so the boosted trees are not better by more than the margin. H1 is also not confirmed, and by a wide margin in the direction opposite to the one hypothesised: the model-class spread 0.1499 exceeds the label spread 0.0123 by -0.1375 , with an interval $[-0.1623, -0.1083]$ that excludes zero.

Read together these are not in conflict; they locate the model-class effect precisely. The large model-class spread is the distance between *families* of estimator, a six-item bedside score against a fitted hazard model, not between one fitted model and another. Within the family of flexible estimators the choice is free (H6), and across families it is the single largest factor in the decomposition (H1). The conventional literature, which fixes the label and compares flexible models to each other, is therefore measuring the one contrast that has almost no room left in it, while the choice that would move the number (fit anything flexible at all, rather than deploy a score) is not in dispute and the choice that moves the cohort, the label, is invisible to the metric being reported.

The H6 result also settles, for this task and this dataset, the narrower version of the statistics-versus-machine-learning question left open in Section 2.2. Christodoulou et al.'s [23] equivalence finding was established for binary outcomes on tabular data, and Fagerström et al. [22] showed it breaking on the temporal early-warning axis, where a deep model detected patients up to twenty hours earlier than a Cox baseline. Here, with the label, the split, the feature set and the evaluation held fixed, an interpretable discrete-time hazard model is not outperformed by gradient-boosted trees, and is ahead of them by 0.0165 AUROC on an interval that excludes zero. That is not a contradiction of Fagerström et al.: their comparison was against a static Cox specification, which is exactly the comparator this study retains as the deliberately weak one, and which loses by a wide margin here too (0.6444 against 0.7590). The result holds between the primary model and the boosted trees; nothing of the kind holds between either of them and the field's default survival specification.

This is also the clearest illustration in the thesis of why the inference layer was

necessary. On point estimates alone, H1 would have been read as confirmed and the study would have told a tidy story in which the label dominates discrimination. With intervals attached that story is not supportable: the label spread is 0.0123 [0.003, 0.032] against a model-class spread of 0.1499 [0.134, 0.171], and the two intervals do not overlap. The two sources are not of comparable size, and the defensible finding is the less convenient one: the label moves discrimination by roughly an order of magnitude less than the model class, while moving the cohort, the event count and the denominator of every clinical quantity computed on it. The label effect is real and consequential, but it is not an effect on learnability, which is the only channel through which the field’s usual metric could have detected it.

6.3 What the External Estimate Can Support

Discrimination is estimated on 27, 29 and 73 external events under Variants A, B and C (Section 5.5). The interpretability floor adopted here is 100 events, following the external-validation sample-size guidance of Riley et al. [46] and Collins et al. [12]: below it the confidence interval on an external AUROC is wider than the degradation the study is trying to detect. **All three variants fall below it.** The degradations are accordingly stated as directions, not as magnitudes, and the H5 interval is the formal expression of that limit.

Two further quantities are unavailable at this event count in any case. AUPRC is dominated by sampling noise at a few dozen positives, and the Utility Score requires enough septic stays for the timeliness reward to be estimated at all. Neither is interpreted externally.

The antibiotic-only arm. Relaxing the culture requirement across the full cohort raises the event count by two orders of magnitude (5,589, 6,140 and 9,886 events across 7,627,689 person-hours), and therefore clears the floor with room to spare. Under that anchor the primary model scores 0.6488, 0.6644 and 0.7209.

The comparators in the same arm carry a result the Sepsis-3 arm is too small to see, and it does not favour the flexible estimator. The boosted model scores 0.5316, 0.5486 and 0.6106, close to chance under the two narrower labels, while NEWS2 scores 0.6121, 0.6071 and 0.6169. A bedside score therefore ranks ahead of the boosted model under every label in this arm. The internal model-class ordering does not transport. That is worth stating plainly for two reasons. It qualifies H6 in a way the internal analysis cannot: the two flexible estimators are indistinguishable inside the development cohort, but only one of them stays ahead of a bedside score outside it, and the one that does is the interpretable model. And it cuts against this study’s own comparators as much as for them, since a rule-based score overtaking a fitted model externally is not the result a modelling

thesis would choose. The event count supports the ordering; it does not license a claim about the size of any gap, for the same reason the Sepsis-3 arm does not, and the arm still labels a different target.

This arm labels a broader and less specific target: treated suspicion of infection, without the organ-dysfunction conjunction’s culture limb. An AUROC computed against it therefore answers a different question and is never compared like-for-like with the internal Sepsis-3 estimates, nor pooled with the primary arm. What it does supply is a consistency check that the primary arm cannot supply for itself. The degradation it shows is of the same sign, the same label ordering and a somewhat larger magnitude than the Sepsis-3 arm’s, on a sample large enough to resolve it. Two estimates that disagree by two orders of magnitude in event count and agree on direction are better evidence of a direction than either is alone, which is the strongest statement the external analysis in this thesis supports, and deliberately weaker than a claim about how much discrimination is lost.

Against the published record assembled in Section 2.5, the direction is unsurprising and the magnitude is unresolvable. Moor et al. [21] report an external penalty of 0.085 AUROC across four international databases; the point estimate here, 0.0630, is about three-quarters of that. Where this study cannot follow Moor et al. is in attributing the loss, because their harmonised databases shared labelling inputs and eICU-CRD does not.

6.4 What Variant D Does and Does Not Establish

Variant D’s high AUROC must be read with the limitation stated in advance in Section 3.6. Its label is a threshold function of recorded physiology, and five of the six SOFA components are themselves model inputs, so its discrimination is partly self-fulfilling. The *level* of 0.8493 (0.9147 for GBT) is not evidence that deterioration prediction is a solved problem, and it is not comparable with Variant B’s 0.7590.

Three things it does establish:

1. **The label sensitivity is not wholly definitional.** At $\kappa = 0.057$ and a 6.5% onset match, a treatment-independent label disagrees with Sepsis-3 about both membership and timing. Had D closely tracked B, the H1 result would have been an artefact of how one definition is read; it is not.
2. **The alert burden is label-invariant; the utility ceiling is not.** Both alternative labels are markedly more scoreable than Sepsis-3. The best achievable Utility Score rises from 0.031 under Variant B to 0.198 under E and 0.271 under D, with the primary model following the same pattern (0.148 and 0.238). What does not improve to a deployable level is the alarm volume:

even D's best configuration carries 12.7 false alerts per true alert, better than Variant B's 51.8 but still nine times the 1.4 benchmark of Moor et al. [21]. Contribution C3 survives the change of label in its alarm-volume form, which the pre-registered variants alone could not show, and does not survive in the form of a utility ceiling near zero. The rise is itself expected from the caveat above: a label built from recorded physiology is easier to alert on than one built from a clinician's decision.

3. **The two metrics order the models differently, and the ordering is stable across labels.** Evaluated at its own best threshold, GBT leads on the Utility Score under B, D and E alike (0.031 against 0.025, 0.271 against 0.238, 0.198 against 0.148). On AUROC the interpretable model leads under B and E, and GBT leads only under D, whose label is a threshold function of the physiology both models are given (Section 5.10.8). Neither ordering is reversed by changing the label; what changes the ordering is changing the metric. A third and different reversal appears if every model is scored at the same 0.5 cut-off, which rewards a model for being too poorly calibrated to reach it. The metric-choice finding (H5) rests on the gap between discrimination and achievable utility, and these three orderings are what that gap looks like when it is written out.

6.5 What the Ablation Establishes

The ablation is a small experiment and should not be asked to carry more than it can. It is exploratory, it is run on one cohort, and with 22 covariates the stability counts on either side are small enough that the comparison is descriptive rather than tested.

What it does establish is a distinction the thesis needs and could not otherwise make. The argument that the sepsis label is constituted by clinician action invites an obvious objection: that the same is true of everything in an ICU dataset, so the observation is unfalsifiable and therefore empty. The ablation answers that objection empirically. The feature set divides into predictors that record the patient and predictors that record the clinician, the division is made in advance rather than fitted, and the two behave differently under a change of label. That is a testable version of the claim, and it is tested here rather than asserted.

It could have failed, and the form of the result matters as much as its direction. Had discrimination fallen substantially, the finding would have been that the models lean on clinician behaviour: supportive of the same argument, but a concern about what reported performance is measuring, belonging as much in the limitations as in the results. Had stability been unchanged, the concentration of

instability in the action-derived features would have been small-sample noise and the reading would have had to be withdrawn. Neither happened. Discrimination is essentially unaffected (-0.0076 for the primary model under the primary label) while the unstable count falls from 6 to 1, which is the one outcome that separates the two hypotheses cleanly: features contributing almost nothing to discrimination cannot be carrying the model’s predictive weight, and features whose removal takes five of six unstable terms with them are where the label sensitivity lives.

The reading this licenses is bounded and worth stating exactly. It is *not* that the action-derived features are unimportant: a feature that moves AUROC by less than 0.01 may still be the reason a particular alert fires. It is that on this cohort the discrimination such a model reports does not depend on them, while its cross-label stability does. A published effect estimate for a treatment indicator from a model of this kind is therefore reporting something about the labelling regime as much as about the patient, and the cheapest available remedy, dropping the explicit action features, costs almost nothing in the metric the field reports.

The corresponding limitation is stated in Section 6.7: the retained laboratory values are themselves forward-filled from tests somebody chose to order, so the boundary drawn here is conservative rather than clean. A model with no trace of clinician behaviour in it is not constructible from this data, and the ablation measures the effect of removing the explicit traces, not all of them.

6.6 Implications for How These Studies Are Reported

The preceding sections bound what this study establishes. This one states what follows for the design of the next one, because the findings bear less on which model to build than on what a sepsis prediction study has to report before its headline number can be read at all.

A label variant is not a sensitivity analysis. Section 2.3 observed that the literature demonstrates *that* the label matters without attaching an interval to how much. The result here sharpens the reason that gap is costly. Label choice moves AUROC by 0.0123 and cohort membership by far more, the two narrowest readings agree at $\kappa = 0.267$ at near-identical prevalence, so a study that reports one definition’s AUROC has reported a number that would barely move under a different definition while every count, rate and denominator beneath it moved substantially. The consequence is that reporting a second variant as a robustness check answers the wrong question. What needs reporting is the agreement between the variants, because that is where the variation is, and it is invisible in the metric

the field publishes.

An onset rule is a claim about what can be predicted, and belongs in the methods. Section 2.4 recorded Kamran et al.’s finding that models partly learn to predict clinician behaviour, and this study locates the mechanism one level down: not in the feature set but in the timestamp. Two studies can share a cohort, a definition and a model and still differ on whether pre-onset prediction is measurable, purely through the choice between t_{susp} and $\min(t_{\text{susp}}, t_{\text{SOFA}})$. That choice is routinely left implicit in a sentence about Sepsis-3. It should be stated explicitly, with its consequence for the evaluation window stated alongside it, because under a conjunction rule an empty pre-treatment window is not evidence of anything (Proposition 5.1).

Discrimination and clinical value must both be reported, and neither substitutes. Section 2.6 set out why AUROC conceals the failure Wang et al. documented: it is monotone-invariant and dominated by true negatives. This study adds the observation that a single threshold conceals it a second way. At the event rate here the conventional 0.5 cut-off separates models by calibration rather than by clinical value, and the primary model’s zero at that cut-off is an artefact of never alerting there. A threshold-dependent metric is interpretable only as a swept ceiling with the attaining operating point reported beside it, and the alert burden at that point (51.8 false alerts per true one here, against the 1.4 benchmark of Moor et al. [21]) is the number that decides deployability. It appears in almost none of the ninety-one models Wang et al. [4] reviewed.

An external cohort should be characterised before it is scored. Section 2.5 assembled the published external penalties into a single frame and treated them as comparable. The experience here is that they are not automatically so. The binding constraint on the external estimate in this study is not sample size but whether the label’s inputs exist in the destination database at all, and that constraint is invisible in a reported AUROC and in a reported cohort size alike. A study transporting a treatment-anchored label should report the coverage of each labelling input in the destination cohort before it reports any performance number, and should state the event count the estimate rests on rather than the stay count.

Model-class comparisons are the least informative axis available. Section 2.2 argued that the statistics-versus-machine-learning partition does not carve the field. The decomposition supports a stronger version: within the family of flexible estimators the choice is free to within 0.0165 AUROC, so a study comparing

one flexible model against another is measuring the axis with the least room left in it, while the label choice that sets its cohort goes unreported. The reporting recommendation follows directly: fit any flexible estimator, report which one and stop defending the choice, and spend the space on the label and the onset rule instead.

A subgroup analysis needs an interpretability floor declared in advance. Section 2.9 noted that the nearest prior work measures subgroup disparity for mortality among already-septic patients rather than for onset. Extending it to onset prediction exposed a problem that is structural rather than particular to this cohort: at a low event rate most demographic strata carry too few events for a subgroup AUROC to mean anything, and an interval reaching the chance line is easily read as a null result rather than as an absent measurement. Declaring the floor before estimation, tabulating the below-floor strata rather than suppressing them, and excluding them from the corrected family is what keeps the distinction visible. Without it, a study underpowered for a fairness question can report one as answered.

6.7 Limitations

1. **Single-centre development.** All models were developed on MIMIC-IV, which originates from a single academic medical centre (Beth Israel Deaconess Medical Center, Boston). Clinical practices, patient demographics, and EHR documentation patterns at BIDMC may not represent those at other institutions. The equity and label-sensitivity findings in particular should not be assumed to transfer to hospitals with different patient populations or antibiotic prescribing cultures. The eICU-CRD analysis is the only external evidence available here, and the limitation below bounds how much it can carry.
2. **External validation is bounded by label transport, not by cohort size.** eICU-CRD contains 181,589 unit stays, but the Sepsis-3 suspicion anchor requires a microbiology culture and `microLab` documents one for only 1.55% of the analysis cohort, and both limbs of the anchor together for 0.128% (232 stays). The primary external analysis is confined to the culture-covered sub-cohort, because in the remaining stays no onset is observable under any variant; the resulting 29 events under the primary label sit below the 100-event interpretability floor (Section 6.3). Adding person-hours from the uncovered stays would enlarge the denominator without adding a single observable onset, so this is not a limitation additional data of the same kind

would remove, it would require a destination database whose microbiology recording matches MIMIC-IV's.

A second consequence follows, and it concerns composition rather than precision. The sub-cohort is defined by a documentation property, and a culture is ordered when infection is already suspected, so the 2,824 covered stays are not a random sample of the 181,589: they are enriched for the patients a clinician had begun to investigate. The external estimate therefore describes transport into a documentation-selected sub-population rather than into eICU-CRD as a whole. The direction of that selection cannot be signed from these data, because the uncovered stays carry no observable onset to compare against, so it stands as an unquantified limit on generalisability rather than a bias with a known sign. The consequence is specific and should not be over-read in either direction: the study establishes that discrimination degrades on transport, and cannot establish by how much. External Utility Score and AUPRC are not interpreted at all. The antibiotic-only sensitivity arm is well powered and agrees on direction, but it labels a different target and cannot stand in for the Sepsis-3 comparison.

3. **The external degradation is not decomposed into its causes.** An AUROC that falls on transport may reflect case-mix differences, documentation and measurement-frequency differences, genuine differences in how sepsis presents across two hundred hospitals, or residual imperfection in the transported label. This study measures the total and attributes none of it. The antibiotic-only arm bounds the label's contribution loosely (degradation persists when the culture limb is removed, so it is not solely an artefact of culture coverage), but a proper attribution would require a destination database with equivalent labelling inputs, which is precisely what was unavailable.
4. **The NEWS2 comparator is not scored identically in the two cohorts.** NEWS2 charges three points for a patient who is not alert and none for a patient whose consciousness was not assessed. In MIMIC-IV the flag is derived from the recorded Glasgow Coma Scale with an unmeasured value treated as "not alert", so the small minority of person-hours carrying no GCS after forward-fill are charged those three points; in eICU-CRD, where the variable is absent from the periodic vitals table altogether, an unmeasured value is propagated as unmeasured and charged nothing. AUROC is a ranking statistic and the internal effect is confined to that minority of rows, but the two cohorts' NEWS2 *thresholds* are not on a common scale, so the internal and external specificity and alert-burden figures for that comparator should be read within a cohort and not across the pair. Nothing in the decomposition, the hypothesis verdicts or the primary-model results depends

on this: NEWS2 enters the study as a deliberately weak reference point, and the internal-versus-external contrast that H5 tests is computed on the primary model.

5. **The feature set cannot be fully purged of clinician action.** The ablation in Section 5.7 removes the 8 predictors that record clinician behaviour explicitly, vasopressor exposure and the per-test order indicators, but the laboratory values it retains are forward-filled from tests somebody chose to order, and even a vital sign is recorded on a schedule that reflects how closely a patient is being watched. The boundary drawn is therefore conservative rather than clean: what is measured is the effect of removing the explicit traces of clinical attention, not of removing all of them. A predictor set with no such trace is not constructible from routinely collected ICU data, which is itself part of the thesis's argument rather than only a limitation of it.
6. **Deep survival model not implemented.** A deep survival comparator was planned but not built. Fagerström et al. [22] report a recurrent architecture detecting septic-shock onset up to twenty hours earlier than a Cox baseline on the same features and target, so the absence of a deep temporal comparator is one of two places where H6's result is incomplete. It establishes non-inferiority against gradient-boosted trees, not against every flexible estimator. The second gap is inferential rather than architectural: the contrast is tested one-sidedly, so what is excluded is the boosted trees outperforming the hazard model, and two-sided equivalence between the two is not established: the interval's lower limit, -0.0308 , falls outside the 0.02 margin (Section 5.10.8). A third qualification is the comparator's fitting procedure. Its boosting-round count is now chosen on a holdout carved out of the training stays (Protocol Amendment 8, Section 3.10.2); under the earlier procedure it was chosen on the test set, which flattered the comparator. Correcting it moved this contrast in the direction that favours the interpretable model, so the result should be read as one the uncorrected fits already supported and the correction sharpened, rather than as one the correction produced. Neither gap nor the correction affects the structural findings about the label.
7. **Limited feature set.** Clinical notes, procedure codes, and ventilator settings were not included. Richer features might improve discrimination but would not change the treatment-constitutionality finding.
8. **Part of the label-variant spread is definitional, not empirical.** Variants A–C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets;

some of the H1 label spread is guaranteed by construction and would appear even if the physiology were identical across patients. Section 5.6 separates the two by labelling each limb of the Sepsis-3 conjunction on its own, but the separation is post-hoc (Protocol Amendment 2) and rests on two variants, not on a systematic survey of alternative definitions.

9. **Discrimination is measured on the current-hour hazard, not on a six-hour horizon.** The models estimate the cause-specific hazard of onset in the current hour, and every AUROC, AUPRC and calibration figure in this thesis scores that hazard against an indicator marking the onset hour among the at-risk hours (Section 3.7). Three consequences follow. The reported discrimination is not comparable, quantity for quantity, with published AUROCs from studies that classify a binary “onset within the next k hours” label, whose positives are k times more numerous and whose task is correspondingly easier to score well on; comparisons with such figures in Chapter 2 are therefore indicative rather than exact. The covariates entering the prediction for the onset hour are contemporaneous with it, so the discrimination figures measure detection at the recorded onset rather than anticipation of it, and the quantity that speaks to anticipation is the lead-time distribution, which Section 5.4 shows to be uninformative at these alert rates. And a cumulative six-hour risk, which is what a deployed instrument would display, is a functional of this hazard together with the competing-cause hazards and is not reported here. A design that estimated and scored both would be stronger; the horizon-free hazard was chosen because the competing-risks formulation makes it the more primitive quantity, not because the six-hour risk is uninteresting.
10. **The primary model’s coefficient standard errors are those of a penalised estimator, and H2 still carries no uncertainty.** A cluster-robust sandwich covariance is computed for the fitted multinomial (Section 3.10.1), but two limits apply. Because the specification requires a ridge penalty to be identified at all, it is the covariance of the penalised estimator rather than of a maximum-likelihood one, so the usual asymptotic interpretation of a Wald interval is approximate here. And it bears on coefficients only: H2’s estimand is a count of covariates crossing a stability threshold across three separate fits, which is not a smooth function of one fit’s parameters and has no analytic variance, so H2 remains a descriptive count with no interval and no test. The Amendment 7 diagnostic narrows what can be said about that count without removing this limit: it establishes that 3 of the 6 flagged covariates move by less than their own estimation noise, so the pre-registered count is a loose upper bound on the instability present, but

the count itself still has no sampling distribution and the diagnostic's own ratio is not a test statistic (Section 5.10.4). Every performance interval in this thesis comes from the stay-level cluster bootstrap and is unaffected by either limit.

11. **The specification requires a penalty to be identified at all.** The unpenalised likelihood is completely separated on this design under every label variant (Section 3.10.1), so the reported fits are ridge-penalised. The penalty is small and no result depends on its value, test AUROC moves by less than 0.02 across a 20-point path, but the coefficients are shrunk estimates, not maximum-likelihood ones, and the H2 stability counts are properties of the penalised solution. Reporting an unpenalised fit here would have been worse, not better: its coefficients have no unique maximiser, so cross-label comparison of them is undefined.
12. **Feature construction and data cleaning are single-analyst work.** The covariate panel is assembled from raw chart and lab events by one pipeline, with no independent re-implementation to check it against. Two classes of defect are known to be possible in that setting and are guarded rather than eliminated: a declared predictor silently failing to reach a model because of a name mismatch, and implausible recorded values entering the fit. The pipeline now errors on any declared-but-absent feature rather than dropping it, and applies an explicit plausibility range to all nineteen physiological variables before forward-fill: MIMIC-IV chartevents contains values several orders of magnitude outside the physiological range, and such rows carry unbounded leverage. Neither guard is a substitute for independent replication of the feature pipeline.
13. **The pre-registered onset rule is not the standard one.** Variants A–C place the onset at t_{susp} and use the SOFA criterion only to gate membership, whereas Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [17] place it at $\min(t_{\text{susp}}, t_{\text{SOFA}})$. The pre-registered family therefore cannot address any question about onset *timing* on its own, H3 among them: under its rule an empty pre-treatment window is a theorem rather than a result (Proposition 5.1). Variant F supplies the measurement the pre-registered design cannot, but it is post-hoc (Protocol Amendment 4), it uses the full SOFA score including the vasopressor terms, so its onsets are free of the antibiotic-and-culture anchor but not of treatment in general, and its t_{SOFA} is searched only inside the 72-hour modelled panel, so a rise occurring earlier in the admission cannot be detected. A design that pre-specified both timing rules across all three variants would be stronger than one that adds the second afterwards.

14. **The treatment-independent label is not a validated construct.** Variant D is an operational deterioration label built for this comparison, not a consensus definition with an evidence base behind it, and its onsets have no external adjudication. Its label is also a threshold function of recorded physiology of which five of six components are model inputs, so its discrimination *level* is not comparable with Variant B's, only its disagreement with Variant B about which patients deteriorate, and when, carries weight.
15. **Specific to Sepsis-3 and to MIMIC-IV.** These findings apply to the Sepsis-3 definition on MIMIC-IV. SEP-1 and the CDC Adult Sepsis Event were not implemented; both are defined on antibiotic days, so neither would have tested treatment-independence, but each would give a different cohort and might behave differently in other respects. Other databases may behave differently again.
16. **Operating points are not externally recalibrated.** The thresholds in Section 5.4 are derived on the test set at which they are evaluated, so the reported PPV and false-alarm rates are optimistic: a prospectively fixed threshold would perform no better. The same applies to the maximum Utility Scores in Table 5.5, which take the best of sixty thresholds on the test set and are therefore upper bounds rather than estimates of what a deployed system would achieve; they are labelled exploratory for that reason. Selecting the threshold on the training years and transferring it would be the honest deployment estimate, and would be lower. The external operating points in Table 5.14 are likewise re-derived on eICU rather than transferred from MIMIC-IV, which isolates the discrimination loss but hides the additional loss a transferred threshold would incur.
17. **Alarm episodes are defined by contiguity, not by acknowledgement.** False-alarm episodes are counted as contiguous runs of alert hours. A real system would apply suppression, snoozing and escalation logic, all of which would reduce the episode counts in Table 5.9. The alert-hour rates are the conservative bound and the episode rates the optimistic one; the true burden lies between them.
18. **The subgroup discrimination analysis is underpowered, and its strata are administrative.** Two distinct limits apply and neither substitutes for the other. On precision: only 2 of the racial strata carry at least 100 sepsis onsets, the floor below which an AUROC is reported but not interpreted here (Section 5.9), and no contrast in family E1 survives false-discovery correction. That is a statement about precision, not about fairness: the data are consistent both with no disparity and with disparities

larger than any effect this thesis reports elsewhere. On construct validity: MIMIC-IV’s **race** field is a single administrative code assigned at registration, so the strata compared are documentation categories rather than populations. Broad and granular codes drawn from the same population appear as separate levels (“White” excludes stays coded “White – Other European”, and “Asian” excludes those coded “Asian – Chinese”), so the reference level is a residual category and the contrasts are not between disjoint groups. “Unknown” and “Unable to obtain” record the absence of an answer. The label-sensitivity result (Table 5.24) is the better powered of the two equity analyses because it is a stay-level proportion rather than a discrimination metric, but it inherits the same construct limitation, and its two significant strata are precisely the two missingness categories, which is why it is read in this thesis as a finding about documentation completeness rather than about race.

19. **The novelty claims rest on a narrative, not systematic, search.** The search protocol in Section 2.1 is documented and every included source was verified against a primary record, but no title-and-abstract screening pass over a full de-duplicated yield was performed and no second reviewer screened independently. The “to the best of our knowledge” claims for contributions C1 and C5 are therefore bounded by that coverage: a systematic review could surface a prior study combining multi-variant label sensitivity with subgroup equity analysis, and this search would not necessarily have found it. Both claims are claims about a *combination* of analyses; each component separately has clear precedent, which Section 2.1 identifies.
20. **Eight protocol amendments are post-hoc.** The inference layer (intervals and multiplicity control, Section 3.14), the alternative label limbs D and E (Section 3.6), the model identification and data-plausibility changes (Section 3.10.1), the standard onset timing of Variant F (Section 3.6.1), the action-derived feature ablation (Section 5.7), the subgroup interpretability floor (Section 5.9.1), the H2 uncertainty diagnostic (Section 5.10.4) and the comparator fit isolation (Section 3.10.2) were all specified after the pre-registered analyses had been run and inspected. None changes an estimand, threshold, direction or hypothesis, and H1–H6 and families E1/E2 are computed from unchanged inputs; but a reader should treat the verdicts as a pre-specified hypothesis set analysed under a retrospectively specified error-control plan, and the amendment results as exploratory. Four of the eight warrant a specific note, and they do not all point the same way. Amendment 4 enlarges family E3, which tightens rather than relaxes the correction applied within it. Amendment 6 *reduces* family E1 from thirty contrasts

to six, which is on its face the manoeuvre the inference plan's own rule against shrinking a family prohibits; it is admitted because the criterion is the stratum event count alone and is blind to outcome, because the reduction removed the family's only surviving discovery rather than creating one, and because nothing in E1 survives correction under either denominator. A reader who rejects that reasoning may read the thirty-contrast correction instead and will reach the same conclusion. Amendment 7 is the one whose result is *favourable* to the thesis's own argument, which is the case for treating it most sceptically. It reports that the covariates whose cross-label movement survives comparison with their standard errors are the action-derived ones, which is what this thesis would like to be true. Three things bound it. It changes no estimand: H2's count is reported first and unmodified, and the diagnostic never replaces it. It is not a test, for the reasons given in Section 5.10.4. And it is not independent evidence: it is computed from the same three fits as the ablation and shares the same flagged set, so it is a second reading of one body of evidence rather than a second experiment agreeing with the first. Amendment 8 is the only one of the eight that changes a reported estimate rather than a family, a diagnostic or a label. It corrects an implementation defect: the gradient-boosted comparator's boosting-round count was selected by early stopping against the test set, so a model-selection choice was being made on the rows the model was then scored on. The correction moves the early-stopping set into the training stays, split on the stay rather than the person-hour. Its direction was recorded before the re-run and is not uniformly convenient. Removing an optimistic bias from the comparator widens H6's gap in the interpretable model's favour, which is the direction this thesis argues for, so H6 is to be read as a finding the uncorrected fits already supported and the correction strengthened, not as one the correction produced. It leaves H1 essentially untouched, because the model-class range is bounded by the primary model above and qSOFA below and the boosted comparator sits at neither extreme.

21. **Two hypotheses admit no test statistic, and a third is tested on a substituted estimand.** H2 is a stability criterion rather than a contrast, and the fitted multinomial carries no variance estimate; H3 is degenerate because its outcome is absent from the evaluation window *by construction* rather than by chance, so no test of it could ever have been informative. Both are reported descriptively while still consuming a slot in the family size, which makes the correction on the remaining four hypotheses stricter, not weaker. H5 is a third and different case: its pre-registered form compares two proportional declines, which cannot be formed against an internal Utility Score of 0.025, so the AUROC limb alone occupies its slot and the adjusted p -

value belongs to that substituted contrast (Section 5.10.7). The substitution is disclosed in the pre-registration rather than reconstructed from the results, but it is a departure from the protocol and not merely a degeneracy in it, and a reader is entitled to treat H5's verdict as weaker evidence than H1's or H4's for that reason. For H1, H4 and H6 the correction determines nothing: their adjusted p -values are 1.000, 1.000 and 1.000, and each fails on its point estimate too. H5 is the exception on precision (its adjusted value is 0.551, and it fails on the width of its interval rather than on its direction), which is precisely why the substitution needed disclosing rather than absorbing.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis asked one question: in real-time sepsis prediction on MIMIC-IV, how much of the reported performance comes from the model versus the researcher’s choices about the label, the data split, and the evaluation metric?

The answer is not the one the study anticipated. Model class dominates discrimination. External transport comes second, moving AUROC several times more than the label definition does, though on an interval too wide to pin down. The label and the split design move it least. What the label determines is not how well a cohort can be predicted but *which cohort exists at all*, and, through its onset rule, whether prediction ahead of clinical action can be measured in the first place.

H1 (Label dominance): Not confirmed, and reversed. Changing the sepsis definition while keeping the model fixed changes AUROC by 0.0123. Changing the model while keeping the label fixed changes it by 0.1499, more than tenfold, and the interval on the difference, $[-0.1623, -0.1083]$, excludes zero. Cohen et al.’s [6] MIMIC-III finding does not reproduce on MIMIC-IV as a *discrimination* effect. It does reproduce as a cohort effect: the two narrowest readings agree on which stays are septic at only $\kappa = 0.267$ despite near-identical prevalence, so the label still determines who is counted, it simply does not determine how learnable they are.

H2 (Coefficient instability): Criterion met, descriptively. 6 of the 22 shared covariates are unstable across label variants: 2 reverse sign (bilirubin and vasopressor exposure) and 5 change magnitude at least twofold, three of the latter being missingness indicators rather than physiological values. Vasopressor exposure meets both criteria, which is why the two counts sum to seven across six

covariates. H2 is a stability criterion rather than a contrast, so no test statistic is defined for it, and the diagnostic of Section 5.10.4 shows the count to be a loose upper bound: 3 of the 6 flagged covariates, both sign reversals among them, move by less than their own estimation noise. What survives that comparison is confined to the order indicators, so the defensible reading is narrower than the count: an effect estimate for whether a test was ordered cannot be interpreted without knowing the exact label definition.

H3 (Treatment leakage): Definitional, plus its empirical counterpart.

The pre-treatment window contains 0 sepsis onset events across 179,003 person-hours, but that zero is entailed by the onset rule the pre-registered variants use, not measured (Proposition 5.1), so the pre-registered contrast was never testable. The empirical question was answered instead by relabelling the identical stays under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset of Seymour et al. and the PhysioNet/CinC 2019 Challenge (Variant F, Protocol Amendment 4): 1,738 onsets move earlier and 71 fall before treatment, identified at 0.8081 AUROC. A published MIMIC-IV real-time sepsis model measures anticipation of physiological deterioration only to the extent its onset rule allows an onset to precede clinical action; under a conjunction rule, it cannot.

H4 (Split optimism): Not confirmed. The random split does not flatter the model. For the primary model it scores 0.7424 against 0.7590 on the temporal split, a difference of -0.0165 (95% CI -0.0372 – 0.0040) in the direction opposite to the hypothesis, and every model fitted under both splits shows the same sign. The two test sets differ in composition as well as in design (the temporal arm is a single admission era, the random arm a sample across all of them), so this is not a clean optimism estimate; what it does establish is that a random split cannot be assumed to be the more permissive choice.

H5 (Metric divergence): Not confirmed, and superseded by a starker finding. Neither arm supports the hypothesis, but the AUROC arm fails narrowly and in an informative way. External discrimination is 0.6960 on eICU-CRD against 0.7590 internally, a degradation of 0.0630: the predicted direction, and roughly three-quarters of the 0.085 anticipated from Moor et al. [21]. The interval (-0.0215 – 0.1563) nonetheless covers zero, because the Sepsis-3 anchor is recoverable in only 1.55% of eICU-CRD stays and the estimate rests on 29 events. This is the one hypothesis a better-powered external cohort might have confirmed, and the antibiotic-only arm (6,140 events, a different target) shows the same degradation with the same ordering across labels. The study establishes that discrimination declines on transport and cannot establish by how much.

The utility arm needs no such qualification, because it is settled internally. The proportional decline the hypothesis asked for cannot be formed at all, there being no internal utility to decline from: swept across sixty thresholds, the best any configuration achieves under the primary label is 0.031, at 51.8 false alerts per true one against a benchmark of 1.4. The utility collapse Wang et al. [4] documented across ninety-one models therefore has its counterpart *internally* on MIMIC-IV under an honest time-based split, there was scarcely any margin left to collapse. Their broader point survives in strengthened form: on the cohort where the label is fully observed, AUROC and clinical utility are close to uncorrelated, discrimination sitting in the high 0.7s while usable clinical value never materialises at any threshold.

H6 (Model-class null): Not rejected; non-inferiority established. The interpretable hazard model is not outperformed by gradient-boosted trees: 0.7590 against 0.7424, a difference of -0.0165 with an interval of $[-0.0308, -0.0012]$. The upper limit sits inside the pre-registered 0.02 margin, so the trees are not better by more than it; the lower limit falls outside, so two-sided equivalence is *not* established and a difference larger than the margin, in the interpretable model's favour, remains consistent with these data. The interval also lies wholly below zero, so the interpretable model is ahead rather than merely not behind, though by an amount of the same order as the label spread. This is the strongest positive result in the study, and it is strongest in exactly the direction that matters: on tabular ICU time-series of this kind the accuracy case for preferring an opaque model is not supported. What little AUROC separates them buys nothing clinically. At their respective best thresholds the two models reach 0.031 and 0.025 on the Utility Score, a gap of well under a hundredth of a point on a scale bounded by 1, and both sit within a few per cent of the no-alert strategy. The two metrics in fact order the pair oppositely, the interpretable model ahead on discrimination and the boosted one ahead on utility, which is the clearest illustration in the study of how little either difference means: neither is large enough to choose on.

7.2 The Central Lesson

The utility collapse documented by Wang et al. [4] is not a model problem. It is a label problem. The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action for membership:** no patient is *labelled* septic without a qualifying antibiotic-culture pair, under any timing rule. The further claim that no onset can *precede* that pair is a property of the

conjunction-based onset rule alone: timed as the standard specifies, Sepsis-3 records 71 pre-treatment onsets on the same stays, and a treatment-independent deterioration label records 1,362. The emptiness belonged to the implementation, not to the patients.

2. **Is sensitive to how it is defined:** the number of sepsis cases nearly doubles across plausible Sepsis-3 interpretations, and the two narrowest readings agree on which stays are septic at only $\kappa = 0.267$. Discrimination is largely insensitive to this (0.0123 of AUROC); every count, rate and denominator built on the label is not.
3. **Produces unusable alert behaviour:** no model, under any pre-registered label variant, has a threshold at which it both beats the no-alert strategy by a clinically meaningful margin and raises an alarm volume a ward could absorb. The best configuration under the primary label reaches 0.031 at 51.8 false alerts per true one. The post-hoc variants reach higher ceilings, Variant D's being the highest, but they label different targets, and each still attains its own optimum at an alert burden several times the benchmark.

These three problems are independent of model complexity. A better model cannot fix a label that is constituted by the very clinical actions it is supposed to predict ahead of. The solution requires changing the label.

Splitting the definition sharpens point (2) and qualifies it (Section 5.6). It is qualified because the onset *time* under every pre-registered Sepsis-3 variant is fixed entirely by the treatment anchor, the organ-dysfunction criterion decides which stays are labelled and moves no onset by an hour, so part of the variation between variants is guaranteed by the definition rather than found in the data. It is sharpened because a label built from physiology alone, owing nothing to treatment timing, agrees with Sepsis-3 barely better than chance ($\kappa = 0.057$). The label effect is therefore real, not merely definitional.

The same comparison relocates the difficulty, and not where the conjunction framing expected. The treatment anchor on its own is about as predictable as Sepsis-3 built on top of it, so requiring both limbs is not what makes the label hard. The strength of that statement depends on which model is asked. The anchor-attributable fraction is 1.041 [0.990, 1.097] for the primary model, an interval that includes 1, and 1.099 [1.037, 1.164] for the gradient-boosted comparator, an interval that does not. What both agree on is the sign and the size: the anchor limb is no harder than the conjunction, and on the boosted model it is slightly easier. Neither supports the reading the conjunction framing predicted, that Sepsis-3 is harder than either of its limbs.

Sepsis-3 with a better clock is easier again (0.7951 against 0.7590). Variant F holds cohort membership fixed and changes only the onset timestamp, which is

what makes that contrast interpretable. It is not computed on an identical event set: moving an onset earlier can move it inside the 72-hour horizon, so F carries 187 stays that conjunction timing discards (Section 5.6.3). Variant D’s higher figure is not offered in support here: its discrimination level is set in part by its own construction and is not comparable with Variant B’s (Section 6.4); what D contributes is its disagreement with Sepsis-3, not its AUROC. What resists prediction is not the conjunction but the moment at which the conjunction is stamped: the clinician’s action rather than the patient’s deterioration.

7.3 Contributions

1. **C1 (Label variance, separated from discrimination):** Three Sepsis-3 readings applied to a discrete-time competing-risks model on MIMIC-IV (65,078 stays, against the roughly two thousand admissions of Cohen et al.’s MIMIC-III comparison), with the label and model-class spreads estimated on the same test rows and each carrying an interval. That combination is, to the best of our knowledge, not previously reported, and it is what separates two effects the literature treats as one: label choice moves cohort membership substantially ($\kappa = 0.267$ between the narrowest readings) while moving AUROC by only 0.0123.
2. **C2 (Treatment leakage separated: membership from onset timing):** A proof that the conjunction-based Sepsis-3 onset rule makes the pre-treatment window empty by construction, paired with the measurement that the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule recovers 71 pre-treatment onsets on the identical stays at 0.8081 AUROC. This confirms Kamran et al.’s [7] concern for label membership and corrects its over-extension to label timing.
3. **C3 (Utility ceiling):** A threshold-swept Utility Score for every model and label variant, establishing that the ceiling, not merely the score at a conventional cut-off, sits within a few per cent of the no-alert strategy under the primary label, and that reaching even that ceiling costs 51.8 false alerts per true one. This extends Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison and removes the operating-point objection that would otherwise apply to it.
4. **C4 (Coefficient instability):** 6 of the 22 shared covariates are unstable across labels (2 by sign and 5 by magnitude, vasopressor exposure meeting both), supporting Lauritsen et al.’s [10] framing-instability claim on MIMIC-IV for specific effect estimates, measured against an identified fit. Referred

to their own cluster-robust standard errors, 3 of the 6 survive, and all 3 are order indicators (Section 5.10.4).

5. **C5 (Equity-by-label):** Subgroup discrimination and subgroup label sensitivity measured on the same strata in the same onset-prediction cohort, both with cluster-bootstrap intervals under false-discovery control: a pairing this study's search did not find reported elsewhere, though each component separately has precedent (Section 2.1). The label-sensitivity contrasts survive correction and the discrimination contrasts do not, so the defensible finding is about the completeness of the demographic record rather than about race; reporting either analysis without the other would have inverted which conclusion the data support.
6. **C6 (Reproducible pipeline):** A pre-registered, TRIPOD+AI-conformant pipeline with fixed seeds and a single orchestration script, which reproduces every reported estimate exactly from the raw databases (with the one documented exception of the gradient-boosted comparator, whose fitting is not bit-reproducible across library builds and whose rows therefore serve as the designated consistency check on any re-run), and which emits the numbers in this document as generated constants so that text and code cannot diverge (Section 4.15.2, Appendix A).
7. **C7 (Definitional vs. empirical label variance separated):** A treatment-independent deterioration label, tested alongside the three Sepsis-3 variants, shows that the label effect is not an artefact of how one definition is read: it agrees with Sepsis-3 at $\kappa = 0.057$ and matches its onset hour in 6.5% of shared stays. The same comparison shows that the treatment anchor alone is about as predictable as Sepsis-3 built on top of it ($\phi_{\text{anchor}} = 1.041 [0.990, 1.097]$ for the primary model, an interval including 1; $1.099 [1.037, 1.164]$ for the boosted comparator, which excludes it), and that the pre-treatment window empty of Sepsis-3 onsets contains 1,362 treatment-independent deterioration onsets identified at 0.8498 AUROC, and 71 Sepsis-3 onsets once the standard onset timing is restored. This relocates the difficulty of sepsis prediction from the physiology to the definition, and within the definition to the onset rule (Section 5.6).

7.4 Future Work

1. **Better sepsis labels.** Section 5.6 takes the first step (a deterioration label built from physiology alone, with no treatment timestamp), and shows it identifies substantially different patients at substantially different hours.

But that label is an operational construct built for this comparison, not a validated definition, and its components overlap the model's own inputs. The work remaining is a sepsis definition grounded in objective measurements such as lactate trajectories and organ-function biomarkers, adjudicated independently of the model's feature set, which would eliminate the treatment-constitutionality problem and allow genuine pre-onset prediction.

2. **Non-Sepsis-3 consensus definitions.** SEP-1 and the CDC Adult Sepsis Event were not implemented. Neither would have tested treatment-independence, both are defined on antibiotic days, but each yields a different cohort, and Dutta et al. [9] report that performance moves substantially between them. Adding them would extend the label-variance decomposition across definitions rather than within one.
3. **Prospective validation.** Test the new label in real clinical settings where sepsis onset can be determined by a protocol-blinded assessor, independent of treatment decisions.
4. **Utility-optimised training.** The thresholds reported here are optimised post hoc, but the models themselves are not: each is fitted to a likelihood or a squared-error surrogate and only afterwards read at the threshold that happens to maximise utility. Training directly against the Utility Score (as a differentiable surrogate loss, or through a cost-sensitive reweighting of the person-hour rows) would change what the model learns rather than only how it is read, and is the natural next step given how shallow the swept optima proved to be. The gap between such a model and the post-hoc ceiling in Table 5.5 would also quantify how much of the failure is representational rather than operational.
5. **Multi-hospital evaluation.** The alert burden, 51.8 false alerts per true alert even at the best model's best operating point, is a deployment problem, not a statistical one. Determining whether any model can be deployed at a practically acceptable burden requires prospective multi-centre testing with clinical-staff feedback.

7.5 Closing Remark

The premise this study was built on did not survive the study. It was designed around the expectation that the analyst's discretionary choices would rival the choice of model as a source of performance variation, and that expectation fails: the label moves discrimination by 0.0123 against the model class's 0.1499, and

the split method moves it not at all. Of the four hypotheses that could be tested, none is confirmed, and two fail in the direction opposite to the one predicted. That outcome is reported here in the same detail it would have received had it gone the other way, which is the reason the pre-registration and the inference layer were worth their cost: they fix what the answer means before the answer is known, and on point estimates alone this thesis would have claimed the opposite of what it can defend.

The premise inverted; the motivating problem did not. What the decomposition establishes is that the label's consequence does not run through discrimination, which is where the field looks for it. It runs through cohort membership, the two narrowest readings agreeing at $\kappa = 0.267$ on which patients are septic, and through onset timing, which fixes what a real-time model can be asked to anticipate at all. Those are the channels the reported metric is blind to, which is why a literature that has optimised AUROC for a decade has not converged on models that help.

The central finding, that the Sepsis-3 label on MIMIC-IV is constituted by clinical action, is not a failure of the model. It is a finding about the label itself, and it is the most important contribution this study makes.

The last part of that finding required looking outside the pre-registered variants. The pre-treatment window is not empty of predictable deterioration, it contains 1,362 onsets under a treatment-independent label at 0.8498 AUROC, and 71 onsets under Sepsis-3 itself once the onset is timed as $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rather than as the suspicion time alone. It was empty here because of an onset rule, and that rule is not peculiar to this thesis. The signal clinicians would want a model to find is present in the data; a widely used way of writing the label down is what makes it unrepresentable.

References

- [1] M. Singer, C. S. Deutschman, C. W. Seymour, et al., “The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3),” *JAMA*, vol. 315, no. 8, pp. 801–810, 2016. DOI: [10.1001/jama.2016.0287](https://doi.org/10.1001/jama.2016.0287).
- [2] K. E. Rudd, S. C. Johnson, K. M. Agesa, et al., “Global, regional, and national sepsis incidence and mortality, 1990–2017: Analysis for the Global Burden of Disease Study 2017,” *The Lancet*, vol. 395, no. 10219, pp. 200–211, 2020. DOI: [10.1016/S0140-6736\(19\)32989-7](https://doi.org/10.1016/S0140-6736(19)32989-7).
- [3] C. W. Seymour, F. Gesten, H. C. Prescott, et al., “Time to treatment and mortality during mandated emergency care for sepsis,” *New England Journal of Medicine*, vol. 376, no. 23, pp. 2235–2244, 2017. DOI: [10.1056/NEJMoa1703058](https://doi.org/10.1056/NEJMoa1703058).
- [4] Z. Wang, W. Wang, C. Sun, J. Li, S. Xie, J. Xu, et al., “A methodological systematic review of validation and performance of sepsis real-time prediction models,” *npj Digital Medicine*, vol. 8, no. 1, p. 190, 2025. DOI: [10.1038/s41746-025-01587-1](https://doi.org/10.1038/s41746-025-01587-1).
- [5] A. Wong et al., “Multicenter prospective validation of an updated proprietary sepsis prediction model,” *JAMA Network Open*, vol. 9, no. 2, e260181, 2026. DOI: [10.1001/jamanetworkopen.2026.0181](https://doi.org/10.1001/jamanetworkopen.2026.0181).
- [6] S. N. Cohen, J. Foster, P. Foster, H. Lou, T. Lyons, S. Morley, et al., “Subtle variation in Sepsis-III definitions markedly influences predictive performance within and across methods,” *Scientific Reports*, vol. 14, no. 1, p. 1920, 2024. DOI: [10.1038/s41598-024-51989-6](https://doi.org/10.1038/s41598-024-51989-6).
- [7] F. Kamran et al., “Evaluation of sepsis prediction models before onset of treatment,” *NEJM AI*, vol. 1, no. 3, AIoa2300032, 2024. DOI: [10.1056/AIoa2300032](https://doi.org/10.1056/AIoa2300032).
- [8] L. L. Guo et al., “Evaluation of domain generalization and adaptation on improving model robustness to temporal dataset shift in clinical medicine,” *Scientific Reports*, vol. 12, no. 1, p. 2726, 2022. DOI: [10.1038/s41598-022-06484-1](https://doi.org/10.1038/s41598-022-06484-1).

- [9] S. Dutta, R. McMurtry, M. C. Tasi, et al., “Performance of a sepsis prediction model across different sepsis definitions,” *JAMA Network Open*, vol. 9, no. 4, e265599, 2026. DOI: [10.1001/jamanetworkopen.2026.5599](https://doi.org/10.1001/jamanetworkopen.2026.5599).
- [10] S. M. Lauritsen, B. Thiesson, M. J. Jørgensen, et al., “The framing of machine learning risk prediction models illustrated by evaluation of sepsis in general wards,” *npj Digital Medicine*, vol. 4, no. 1, p. 158, 2021. DOI: [10.1038/s41746-021-00529-x](https://doi.org/10.1038/s41746-021-00529-x).
- [11] H. Wang, Y. Li, A. Naidech, and Y. Luo, “Comparison between machine learning methods for mortality prediction for sepsis patients with different social determinants,” *BMC Medical Informatics and Decision Making*, vol. 22, no. S2, p. 156, 2022. DOI: [10.1186/s12911-022-01871-0](https://doi.org/10.1186/s12911-022-01871-0).
- [12] G. S. Collins, K. G. M. Moons, P. Dhiman, et al., “TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ*, vol. 385, e078378, 2024. DOI: [10.1136/bmj-2023-078378](https://doi.org/10.1136/bmj-2023-078378).
- [13] K. G. M. Moons, J. A. A. Damen, T. Kaul, L. Hooft, C. Andaur Navarro, P. Dhiman, et al., “PROBAST+AI: An updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods,” *BMJ*, vol. 388, e082505, 2025. DOI: [10.1136/bmj-2024-082505](https://doi.org/10.1136/bmj-2024-082505).
- [14] M. J. Page et al., “The PRISMA 2020 statement: An updated guideline for reporting systematic reviews,” *BMJ*, vol. 372, n71, 2021. DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71).
- [15] A. E. W. Johnson, J. Aboab, J. D. Raffa, et al., “A comparative analysis of sepsis identification methods in an electronic database,” *Critical Care Medicine*, vol. 46, no. 4, pp. 494–499, 2018. DOI: [10.1097/CCM.0000000000002965](https://doi.org/10.1097/CCM.0000000000002965).
- [16] A. Johnson et al., “MIMIC-IV,” *PhysioNet*, Oct. 2024, Version 3.1. DOI: [10.13026/kpb9-mt58](https://doi.org/10.13026/kpb9-mt58).
- [17] M. A. Reyna, C. S. Josef, R. Jeter, et al., “Early prediction of sepsis from clinical data: The PhysioNet/Computing in Cardiology Challenge 2019,” *Critical Care Medicine*, vol. 48, no. 2, pp. 210–217, 2020. DOI: [10.1097/CCM.0000000000004145](https://doi.org/10.1097/CCM.0000000000004145).
- [18] S. Nemati, A. Holder, F. Razmi, M. D. Stanley, G. D. Clifford, and T. G. Buchman, “An interpretable machine learning model for accurate prediction of sepsis in the ICU,” *Critical Care Medicine*, vol. 46, no. 4, pp. 547–553, 2018. DOI: [10.1097/CCM.0000000000002936](https://doi.org/10.1097/CCM.0000000000002936).

- [19] K. E. Henry, D. N. Hager, P. J. Pronovost, and S. Saria, “A targeted real-time early warning score (TREWScore) for septic shock,” *Science Translational Medicine*, vol. 7, no. 299, 299ra122, 2015. DOI: [10.1126/scitranslmed.aab3719](https://doi.org/10.1126/scitranslmed.aab3719).
- [20] R. Adams, K. E. Henry, A. Sridharan, et al., “Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis,” *Nature Medicine*, vol. 28, no. 7, pp. 1455–1460, 2022. DOI: [10.1038/s41591-022-01894-0](https://doi.org/10.1038/s41591-022-01894-0).
- [21] M. Moor et al., “Predicting sepsis using deep learning across international sites: A retrospective development and validation study,” *eClinicalMedicine*, vol. 62, p. 102124, 2023. DOI: [10.1016/j.eclinm.2023.102124](https://doi.org/10.1016/j.eclinm.2023.102124).
- [22] J. Fagerström, M. Bång, D. Wilhelms, and M. S. Chew, “LiSep LSTM: A machine learning algorithm for early detection of septic shock,” *Scientific Reports*, vol. 9, p. 15132, 2019. DOI: [10.1038/s41598-019-51219-4](https://doi.org/10.1038/s41598-019-51219-4).
- [23] E. Christodoulou, J. Ma, G. S. Collins, et al., “A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models,” *Journal of Clinical Epidemiology*, vol. 110, pp. 12–22, 2019. DOI: [10.1016/j.jclinepi.2019.02.004](https://doi.org/10.1016/j.jclinepi.2019.02.004).
- [24] M. Moor, B. Rieck, M. Horn, C. R. Jutzeler, and K. Borgwardt, “Early prediction of sepsis in the ICU using machine learning: A systematic review,” *Frontiers in Medicine*, vol. 8, p. 607952, 2021. DOI: [10.3389/fmed.2021.607952](https://doi.org/10.3389/fmed.2021.607952).
- [25] L. M. Fleuren, T. L. T. Klausch, C. L. Zwager, et al., “Machine learning for the prediction of sepsis: A systematic review and meta-analysis of diagnostic test accuracy,” *Intensive Care Medicine*, vol. 46, no. 3, pp. 383–400, 2020. DOI: [10.1007/s00134-019-05872-y](https://doi.org/10.1007/s00134-019-05872-y).
- [26] C. W. Seymour, V. X. Liu, T. J. Iwashyna, et al., “Assessment of clinical criteria for sepsis: For the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3),” *JAMA*, vol. 315, no. 8, pp. 762–774, 2016. DOI: [10.1001/jama.2016.0288](https://doi.org/10.1001/jama.2016.0288).
- [27] A. Wong, E. Otles, J. P. Donnelly, A. Krumm, J. McCullough, O. DeTroyer-Cooley, et al., “External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients,” *JAMA Internal Medicine*, vol. 181, no. 8, pp. 1065–1070, 2021. DOI: [10.1001/jamainternmed.2021.2626](https://doi.org/10.1001/jamainternmed.2021.2626).

- [28] D. G. Ostermayer, B. Braunheim, A. M. Mehta, J. Ward, S. Andrabi, and A. M. Sirajuddin, “External validation of the Epic sepsis predictive model in 2 county emergency departments,” *JAMIA Open*, vol. 7, no. 4, ooa133, 2024. DOI: [10.1093/jamiaopen/ooae133](https://doi.org/10.1093/jamiaopen/ooae133).
- [29] B. Van Calster, D. J. McLernon, M. van Smeden, et al., “Calibration: The Achilles heel of predictive analytics,” *BMC Medicine*, vol. 17, no. 1, p. 230, 2019. DOI: [10.1186/s12916-019-1466-7](https://doi.org/10.1186/s12916-019-1466-7).
- [30] J. Kasal, Z. Jovanovic, G. Clermont, et al., “Comparison of Cox and Gray’s survival models in severe sepsis,” *Critical Care Medicine*, vol. 32, no. 3, pp. 700–707, 2004. DOI: [10.1097/01.CCM.0000114819.37569.4B](https://doi.org/10.1097/01.CCM.0000114819.37569.4B).
- [31] M. Resche-Rigon, E. Azoulay, and S. Chevret, “Evaluating mortality in intensive care units: Contribution of competing risks analyses,” *Critical Care*, vol. 10, no. 1, R5, 2006. DOI: [10.1186/cc3921](https://doi.org/10.1186/cc3921).
- [32] J. P. Fine and R. J. Gray, “A proportional hazards model for the subdistribution of a competing risk,” *Journal of the American Statistical Association*, vol. 94, no. 446, pp. 496–509, 1999. DOI: [10.1080/01621459.1999.10474144](https://doi.org/10.1080/01621459.1999.10474144).
- [33] D. Schoenfeld, “Survival methods, including those using competing risk analysis, are not appropriate for intensive care unit outcome studies,” *Critical Care*, vol. 10, no. 1, p. 103, 2006. DOI: [10.1186/cc3949](https://doi.org/10.1186/cc3949).
- [34] H. C. van Houwelingen, “Dynamic prediction by landmarking in event history analysis,” *Scandinavian Journal of Statistics*, vol. 34, no. 1, pp. 70–85, 2007. DOI: [10.1111/j.1467-9469.2006.00529.x](https://doi.org/10.1111/j.1467-9469.2006.00529.x).
- [35] D. Rizopoulos, G. Molenberghs, and E. M. E. H. Lesaffre, “Dynamic predictions with time-dependent covariates in survival analysis using joint modeling and landmarking,” *Biometrical Journal*, vol. 59, no. 6, pp. 1261–1276, 2017. DOI: [10.1002/bimj.201600238](https://doi.org/10.1002/bimj.201600238).
- [36] R. B. D’Agostino, M.-L. Lee, A. J. Belanger, et al., “Relation of pooled logistic regression to time dependent Cox regression analysis: The Framingham Heart Study,” *Statistics in Medicine*, vol. 9, no. 12, pp. 1501–1515, 1990. DOI: [10.1002/sim.4780091214](https://doi.org/10.1002/sim.4780091214).
- [37] J. S. Ngwa, H. J. Cabral, D. M. Cheng, et al., “A comparison of time dependent Cox regression, pooled logistic regression and cross sectional pooling with simulations and an application to the Framingham Heart Study,” *BMC Medical Research Methodology*, vol. 16, no. 1, p. 148, 2016. DOI: [10.1186/s12874-016-0248-6](https://doi.org/10.1186/s12874-016-0248-6).

- [38] R. Heyard, J.-F. Timsit, W. I. Essaied, and L. Held, “Dynamic clinical prediction models for discrete time-to-event data with competing risks — a case study on the OUTCOMEREA database,” *Biometrical Journal*, vol. 61, no. 3, pp. 514–534, 2019. DOI: [10.1002/bimj.201700259](https://doi.org/10.1002/bimj.201700259).
- [39] R. Heyard, J.-F. Timsit, and L. Held, “Validation of discrete time-to-event prediction models in the presence of competing risks,” *Biometrical Journal*, vol. 62, no. 3, pp. 643–657, 2020. DOI: [10.1002/bimj.201800293](https://doi.org/10.1002/bimj.201800293).
- [40] L. Evans, A. Rhodes, W. Alhazzani, et al., “Surviving Sepsis Campaign: International guidelines for management of sepsis and septic shock 2021,” *Intensive Care Medicine*, vol. 47, no. 11, pp. 1181–1247, 2021. DOI: [10.1007/s00134-021-06506-y](https://doi.org/10.1007/s00134-021-06506-y).
- [41] L. Mellhammar et al., “Scores for sepsis detection and risk stratification — construction of a novel score using a statistical approach and validation of RETTS,” *PLOS ONE*, vol. 15, no. 2, e0229210, 2020. DOI: [10.1371/journal.pone.0229210](https://doi.org/10.1371/journal.pone.0229210).
- [42] A. E. W. Johnson et al., “MIMIC-IV, a freely accessible electronic health record dataset,” *Scientific Data*, vol. 10, no. 1, 2023. DOI: [10.1038/s41597-022-01899-x](https://doi.org/10.1038/s41597-022-01899-x).
- [43] T. Pollard et al., “PhysioNet as a global platform for biomedical research,” *Nature Health*, 2026. DOI: [10.1038/s44360-026-00096-z](https://doi.org/10.1038/s44360-026-00096-z).
- [44] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi, “The eICU Collaborative Research Database, a freely available multi-centre database for critical care research,” *Scientific Data*, vol. 5, p. 180178, 2018. DOI: [10.1038/sdata.2018.178](https://doi.org/10.1038/sdata.2018.178).
- [45] A. J. Vickers and E. B. Elkin, “Decision curve analysis: A novel method for evaluating prediction models,” *Medical Decision Making*, vol. 26, no. 6, pp. 565–574, 2006. DOI: [10.1177/0272989X06295361](https://doi.org/10.1177/0272989X06295361).
- [46] R. D. Riley et al., “Minimum sample size for developing a multivariable prediction model: PART II — binary and time-to-event outcomes,” *Statistics in Medicine*, vol. 38, no. 7, pp. 1276–1296, 2019. DOI: [10.1002/sim.7992](https://doi.org/10.1002/sim.7992).

Appendix A

Source Code and Reproduction

A.1 Repository

The complete implementation of this study is held in a version-controlled public repository:

Source code repository

<https://github.com/ahadarif007/RESEARCH>

Source code licensed under the Apache License 2.0.

No patient data is included.

The Apache-2.0 grant covers the pipeline source written for this study. It does not extend to material this work does not own, and the repository’s **NOTICE** file draws that boundary explicitly: the institutional thesis template and its brand assets are reproduced under their owner’s rights and are excluded; the two clinical databases are excluded absolutely, being redistributed by neither this repository nor its licence; and the thesis text and its figures, being scholarly prose rather than software, are made available for citation and academic reuse with attribution rather than under a software licence.

Where a routine implements a published method (the SOFA score, the Sepsis-3 operational criteria, NEWS2, qSOFA, SIRS, the PhysioNet Utility Score, the sample-size criteria, pooled logistic regression, decision-curve analysis) the implementation is original to this work and written from the published definition, and the method is attributed to its authors both in **NOTICE** and at the point of use in this thesis. No implementation was copied or adapted from a third party’s source.

The repository contains **code only**. No MIMIC-IV or eICU-CRD record, derived person-hour table, or fitted model object is included: the **data/**, **processed_data/** and **output/** directories are excluded from version control by

design, because the PhysioNet Credentialed Data Use Agreement (Section 3.3) prohibits redistribution of the underlying data. Exclusion is enforced by `.gitignore` and was verified against the tracked file list before release.

A.2 Repository Layout

Table A.1: Repository layout.

Path	Contents
<code>00_preregistration.md</code>	The pre-registration, including the G0 analysis gate and Protocol Amendments 1–8, each dated and disclosed as post-hoc
<code>config.R</code>	All constants: label-variant parameters, feature lists, plausibility ranges, seeds, bootstrap sizes, multiplicity families, event floors. No numeric constant is defined anywhere else
<code>utils.R</code>	Shared helpers, including <code>require_features()</code> (errors on a declared-but-absent predictor) and <code>check_multinom_fit()</code>
<code>clinical_scores.R</code>	SOFA, NEWS2, qSOFA and SIRS implementations
<code>01_setup.Rmd</code> <code>12_inference.Rmd</code>	... The twelve pipeline stages described in Chapter 4, in execution order
<code>run_pipeline.sh</code> , <code>run_pipeline.R</code>	Orchestration. Runs all stages, or a named subset, and regenerates the thesis constants on success
<code>thesis_constants.R</code>	Reads the result tables and emits every numerical value used in this document as a $\text{\LaTeX}$ macro (Section A.3)
<code>utility_score.py</code>	Reference implementation of the Utility Score, maintained as a mirror of the R scoring rule and cross-checked against it
<code>check_consistency.R</code>	Cross-file assertions over the result tables: arithmetic identities that must hold between quantities written by different stages, and predicates for the comparative claims this document states in words

Path	Contents
<code>check_thesis_numbers.R</code>	Checks that every result-shaped number in the thesis sources is either generated, cited to prior work, or recorded as a design constant
<code>check_thesis_margins.R</code>	Measures the built PDF for content passing the text block, which $\text{\LaTeX}$ does not report for a table
<code>output/</code>	Created at run time: <code>processed_data/</code> , <code>figures/</code> , <code>logs/</code> . Not tracked

A.3 Reproducing the Results

Prerequisites. R 4.5 or later with the packages listed in the repository, Python 3.14 with `pandas` 3.0.4 for the Utility Score mirror, and a $\text{\TeX}$ distribution providing `xelatex` and `pdflatex/biber`. The two source databases must be obtained independently through PhysioNet credentialing: MIMIC-IV v3.1 and eICU-CRD v2.0. They are placed under `data/` and located automatically, or pointed at with the `MIMIC_RESEARCH_ROOT` environment variable.

Full run. A complete execution is a single command,

```
./run_pipeline.sh
```

which runs stages 01–12 in order and, on success, regenerates the thesis constants and figures. Individual stages or phases can be run by number. Every stochastic step is seeded, so a repeated run reproduces the reported estimates exactly; the one exception is noted below.

How the thesis stays aligned with the pipeline. No quantity estimated by this study is typed by hand into this document; the exceptions are figures quoted from prior work, constants fixed by the pre-registration, and prose roundings of a macro stated exactly nearby. On completion, `thesis_constants.R` reads the result tables and writes every reported quantity as a macro into `pipeline_constants.tex`, which the thesis includes; it also copies the generated figures into the document’s image directory. A quantity the pipeline did not produce typesets as a conspicuous marker rather than as a stale number, so a partial run cannot silently leave an outdated value in the text. This is the mechanism by which the numbers in Chapters 5 and 6 are guaranteed to be the numbers the code produced.

Determinism. All seeds are fixed in `config.R` and the train/test partition is defined by admission year rather than by sampling, so it is invariant. Gradient-boosted tree fitting is the one step that is not bit-reproducible across differing library builds; the pipeline therefore reports the gradient-boosted results as an explicit consistency check on any re-run, and every other estimate is invariant to it.

A.4 Ethical and Data-Access Statement

Both databases are de-identified and publicly available to credentialed researchers under the PhysioNet Credentialed Health Data Use Agreement. The analysis is retrospective and observational, involves no patient contact and no intervention, and required no additional ethical approval beyond the data use agreement, whose terms, including the prohibition on redistribution and on attempted re-identification, were observed throughout. Section 3.3 sets out the full statement.

Appendix B

Pipeline Stage Reference

Chapter 4 describes what each stage does and why the architecture takes the shape it does. This appendix records the file-level interface: what each stage reads and what it writes; the mapping from an analysis to its inference status is in the analysis register (Section 5.11) rather than repeated here. Paths are relative to `sepsis/project/`; all intermediate tables are Parquet, with a CSV mirror written for any result table small enough to read directly.

Stage	Reads	Writes
01	MIMIC-IV source tables	n/a (validation only; halts on a missing table)
02	hosp/, icu/ source tables	02_cohort_{A..F}, 02_label_variance_table, 02_alt_label_variance_table
03	02_cohort_*; chartevents, labevents, inpatientevents, outputevents (streamed)	03_person_hours_{A..F}, 03_early_onset_shift
04	03_person_hours_* (three columns)	04_sample_size
05	03_person_hours_*	05_model_primary_*, 05_coef_all_variants, 05_predictions_primary_*, 05_h2_stability_*, *_ablated counterparts
06	03_person_hours_*	06_model_gbt_*, 06_predictions_comparators_*, 06_gbt_importance_*, 06_schoenfeld_*; ablated GBT counterparts
07	05_*, 06_* predictions	07_metric_results, _random, _altlabel, 07_utility_sweep, 07_utility_max, 07_ablation_metrics
08	eICU-CRD source tables; frozen 05_/06_ models	08_external_validation_results, _abxonly, 08_eicu_predictions_*, 08_eicu_coverage, 08_event_rate_reconciliation, 08_h5_internal_external
09	07_metric_results*	09_variance_decomposition_table, 09_hypothesis_verdicts, 09_label_anchor_attribution, 09_label_agreement, 09_pretreatment_onsets
10	03_person_hours_*, 05_/06_ predictions	Per-stratum performance and label-sensitivity tables
11	05_/06_/08_ predictions	11_calibration_summary, 11_operating_points, 11_net_benefit_at_prevalence, 11_patient_level_metrics, 11_lead_time_summary, external counterparts
12	Stored predictions only (refits nothing)	12_confirmatory_tests, 12_internal_ci, 12_subgroup_auroc_ci, _contrasts, 12_subgroup_labelsens_*, 12_altlabel_*, 12_h2_*, 12_analysis_register

Table B.1: Stage inputs and outputs. Which result each stage supports is recorded once, in the analysis register of Section 5.11, and is not repeated here. Stage 03 is the only stage that reads the full-size source tables, and Stage 12 is the only one that reads nothing but stored predictions, which is what allows the inference layer to be re-run and checked without refitting any model.

Naming conventions. A file’s numeric prefix is the stage that wrote it. A `_A..F` suffix denotes the label variant; `_ablated` denotes the Protocol Amendment 5 feature arm; `_random` denotes the random-split arm used only for H4. Files without a variant suffix are pooled across variants by construction. No stage overwrites a file written by another stage.